

Lin2-Xor Lemma: an OR-proof that leads to the membership proof and signature

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Abstract

This paper introduces **Lin2Selector**, a logarithmic-size membership proof for the relation $Z \in \text{Span}\{G_{2s}, G_{2s+1}\}$ over a public array of n linearly independent generator pairs $\{G_i\}_{i=0}^{2n-1}$. The protocol is obtained by iterating a new two-pair OR-reduction step, formalized as a standalone argument **Lin2Xor**. Our main technical contribution is the Lin2-Xor lemma, which establishes the required special soundness for this reduction and thereby enables a clean proof that **Lin2Selector** is a zero-knowledge argument of knowledge.

As an application, we instantiate two setup-free linkable ring signatures via the Fiat–Shamir transform in the random-oracle model: minimalistic **L2S-LRS**, of length $2\lceil\log_2(n)\rceil + 4$ group/scalar elements, and multisignature-friendly **L2S-LRS-MS**, of length $2\lceil\log_2(n)\rceil + 5$, supporting an LSAG-style key image. Both schemes rely only on the standard DLR/DDH assumptions in a black-box prime-order group and allow efficient verification dominated by a single multi-exponentiation, enabling standard batching optimizations.

The OR-reduction step **Lin2Xor** suggests a natural generalization to higher arity. This may enable even more compact one-out-of-many proofs/signatures with lengths approaching $\log_2(n) + O(\log_2 \log_2(n))$, as well as compact k -out-of-many variants; we leave these as open directions.

I. INTRODUCTION

A *proof of membership* convinces a verifier that a commitment opens to an element of a public reference set, without revealing which element is opened. In the special case where the set has two elements, the primitive is often referred to as an *OR-proof*. Membership proofs are central to anonymity mechanisms: they allow a party to prove that it possesses at least one valid secret key in a public *ring* of candidate keys while keeping the signing index hidden.

A closely related and widely deployed application is the *ring signature*. A ring signature on a message m with respect to a ring (or anonymity set) $\mathbf{K} = \{K_0, \dots, K_{n-1}\} \subset \mathbb{G}^*$ convinces a verifier that some member of $\mathbf{K}$ signed m , while hiding the signer’s identity. In many systems, it is additionally required that a secret key can sign at most once without being detected. Such schemes are called *linkable ring signatures* (LRS), where each signature contains a deterministic *key image* (a.k.a. linking tag). This tag remains invariant across all signatures produced by the same secret key, enabling public detection of double-signing.

Early formalization and constructions of OR-proofs date back to the work of Cramer, Damgård, and Schoenmakers [1]. While the survey in [2] outlines the state of the art, the tutorial in [3] illustrates a particularly elegant approach. Ring signatures were introduced by Rivest, Shamir, and Tauman [4] and refined later by Abe, Ohkubo, and Suzuki [5]. Practical linkable variants, such as CLSAG [6], stem from the LSAG signature by Liu, Wei, and Wong [7] and have become a backbone of privacy-oriented cryptocurrencies. A key appeal of these constructions is their *ad hoc* nature: there is no trusted setup, and a signer can form a ring without coordination with other ring members. However, in these classic schemes the proof (and hence the signature) grows linearly with n .

Over the past decade, a line of research has shown that *setup-free logarithmic-size* membership proofs and ring signatures are achievable in prime-order groups under standard assumptions. Two representative base techniques are the one-out-of-many proof by Groth and Kohlweiss [8] and the Bulletproofs inner-product argument by Bünz, Bootle, Boneh, Poelstra, Wuille, and Maxwell [9]. The recent efficient signatures [10]–[14] are built upon these techniques under the standard DLR/DDH assumptions in *black-box pairings-free* prime-order groups. Despite this progress, the question of which approach yields the most efficient LRS under *only* these assumptions in the simplest group remains open.

II. CONTRIBUTION

Our contribution is a novel base technique, distinct from the two widely used lines of work [8], [9], although partially inspired by them and other primitives such as Okamoto protocol [15] and Merkle tree [16]. Based on our technique, we construct a logarithmic-size proof of membership, **Lin2Selector**, along with two derived LRS schemes. At a high level, the technique can be viewed as two rearranged Okamoto-style protocols interlocked into a two-branch OR-proof, **Lin2Xor**. The latter is proven sound in our Lin2-Xor lemma and serves as a Bulletproofs-inspired folding reduction step; it recursively halves the anonymity set, effectively traversing a binary tree over the ring and retaining only the left or right branch at each level in a Merkle-tree-like manner.

We note that modestly stronger assumptions or additional algebraic structure can enable substantially more compact schemes, for example, a constant-size LRS under bilinear pairings [17], or an efficient scheme in a *non-black-box*

Scheme	Size (for n keys)	Verifier (dominant)	Key image	Multisig-friendly
CLSAG [6]	$n + 2$	$\text{mexp}(2n) + \text{hg}(n)$	$x\text{H}_{\text{gr}}(xG)$	✓
Triptych [10]	$3\lceil \log_2 n \rceil + 8$	$\text{mexp}(2n)$	U/x	✓ with Chaum-Pedersen
Lelantus Spark [13]	$3\lceil \log_2 n \rceil + 5$	$\text{mexp}(2n)$	U/x	
RingCT3.0 [11] (acc. [20])	$2\lceil \log_2 n \rceil + 14$	$\text{mexp}(2n)$	U/x	
Omniring [12] (acc. [20])	$2\lceil \log_2(n+2) \rceil + 9$	$\text{mexp}(2n) [+ \text{hg}(n)]$	$U/x, [x\text{H}_{\text{gr}}(xG)]$	[✓]
LLRing-DL [14]	$4\lceil \log_2 n \rceil$	$\text{mexp}(n) + \text{hg}(n)$	$x\text{H}_{\text{gr}}(xG)$	✓
L2S-LRS (this work)	$2\lceil \log_2 n \rceil + 4$	$\text{mexp}(3n) + \text{hg}(2n)$	$x^{-1}\text{H}_{\text{gr}}(xG)$	✓
L2S-LRS-MS (this work)	$2\lceil \log_2 n \rceil + 5$	$\text{mexp}(3n) + \text{hg}(2n)$	$x\text{H}_{\text{gr}}(xG)$	

TABLE I: Indicative comparison of DLR-based setup-free LRS in a black-box prime-order group: $\text{mexp}(n)$ denotes multi-exponentiation of n summands; $\text{hg}(n)$ represents n hash-to-group calls; H_{gr} is a hash-to-group function.

prime-order group model [18]. These directions are beyond the scope of our paper: we focus specifically on setup-free LRS under the DLR/DDH assumptions in the simplest setting of a black-box pairings-free prime-order group.

While this paper introduces the core technique, several directions regarding generalization, extensions, and post-quantum variants are left open. We present an one-out-of-many proof and its corresponding LRS, although a k -out-of- n extension appears feasible. We set a practical requirement for our LRS to meet a security model at least as strong as that of LSAG [7], alongside supporting an LSAG-style multisignature-friendly linking tag [19]. This allows the LRS to be viewed as a viable drop-in replacement for LSAG-type schemes.

Table I compares the state-of-the-art DLR-based setup-free LRS in a black-box pairings-free prime-order group with our proposed ones. Group elements and scalars are counted as one element each for a rough size comparison. Verification costs are presented at a high level (multi-exponentiations and hash calls), omitting lower-order terms. Our schemes achieve $2\lceil \log_2 n \rceil + O(1)$ space-efficiency while offering two distinct design points: the minimalistic L2S-LRS and the multisignature-friendly L2S-LRS-MS with an LSAG-style linking tag.

III. METHOD OVERVIEW

We provide the intuition behind the rearrangement of the Okamoto protocol and describe the derivation of the core reduction Lin2Xor. Then, we briefly explain how this leads to the membership proof Lin2Selector and its application as linkable ring signatures.

Recap: Okamoto protocol. Given two linearly independent generators $G_0, G_1 \in \mathbb{G}^*$ and a statement $Z \in \mathbb{G}$, the Okamoto identification protocol [15], [21] proves knowledge of (z_0, z_1) such that $Z = z_0G_0 + z_1G_1$. For convenience, the protocol steps are depicted in Figure 1. The verifier checks a single equation of the form

$$uG_0 + vG_1 = A + cZ,$$

with $A \in \mathbb{G}$ the initial message, $c \in \mathbb{Z}_p$ the public-coin challenge, and $(u, v) \in \mathbb{Z}_p^2$ the response. Once this equation is satisfied, the verifier is convinced that $\{A, Z\} \subseteq \text{Span}(G_0, G_1)$, a result that subsumes the sought identification $Z = z_0G_0 + z_1G_1$.

Warm-up: Lin2 as a rearranged Okamoto protocol. Note that for any linearly independent $H, Z \in \mathbb{G}^*$, by linear algebra $\{H, Z\} \subseteq \text{Span}(G_0, G_1)$ holds if and only if $\{G_0, G_1\} \subseteq \text{Span}(H, Z)$. Therefore, the Okamoto protocol can be viewed as operating on H, Z as the basis, with G_0 and G_1 acting as the statement and initial message, respectively. In this case, the verification equation is

$$uH + vZ = G_0 + cG_1. \quad (1)$$

By symbolically rewriting eq. (1) with $a := 1/v$ and $r := u/v$, we obtain the modified verification equation

$$a(G_0 + cG_1) = Z + rH, \quad (2)$$

where the prover provides (a, r) as the response instead of (u, v) . The new response (a, r) constitutes a reparameterization of the original protocol. Provided $a \neq 0$, it allows the verifier to recover (u, v) ; consequently, fulfillment of eq. (2) implies that eq. (1) holds, thus convincing the verifier that $\{G_0, G_1\} \subseteq \text{Span}(H, Z)$ and, hence, $\{H, Z\} \subseteq \text{Span}(G_0, G_1)$.

Building on these observations, our rearranged Okamoto protocol, shown in Figure 2 and denoted Lin2, proves knowledge of (z_0, z_1) such that $Z = z_0G_0 + z_1G_1$. The protocol requires the prover to send $H \in \mathbb{G}$ and, upon receiving c , provide (a, r) . The verifier checks that $a \neq 0$ and eq. (2) holds, and concludes that $\{H, Z\} \subseteq \text{Span}(G_0, G_1)$, particularly that the sought relation $Z = z_0G_0 + z_1G_1$ is satisfied.

Overall, the **Lin2** protocol is defined for the NP relation detailed in (11):

$$\mathcal{R}_{\text{Lin2}} = \{ (G_0, G_1, Z) \in \mathbb{G}^{*3} ; (z_0, z_1) \in \mathbb{Z}_p^{2*} \mid Z = z_0 G_0 + z_1 G_1 \}.$$

In Section VIII, we formally prove that **Lin2** has special soundness and, for H sampled independently and uniformly from $\mathbb{G} \setminus \text{Span}(Z)$, is complete and special honest-verifier zero-knowledge (sHVZK). Crucially, all three properties are established exclusively over the domain $Z \in \mathbb{G}^*$.

Informally, completeness of **Lin2** over this domain is straightforward: for any $Z \neq 0$, it suffices to choose $H \notin \text{Span}(Z)$ to satisfy eq. (2) by solving for appropriate $a \in \mathbb{Z}_p^*$ and $r \in \mathbb{Z}_p$. Soundness for $Z, H \in \mathbb{G}^*$ such that $H \notin \text{Span}(Z)$ is also direct; the validity of eq. (2) with $a \neq 0$ implies the original Okamoto protocol accepts, ensuring $\{H, Z\} \subseteq \text{Span}(G_0, G_1)$.

Simultaneously, the cases where $Z = 0$, $H = 0$, or $H \in \text{Span}(Z)$ have only negligible probability of satisfying eq. (2) with $a \neq 0$. Thus, an accepting verifier is convinced with overwhelming probability that $Z \neq 0$, $H \neq 0$, and $H \notin \text{Span}(Z)$. This ensures that the protocol is ultimately sound for $Z \in \mathbb{G}^*$. Notably, if the verifier accepts, it is convinced that both $Z \in \mathbb{G}^*$ and $Z \in \text{Span}(G_0, G_1)$ hold.

The Lin2Xor construction: two instances of Lin2 interlocked. We have already introduced a two-generator protocol, **Lin2**, defined for $Z \in \mathbb{G}^*$, whose verifier checks a single relation of the form eq. (2). There, (G_0, G_1) is the basis, $Z \in \mathbb{G}^*$ is the statement, $H \in \mathbb{G}$ is the initial message, c is public-coin, and (a, r) is the response such that $a \neq 0$. This form is particularly convenient for constructing an OR-protocol.

The two-pair protocol **Lin2Xor** in Figure 3 lifts this form to a four-generators basis (G_0, G_1, G_2, G_3) : the verifier checks

$$a_0(G_0 + c G_1) + a_1(G_2 + c^2 G_3) = Z + rH, \quad (3)$$

where G_0, G_1, G_2, G_3 are linearly independent, $Z \in \mathbb{G}^*$, $H \in \mathbb{G}$ as before, and (a_0, a_1, r) is the updated response. The critical design choice is that the two summands use distinct challenge monomials (c vs. c^2), which enables the **Lin2-Xor** lemma: except with negligible probability, any accepting execution behaves as if exactly one summand is active, yielding an OR-proof over the two generator pairs, namely, convincing the verifier that $Z \in \text{Span}(G_{2s}, G_{2s+1})$ for some $s \in \{0, 1\}$.

Recalling that **Lin2** requires $a \neq 0$, the mechanism for such an activation becomes clear: for $s \in \{0, 1\}$, if $a_s \neq 0$, the branch defined by the summand with (G_{2s}, G_{2s+1}) is active; otherwise, it is inactive. To exclude the case where both a_0 and a_1 vanish, the verifier also checks that

$$Z + rH \neq 0. \quad (4)$$

Thus, the objective of the **Lin2-Xor** lemma mainly reduces to showing that the case $a_0, a_1 \neq 0$ is impossible. Once the condition $a_0 = 0 \oplus a_1 = 0$ is established, soundness of the **Lin2Xor** protocol immediately follows from soundness of **Lin2**; namely, if the **Lin2Xor** verifier accepts, it is convinced that both $Z \in \mathbb{G}^*$ and $Z \in \text{Span}(G_{2s}, G_{2s+1})$, $s \in \{0, 1\}$ hold.

Suppose, toward a contradiction, that $a_0, a_1 \neq 0$ and yet the verifier accepts. Using a standard rewinding argument [22], [23] to eliminate H from two accepting transcripts, one finds that $Z \in \text{Span}(G_0, G_1, G_2, G_3)$, which in turn implies $H \in \text{Span}(G_0, G_1, G_2, G_3)$. Therefore, we can consider eq. (3) componentwise. By introducing the unique decompositions

$$Z = Z_{01} + Z_{23}, \quad \text{where } Z_{01} \in \text{Span}(G_0, G_1) \text{ and } Z_{23} \in \text{Span}(G_2, G_3),$$

$$H = H_{01} + H_{23}, \quad \text{where } H_{01} \in \text{Span}(G_0, G_1) \text{ and } H_{23} \in \text{Span}(G_2, G_3),$$

and substituting them into eq. (3), we obtain the following pair of equalities:

$$a_0(G_0 + c G_1) = Z_{01} + rH_{01}, \quad (5)$$

$$a_1(G_2 + c^2 G_3) = Z_{23} + rH_{23}, \quad (6)$$

each representing an accepting **Lin2** instance, notably constrained by the shared response r .

Intuitively, as the challenge monomials c and c^2 exhibit probabilistic independence (cf. [24]), it seems unlikely that the same response r can fit both eqs. (5) and (6) simultaneously. It follows that, except with negligible probability, at most one of the branches can be active. We implement this intuition through a technically more direct derivation, without explicitly appealing to the distribution of r . Below is an overview of the formal approach.

Lin2-Xor/Lemma 11 formal proof method. In Section IX, we formally define **Lin2Xor** as a protocol for the NP relation given in (15):

$$\mathcal{R}_{\text{Lin2xor}} = \{ (G_0, G_1, G_2, G_3, Z) \in \mathbb{G}^{*5} ; (z_0, z_1, s) \in \mathbb{Z}_p^{2*} \times \{0, 1\} \mid Z = z_0 G_{2s} + z_1 G_{2s+1} \}.$$

Completeness of this protocol is trivial.

Our core technical result, Lemma 12 (**Lin2-Xor**), establishes that **Lin2Xor** is sound (specifically, has special soundness). Its most complex part, showing that the case $a_0, a_1 \neq 0$ in eq. (3) is infeasible, is presented in Lemma 11. The proof

of Lemma 11 demonstrates that a prover is unable to satisfy the Lin2Xor verification equation eq. (3) with $a_0, a_1 \neq 0$ except by breaking the DLR assumption for (G_0, G_1, G_2, G_3) .

Using a standard rewinding argument, a DLR-breaker algorithm $\mathcal{B}$ obtains three accepting Lin2Xor transcripts τ', τ'', τ sharing the prefix $(G_0, G_1, G_2, G_3, Z, H)$ such that $a_0, a_1 \neq 0$ in each. By picking weights $\mu', \mu'' \in \mathbb{Z}_p$ such that $\mu' + \mu'' = 1$ and $\mu' r' + \mu'' r'' = r$, $\mathcal{B}$ eliminates Z and H from the weighted sum of the verification equations. This yields a candidate DLR solution $\sum_{i=0}^3 \gamma_i G_i = 0$, where the coefficients γ_i are defined as:

$$\begin{cases} \gamma_0 &:= \mu' a'_0 + \mu'' a''_0 - a_0, \\ \gamma_1 &:= \mu' a'_0 c' + \mu'' a''_0 c'' - a_0 c, \\ \gamma_2 &:= \mu' a'_1 + \mu'' a''_1 - a_1, \\ \gamma_3 &:= \mu' a'_1 c'^2 + \mu'' a''_1 c''^2 - a_1 c^2. \end{cases} \quad (7)$$

We can estimate the probability of the event $\gamma_0 = \gamma_1 = \gamma_2 = \gamma_3 = 0$ in (7). When all γ_i vanish, it is possible to simplify (7) by eliminating the weights μ' and μ'' .

Indeed, the conditions $\gamma_0 = \gamma_1 = 0$ in (7) correspond to the weighted sum of equations of type eq. (5), where the resulting right-hand side vanishes due to the choice of (μ', μ'') . Simultaneously, the conditions $\gamma_2 = \gamma_3 = 0$ correspond to the weighted sum of equations of type eq. (6), whose right-hand side similarly vanishes under the same weights. Consequently, (7) rewrites as the following two 2×2 linear systems:

$$\underbrace{\begin{bmatrix} a'_0 & a''_0 \\ a'_0 c' & a''_0 c'' \end{bmatrix}}_{=:V} \begin{bmatrix} \mu' \\ \mu'' \end{bmatrix} = \begin{bmatrix} a_0 \\ a_0 c \end{bmatrix}, \quad \underbrace{\begin{bmatrix} a'_1 & a''_1 \\ a'_1 c'^2 & a''_1 c''^2 \end{bmatrix}}_{=:W} \begin{bmatrix} \mu' \\ \mu'' \end{bmatrix} = \begin{bmatrix} a_1 \\ a_1 c^2 \end{bmatrix}, \quad (8)$$

where V and W are invertible with overwhelming probability. By eliminating the vector $[\mu', \mu'']^\top$ from the systems (8), we obtain the substantially simplified (7):

$$a_1 \begin{bmatrix} 1 \\ c^2 \end{bmatrix} = a_0 \cdot T \begin{bmatrix} 1 \\ c \end{bmatrix}, \quad \text{where } T = WV^{-1} \text{ is an invertible matrix that is invariant across all possible } \tau. \quad (9)$$

As a_0, a_1 are both nonzero and T is non-singular and independent of the third challenge c , eq. (9) inevitably reduces to a polynomial in c of degree at most 3, whose coefficients are independent of a_0, a_1 . Such a polynomial has at most three roots in $\mathbb{Z}_p$, meaning that eq. (9) holds with probability $\leq 3/p$. Thus, the event $\gamma_0 = \gamma_1 = \gamma_2 = \gamma_3 = 0$ occurs only with negligible probability.

Therefore, with non-negligible probability, the sum $\sum_{i=0}^3 \gamma_i G_i = 0$ represents a non-trivial linear combination, providing a solution to the DLR for (G_0, G_1, G_2, G_3) and, thus, concluding the proof.

The Lin2Selector construction: a Merkle tree composed of Lin2Xor's. Thus far, we have constructed the Lin2Xor protocol over a four-generator basis (G_0, G_1, G_2, G_3) . It is complete for $Z \in \mathbb{G}^*$ and, as demonstrated, sound for the relation $Z \in \text{Span}(G_{2s}, G_{2s+1})$ for some $s \in \{0, 1\}$, regardless of any assumptions about Z . However, the prover's response (a_0, a_1, r) to the challenge c reveals the witness index s , as one of a_0, a_1 , concretely, a_{1-s} , is zero. Hence, the protocol is not zero-knowledge and may appear of limited use in its raw form.

To achieve zero-knowledge, we can modify the protocol such that the prover withholds (a_0, a_1) and instead provides r alongside a zero-knowledge proof of knowledge of (a_0, a_1) . For instance, we can use Lin2 for this purpose. In such a combined construction, the Lin2Xor component provides the statement $Z \in \mathbb{G}^*$, first message $H_0 \in \mathbb{G}$, challenge c_0 , and response part r_0 . To these, we append the Lin2 components: first message $H_1 \in \mathbb{G}$, challenge c_1 , and full response (a, r_1) . The verifier checks that the unfolded eq. (2), together with the condition on a , holds:

$$a(G_0 + c_0 G_1 + c_1(G_2 + c_0^2 G_3)) = Z + r_0 H_0 + r_1 H_1, \quad \text{where } a \neq 0.$$

Due to the soundness of Lin2, as previously shown, this convinces the verifier of the existence of two scalars, which we call a_0, a_1 here, such that:

$$a_0(G_0 + c_0 G_1) + a_1(G_2 + c_0^2 G_3) = Z + r_0 H_0, \quad \text{where } (Z + r_0 H_0) \neq 0.$$

Notably, this is the exact acceptance criterion of Lin2Xor. Consequently, by the soundness of Lin2Xor, the verifier is convinced that

$$Z = z_0 G_{2s} + z_1 G_{2s+1}, \quad \text{where } s \in \{0, 1\} \text{ and } Z \in \mathbb{G}^*.$$

Thus, we have obtained a 2-round zero-knowledge OR-proof, where Lin2Xor plays the first round, and Lin2 the second.

For an eight-generator basis $(G_0, \dots, G_7)$, we obtain a 3-round zero-knowledge proof of membership with the round schedule Lin2Xor-Lin2Xor-Lin2. The combined verification equation is:

$$a \left[(G_0 + c_0 G_1 + c_1 (G_2 + c_0^2 G_3)) + c_2 (G_4 + c_0 G_5 + c_1^2 (G_6 + c_0^2 G_7)) \right] = Z + r_0 H_0 + r_1 H_1 + r_2 H_2,$$

where $a \neq 0$. By the soundness of Lin2, this convinces the verifier of the existence of scalars $(a_0^{(1)}, a_1^{(1)}) \in \mathbb{Z}_p^2$ such that:

$$a_0^{(1)} (G_0 + c_0 G_1 + c_1 (G_2 + c_0^2 G_3)) + a_1^{(1)} (G_4 + c_0 G_5 + c_1^2 (G_6 + c_0^2 G_7)) = Z + r_0 H_0 + r_1 H_1,$$

with a nonzero right-hand side. Applying the soundness of Lin2Xor iteratively, the verifier is convinced that only one branch remains active at each level. From the above equality, where only one of $a_0^{(1)}, a_1^{(1)}$ is nonzero, for some $(a_0^{(0)}, a_1^{(0)}) \in \mathbb{Z}_p^2$, $s_1 \in \{0, 1\}$, the verifier is convinced that

$$a_0^{(0)} (G_{4s_1} + c_0 G_{4s_1+1}) + a_1^{(0)} (G_{4s_1+2} + c_0^2 G_{4s_1+3}) = Z + r_0 H_0.$$

Finally, a second application of Lin2Xor soundness ensures that $Z = z_0 G_{2s} + z_1 G_{2s+1}$ for a hidden index $s = 2s_1 + s_0$ and $Z \in \mathbb{G}^*$. Thus, we have obtained a zero-knowledge membership proof for a set of four generator pairs.

The pattern becomes apparent: an accepting verifier effectively traverses the tree from the root. At each step, the top-level term $r_L H_L$ is removed while the set of possible generator pairs is halved, with the verifier remaining unaware of which branch is active. This structure is reminiscent of a Merkle tree, albeit performed in zero-knowledge. Ultimately, this process isolates the pair (G_{2s}, G_{2s+1}) , convincing the verifier of the existence of the witness index s and scalar coefficients (z_0, z_1) such that $Z = z_0 G_{2s} + z_1 G_{2s+1}$.

In Section X, we formally develop this construction into a one-out-of- n proof among n generator pairs. The resulting protocol, shown in Figure 4, is proven to be complete, sound, and sHVZK over the domain $Z \in \mathbb{G}^*$ with respect to the NP relation in (16):

$$\mathcal{R}_{\text{Lin2sel}} = \left\{ ((G_0, \dots, G_{2n-1}), Z) \in \mathbb{G}^{*2n} \times \mathbb{G}^* ; (z_0, z_1, s) \in \mathbb{Z}_p^{2*} \times [0, n-1] \mid Z = z_0 G_{2s} + z_1 G_{2s+1} \right\}.$$

Instantiating linkable ring signatures. We obtain LRS schemes by applying the Fiat–Shamir transform [22], [25] to Lin2Selector in the random-oracle model (ROM) [26]. Given a ring of public keys $\mathbf{K} = (K_0, \dots, K_{n-1}) \subset \mathbb{G}^*$ and message m , the signer and verifier derive a public generator array $\mathbf{G} = (P_0, Q_0, \dots, P_{n-1}, Q_{n-1})$ from $\mathbf{K}$ and random-oracle outputs. The verifier then verifies a non-interactive Lin2Selector transcript proving that a public statement Z decomposes over a single pair (P_s, Q_s) for some hidden index s .

Our basic LRS scheme L2S-LRS, shown in Figure 5, uses a linking tag $\text{Kl} := x^{-1} \text{H}_{\text{gr}}(K_s)$. With the random oracles H_{sc} (hash-to-scalar) and H_{gr} (hash-to-group), for all $i \in [0, n-1]$, the scheme defines the public generators P_i, Q_i and statement Z as:

$$\begin{aligned} P_i &:= K_i + \xi \cdot \text{H}_{\text{gr}}(K_i), \\ Q_i &:= \text{H}_{\text{gr}}(\text{Kl}, K_i), \\ Z &:= G + \xi \text{Kl}, \end{aligned}$$

where $\xi := \text{H}_{\text{sc}}(m, G, \mathbf{K}, \text{Kl})$. The element $G \in \mathbb{G}^*$ is a predefined group generator such that $K_s = xG$ for the signing index s and corresponding private-public key pair (x, K_s) .

Having checked the non-interactive Lin2Selector transcript, the verifier is convinced that $Z = z_0 P_s + z_1 Q_s \in \mathbb{G}^*$ for some hidden index s . Crucially, the properties of the hash-to-group random oracle (denoted as H_{gr}) preclude Q_i from fitting this relation; otherwise, an H_{gr} output would be representable as a linear combination of group elements known before the query was made. Thus, the verifier is convinced that $z_1 = 0$, which implies $Z = z_0 P_s$, or equivalently:

$$G + \xi \text{Kl} = z_0 (K_s + \xi \cdot \text{H}_{\text{gr}}(K_s)).$$

Because $\text{H}_{\text{gr}}(K_s)$ is linearly independent of K_s owing to the same properties of the hash-to-group random oracle, the verifier is convinced that both

$$G = z_0 K_s \quad \text{and} \quad \text{Kl} = z_0 \text{H}_{\text{gr}}(K_s)$$

hold. This convinces the verifier that the signer knows a private key $x = z_0^{-1}$ such that K_s is the corresponding public key, while the linking tag Kl is correctly computed as $x^{-1} \text{H}_{\text{gr}}(K_s)$.

Our second LRS scheme L2S-LRS-MS, shown in Figure 6, uses a multisignature-friendly [19] LSAG-style [6], [7] linking tag $\text{KI} := x\text{H}_{\text{gr}}(K_s)$. For all $i \in [0, n-1]$, the scheme defines the public generators P_i, Q_i as:

$$\begin{aligned} P_i &:= \text{H}_{\text{gr}}(\text{KI}, K_i) + \xi K_i + \xi^2 \text{KI}, \\ Q_i &:= \xi G + \xi^2 \text{H}_{\text{gr}}(K_i), \end{aligned}$$

where $\xi := \text{H}_{\text{sc}}(m, G, \mathbf{K}, \text{KI}, Z)$; note that G, K_s, KI , and Z are included into this query for ξ . By the properties of the hash-to-group oracle, $\text{H}_{\text{gr}}(\text{KI}, K_s)$ is linearly independent of $\{G, K_s, \text{KI}, \text{H}_{\text{gr}}(K_s)\}$, and $\text{H}_{\text{gr}}(K_s)$ is linearly independent of $\{G, K_s, \text{H}_{\text{gr}}(\text{KI}, K_s)\}$. Using rewindings in the ROM we show that, informally, Z must be proportional to $\text{H}_{\text{gr}}(\text{KI}, K_s)$, whereas $z_1 Q_s$ must annihilate $z_0(\xi K_s + \xi^2 \text{KI})$. Furthermore, we show that the latter simultaneously implies:

$$K_s = xG \quad \text{and} \quad \text{KI} = x\text{H}_{\text{gr}}(K_s), \quad \text{for } x = -z_1/z_0,$$

ensuring that the signer knows the private key x such that K_s is the corresponding public key and KI is a correctly computed linking tag.

The details of these LRS constructions are formalized in Sections XII and XIII. Specifically, Lemmas 21 and 26 establish rigorous witness-binding for L2S-LRS and L2S-LRS-MS, respectively. These sections also contain formal proofs that both schemes are correct, linkable, EUF-CMA secure, and signer-ambiguous with respect to the NP relations (26) and (32) in the ROM.

IV. COMPARISON WITH PRIOR WORK

In the setting of a simple prime-order group, we position Lin2Xor, Lin2Selector, and the resulting signatures relative to existing membership-proof and ring-signature methods. A quantitative comparison is already given in Table I.

Comparison with classical schemes. Our Lin2Xor OR-proof mechanism appears structurally distinct from both the classical parallel OR-proofs [1]–[3], which typically split or correlate challenges across branches, and the sequential OR-proof style used in early ring signatures [5], [7]. In our protocols, challenges are handled in a direct Schnorr-like manner, and the Fiat–Shamir transform is applied over public-coin transcripts without circular dependencies.

Comparison with Bulletproofs technique. The closest high-level reduction paradigm to our Lin2Xor/Lin2Selector construction is the Bulletproofs reduction by Bünz et al. [9]. For example, in the 2×2 inner-product setting, omitting the inner-product-carrying term, the Bulletproofs core reduction can be written as

$$Z + c^2 L + c^{-2} R = w_0(cP_0 + c^{-1}P_1) + w_1(c^{-1}Q_0 + cQ_1),$$

where L, R are prover messages and c is a challenge. At a superficial level, one might attempt to emulate our Lin2Xor behavior by replacing (L, R) with (H, r) so as to obtain an equation of the form $Z + rH = \dots$ while hiding which “degree” is active. However, this does not enforce the XOR structure: two random transformations with dependent randomness (equivalent to c^2) are applied to both pairs, and the resulting equation admits multiple solutions where all coefficients w_0, w_1 are nonzero.

The essential distinction in our Lin2Xor step is that the two branches are weighted by distinct challenge monomials (c vs. c^2) exhibiting probabilistic independence, which forces one branch to become inactive in any accepting transcript. This asymmetry is precisely what enables the Lin2-Xor lemma and, when combined with Bulletproofs-style folding (17), leads to the compact Lin2Selector one-out-of-many proof.

Comparison with Inner-product based schemes. The primary feature of Bulletproofs by Bünz et al. [9] is the succinct inner-product argument, which enables proving knowledge of a solution to a large system of degree-2 multivariate equations, specifically a secret input to an arithmetic circuit. Logarithmic-size LRS in a black-box prime-order group, such as Omniring [12], RingCT 3.0 [11], or LLRing-DL [14], rely entirely on the inner-product backend, particularly to prove that a vector commitment opening is a one-hot vector.

By contrast, our Lin2Selector membership proof does not rely on any inner-product backend and is independent of such techniques even in the broad sense of [9], [27], [28], using the Lin2Xor OR-proof instead.

Comparison with Groth–Kohlweiss based schemes. Another line of setup-free logarithmic-size LRS in a black-box prime-order group is represented by the Triptych [10] and Lelantus Spark [13] schemes, which originate from the technique of Groth and Kohlweiss [8].

The work [8] introduced a mechanism resembling the Kronecker delta to select a member of an anonymity set without revealing it. Our signatures use Lin2Xor and, consequently, Lin2Selector in a functionally equivalent role. However, there is a fundamental difference in how the anonymity sets are constructed. While the anonymity sets in [8] are defined by linear combinations of two linearly independent generators, the sets for Lin2Selector are themselves composed of linearly independent generators.

Note on DualRing based schemes. The DualRing framework by Yuen et al. [20] introduced a different paradigm for compact setup-free ring signatures, featuring a dual-ring structure of commitments and challenges. While this generic construction is promising, particularly for post-quantum instantiations [29], the discrete-logarithm variants (e.g., DualRing-EC [20]) impose restrictive assumptions on the public keys. Specifically, they require the absence of any non-trivial relations between the ring keys and the generator G , beyond the standard $pk = sk \cdot G$.

To illustrate, the DualRing-EC signature can be successfully forged for a ring $\{pk_0, pk_1\}$ without knowledge of the corresponding secret keys, provided the signer knows a non-trivial DLR relation such as $pk_0 + pk_1 = \alpha G$. By setting $R = rG$, $c = H_Z(m, \mathbf{pk}, R)$, $z = r - c\alpha/2$, and $\{c_0, c_1\} = \{c/2, c/2\}$, a valid signature is produced.

We view such an assumption regarding the independence of ring keys as a significant "efficiency booster" and therefore exclude DualRing-EC and its derivatives from our performance comparison. Notably, our LRS schemes are proven sound for relations (26) and (32) without restrictive assumptions on the ring keys $(K_0, \dots, K_{n-1}) \in \mathbb{G}^{*n}$ (requiring only that they are distinct), thereby allowing the signer to know linear relations between the ring keys and other group elements.

Note on Curve Trees and a non-black-box group. While our primary focus is on methods in black-box prime-order groups, the Curve Trees membership proof [18] warrants a brief comparison despite its non-black-box use of the underlying group.

Curve Trees target extremely large anonymity sets and achieve impressive efficiency at that scale. This approach fundamentally relies on non-black-box access to a group of elliptic curve points, which are represented by coordinates over a field, and the prover/verifier reason about curve equations using an inner-product-based circuit. The scheme requires specific elliptic curve families (e.g., cycles of curves) to compose commitments across its layers.

In contrast, Lin2Selector is designed for the black-box prime-order group model used by Schnorr, Okamoto, and Bulletproofs-style protocols; it requires only group operations and random-oracle sampling into the group and field. This makes Lin2Selector directly portable to any suitable group abstraction, regardless of the underlying group structure and independent of specific curve implementations.

V. PRELIMINARIES

In what follows, we adopt the standard notation and security definitions of Bünz et al. [9], Bootle et al. [23], Lindell [30], and Groth [31], except where noted.

All algorithms and adversaries are restricted to probabilistic polynomial time (PPT) in the security parameter λ . We often omit explicit mention of λ when it is clear from context.

Let $\mathbf{GGen}$ be a PPT algorithm that, on input 1^λ , returns a group description $\mathbf{GGD} = (\mathbb{G}, p)$, where $\mathbb{G}$ is a cyclic group of prime order $p \geq 2^\lambda$ written additively, with the corresponding scalar field $\mathbb{Z}_p$.

VI. DEFINITIONS

Definition 1 (Discrete Logarithm (DL)). For every PPT adversary $\mathcal{A}$,

$$\Pr \left[\mathbf{GGD} \leftarrow \mathbf{GGen}(\{1\}^\lambda); G, G_0 \xleftarrow{\$} \mathbb{G}; a_0 \leftarrow \mathcal{A}(\mathbf{GGD}, G, G_0) : a_0 G = G_0 \right] \approx 0.$$

Definition 2 (Discrete Logarithm Relation (DLR)). For every $n \geq 2$ and every PPT adversary $\mathcal{A}$,

$$\Pr \left[\mathbf{GGD} \leftarrow \mathbf{GGen}(\{1\}^\lambda); G_1, \dots, G_n \xleftarrow{\$} \mathbb{G}; (a_1, \dots, a_n) \leftarrow \mathcal{A}(\mathbf{GGD}, G_1, \dots, G_n) : \exists i (a_i \neq 0) \wedge \sum_{i=1}^n a_i G_i = 0 \right] \approx 0.$$

It is a standard result (cf. [23]) that the DL and DLR assumptions are equivalent in cyclic groups of prime order. Consequently, we use the term DLR to denote both assumptions throughout this paper.

Linearly independent group elements. For a set of group elements $\{G_1, \dots, G_n\}$ subject to the DLR assumption, we write $\text{LI}(G_1, \dots, G_n)$ as a shorthand and refer to such elements as linearly independent.

Commitments and their properties. We utilize standard commitment schemes and properties from [9], specifically the Pedersen Commitment, based on the DLR; explicit definitions are omitted to conserve space.

Definition 3 (Public-coin protocol). A triple $(\text{Setup}, \mathcal{P}, \mathcal{V})$ is *public coin* if every message sent from $\mathcal{V}$ to $\mathcal{P}$ is sampled uniformly at random and independently of $\mathcal{P}$'s messages; equivalently, verifier challenges are determined solely by $\mathcal{V}$'s randomness ρ .

Definition 4 (Completeness). A protocol $(\text{Setup}, \mathcal{P}, \mathcal{V})$ has *perfect completeness* for a relation $\mathcal{R}$ if for all PPT adversaries $\mathcal{A}$,

$$\Pr \left[\sigma \leftarrow \text{Setup}(\{1\}^\lambda); (u, w) \leftarrow \mathcal{A}(\sigma) : (u, w) \in \mathcal{R} \wedge \langle \mathcal{P}(\sigma, u; w), \mathcal{V}(\sigma, u) \rangle = 0 \right] = 0.$$

If the probability above is negligible, the protocol is said to have *completeness* rather than perfect completeness.

Definition 5 (Computational Witness-Extended Emulation (cWEE)). A public-coin protocol $(Setup, \mathcal{P}, \mathcal{V})$ for relation $\mathcal{R}$ has *computational witness-extended emulation* if for every deterministic polynomial-time prover strategy $\mathcal{P}^*$, there exists an expected polynomial-time emulator $\mathcal{E}$ such that for all interactive PPT adversaries $\mathcal{A}_1, \mathcal{A}_2$,

$$\Pr \left[\sigma \leftarrow Setup(\{1\}^\lambda); (u, s) \leftarrow \mathcal{A}_2(\sigma); tr \leftarrow \langle \mathcal{P}^*(\sigma, u, s), \mathcal{V}(\sigma, u) \rangle : \mathcal{A}_1(tr) = 1 \right] \\ \approx \Pr \left[\sigma \leftarrow Setup(\{1\}^\lambda); (u, s) \leftarrow \mathcal{A}_2(\sigma); (tr, w) \leftarrow \mathcal{E}^\mathcal{O}(\sigma, u) : \mathcal{A}_1(tr) = 1 \wedge (tr \text{ accepting} \Rightarrow (u, w) \in \mathcal{R}) \right],$$

where $\mathcal{O}$ is a rewinding oracle for interactions with $\mathcal{P}^*$ on input (σ, u, s) , allowing $\mathcal{E}$ to rewind to a chosen point and resume with fresh verifier randomness thereafter.

Following standard practice (e.g., [9], [23]), we do not prove cWEE directly; rather, we establish the existence of an efficient witness extractor (or prove *k-special soundness* [8]) and invoke the Forking Lemma [9], [22], [26], [32] to conclude this property.

Definition 6 (Witness extractor). Let $(Setup, \mathcal{P}, \mathcal{V})$ be a public-coin $(2k - 1)$ -move interactive protocol for relation $\mathcal{R}$. A PPT algorithm $\mathcal{X}$ is a *witness extractor* if, given an accepting transcript tree of branching profile $(n_1, \dots, n_k)$ with $\prod_{i=1}^k n_i$ polynomially bounded in λ , $\mathcal{X}$ outputs a witness w such that $(u, w) \in \mathcal{R}$ except with negligible probability.

For 3-move protocols, this definition reduces to *k-special soundness* (cf. [8], [21]), where the branching profile is simply (k) . Since the construction of an efficient witness extractor $\mathcal{X}$ (or, equivalently, proving *k-special soundness*) implies cWEE via standard reduction [8], [9], [23], we henceforth conclude the cWEE property directly from the existence of $\mathcal{X}$.

Definition 7 (Special Honest-Verifier Zero-Knowledge (sHVZK)). A public-coin protocol $(Setup, \mathcal{P}, \mathcal{V})$ for relation $\mathcal{R}$ is *special honest-verifier zero-knowledge* if there exists a PPT simulator $\mathcal{S}$ such that for all (non-uniform) PPT adversaries $\mathcal{A}$, it holds that

$$\Pr \left[\sigma \leftarrow Setup(\{1\}^\lambda); (u, w, \rho) \leftarrow \mathcal{A}(\sigma); tr \leftarrow \langle \mathcal{P}(\sigma, u; w), \mathcal{V}(\sigma, u; \rho) \rangle : (u, w) \in \mathcal{R} \wedge \mathcal{A}(tr) = 1 \right] \\ \approx \Pr \left[\sigma \leftarrow Setup(\{1\}^\lambda); (u, w, \rho) \leftarrow \mathcal{A}(\sigma); tr \leftarrow \mathcal{S}(\sigma, u, \rho) : (u, w) \in \mathcal{R} \wedge \mathcal{A}(tr) = 1 \right],$$

where ρ denotes the verifier's randomness (chosen adversarially in the sHVZK game, as in [31]).

When a protocol satisfies sHVZK, we call it a zero-knowledge protocol, as a shorthand.

Definition 8 (Argument of knowledge). A public-coin protocol $(Setup, \mathcal{P}, \mathcal{V})$ is an *argument of knowledge* for relation $\mathcal{R}$ if it has completeness (Definition 4) and cWEE (Definition 5).

We do not require perfect completeness in this definition; according to Groth [31], completeness suffices.

Auxiliary shorthand for a line in the field. We define the following shorthand for a line in $\mathbb{Z}_p^2$ passing through the origin, where $(\alpha, \beta) \in \mathbb{Z}_p^{2*} := \mathbb{Z}_p^2 \setminus \{(0, 0)\}$:

$$\text{Line}(\alpha, \beta) = \{ (x, y) \in \mathbb{Z}_p^2 \mid \beta x = \alpha y \} \subset \mathbb{Z}_p^2.$$

VII. OKAMOTO IDENTIFICATION PROTOCOL RECAP

The Okamoto identification protocol [15], [21] is a 3-move, public-coin argument of knowledge of two discrete logarithms with respect to two generators. We outline its salient properties below.

Let $G_0, G_1 \in \mathbb{G}^*$ be two independently sampled group elements. According to the DLR, they are linearly independent, which we denote as $\text{LI}(G_0, G_1)$. We write the protocol's statement-witness relation in the standard NP form:

$$\mathcal{R}_{\text{Oka}} = \{ (G_0, G_1, Z) \in \mathbb{G}^{*2} \times \mathbb{G} ; (z_0, z_1) \in \mathbb{Z}_p^2 \mid Z = z_0 G_0 + z_1 G_1 \}. \quad (10)$$

An interactive scheme of the protocol is shown in Figure 1. Its common input is (G_0, G_1, Z) with GGD implicitly included. Prover has the witness (z_0, z_1) . The standard completeness, cWEE (special soundness), and sHVZK properties of this protocol are established by the following claims.

Lemma 1 (Oka perfect completeness). *For any $(G_0, G_1, Z; z_0, z_1) \in \mathcal{R}_{\text{Oka}}$, for the honest prover, the verifier in Figure 1 accepts with probability 1.*

Lemma 2 (Oka 2-special soundness). *Assume $\text{LI}(G_0, G_1)$. Given two accepting Oka transcripts with the same G_0, G_1, A and two distinct challenges $c \neq c'$,*

$$(A, c, u, v) \quad \text{and} \quad (A, c', u', v'),$$

an extractor unconditionally recovers a witness (z_0, z_1) such that $Z = z_0 G_0 + z_1 G_1$.

Prover $\mathcal{P}$	Verifier $\mathcal{V}$	
Common input: statement $(G_0, G_1, Z) \in \mathbb{G}^{*2} \times \mathbb{G}$ with $Z = z_0 G_0 + z_1 G_1$.		
$\mathcal{P}$'s private input: witness $(z_0, z_1) \in \mathbb{Z}_p^2$.		
1: $(\alpha_0, \alpha_1) \xleftarrow{\$} \mathbb{Z}_p^2$		
2: $A \leftarrow \alpha_0 G_0 + \alpha_1 G_1$	$\xrightarrow{A}$	
3:	$\xleftarrow{c}$	$c \xleftarrow{\$} \mathbb{Z}_p$
4: $u \leftarrow \alpha_0 + cz_0$		
5: $v \leftarrow \alpha_1 + cz_1$	$\xrightarrow{(u,v)}$	
6:		Accept iff $uG_0 + vG_1 = A + cZ$.

Fig. 1: Oka, Okamoto protocol for $\mathcal{R}_{\text{Oka}}$.

Lemma 3 (Oka sHVZK). *There exists a PPT simulator that, on input $(\mathbb{G}, p, G_0, G_1, Z)$ and a challenge $c \xleftarrow{\$} \mathbb{Z}_p$, outputs an Oka transcript (A, c, u, v) distributed identically to the view of an honest verifier interacting with an honest prover according to Figure 1.*

The proofs for the above lemmas are well-established in the literature (e.g., [21], [30]); we provide them in Appendix sections A-1, A-2, A-3 for the reader's convenience.

Theorem 1 (Oka is a zero-knowledge argument of knowledge). *Under $\text{LI}(G_0, G_1)$, the Oka protocol in Figure 1 is a public-coin, 3-move zero-knowledge argument of knowledge for the relation $\mathcal{R}_{\text{Oka}}$.*

Proof. Perfect completeness follows from Lemma 1, and special soundness from Lemma 2. By Definition 8, the protocol is an argument of knowledge, and Lemma 3 establishes that the protocol is sHVZK, which we refer to as zero-knowledge. $\square$

NIZK counterpart. A non-interactive zero-knowledge (NIZK) counterpart to this protocol is obtained via the Fiat–Shamir transform [22]. In the random-oracle model [26], the challenge c is computed as

$$c \leftarrow \mathcal{H}(\text{GGD}, G_0, G_1, Z, A, m),$$

where m is the message being signed or bound to the proof, and $\mathcal{H}: \{0, 1\}^* \rightarrow \mathbb{Z}_p$ is a hash function modeled as a random oracle. The resulting non-interactive transcript is (A, u, v) .

This yields a Schnorr-style scheme [33] with the usual ROM security reduction [34]. The latter relies on programming the random oracle on the relevant transcript prefix and rewinding/forking to obtain two accepting transcripts with the same first message for distinct challenges, thus enabling extraction via special soundness. Omitting details, we refer to the classic treatments in [22], [25], [26], [33]–[35].

VIII. LIN2: A REARRANGED OKAMOTO PROTOCOL

The relation $\mathcal{R}_{\text{Lin2}}$ for our rearranged Okamoto protocol, denoted by Lin2, is a slightly modified $\mathcal{R}_{\text{Oka}}$. The only change is that the case of the trivial statement $Z = 0$ and witness $(z_0, z_1) = (0, 0)$ is excluded from the domain.

$$\mathcal{R}_{\text{Lin2}} = \{ (G_0, G_1, Z) \in \mathbb{G}^{*3} ; (z_0, z_1) \in \mathbb{Z}_p^{2*} \mid Z = z_0 G_0 + z_1 G_1 \}. \quad (11)$$

The protocol is shown in Figure 2. Its common input is (G_0, G_1, Z) , and $\mathcal{P}$'s private input is (z_0, z_1) . We prove completeness, special soundness, and special honest-verifier zero-knowledge for Lin2. Throughout, we assume that $\text{LI}(G_0, G_1)$ holds and that $Z \neq 0$, as required by $\mathcal{R}_{\text{Lin2}}$ (11).

Observe that, according to Figure 2, $H := \eta_0 G_0 + \eta_1 G_1$ is the first message. The verifier's acceptance condition

$$a(G_0 + cG_1) = Z + rH \quad (12)$$

rewrites, by collecting coefficients of G_0 and G_1 , as the following linear system in $\mathbb{Z}_p$

$$\begin{bmatrix} 1 & -\eta_0 \\ c & -\eta_1 \end{bmatrix} \cdot \begin{bmatrix} a \\ r \end{bmatrix} = \begin{bmatrix} z_0 \\ z_1 \end{bmatrix}, \quad (13)$$

where $Z = z_0 G_0 + z_1 G_1$, and the matrix determinant is

$$\det \left(\begin{bmatrix} 1 & -\eta_0 \\ c & -\eta_1 \end{bmatrix} \right) = c\eta_0 - \eta_1. \quad (14)$$

Prover $\mathcal{P}$	Verifier $\mathcal{V}$
Common input: statement $(G_0, G_1, Z) \in \mathbb{G}^{*3}$ with $Z = z_0 G_0 + z_1 G_1$. $\mathcal{P}$'s private input: witness $(z_0, z_1) \in \mathbb{Z}_p^{2*}$.	
1: $(\eta_0, \eta_1) \xleftarrow{\$} \mathbb{Z}_p^2 \setminus \text{Line}(z_0, z_1)$	
2: $H \leftarrow \eta_0 G_0 + \eta_1 G_1$	$\xrightarrow{H}$
3:	$\xleftarrow{c} \quad c \xleftarrow{\$} \mathbb{Z}_p$
4: Halt if $z_1 = cz_0 \vee \eta_1 = c\eta_0$	
5: $\begin{bmatrix} a \\ r \end{bmatrix} \leftarrow \begin{bmatrix} 1 & -\eta_0 \\ c & -\eta_1 \end{bmatrix}^{-1} \cdot \begin{bmatrix} z_0 \\ z_1 \end{bmatrix}$	$\xrightarrow{(a,r)}$
6:	Reject if $Z + rH = 0$
7:	Accept iff $a(G_0 + cG_1) = Z + rH$.

Fig. 2: Lin2 protocol for $\mathcal{R}_{\text{Lin2}}$. Note, $\mathcal{V}$ rejects if $a = 0$.

As follows from eq. (14), the matrix in eq. (13) is invertible iff $\eta_1 \neq c\eta_0$. Thus, for any input satisfying $\mathcal{R}_{\text{Lin2}}$ (i.e., $(z_0, z_1) \neq (0, 0)$), an honest prover generates a valid response from the uniformly sampled (η_0, η_1) , and the verifier accepts the resulting transcript with overwhelming probability. The negligible failure probability arises only upon a 'bad' challenge that either forces the prover to halt or the verifier to falsely reject.

Conversely, any two accepting transcripts sharing the same first message H and having distinct challenges c allow for the extraction of witness (z_0, z_1) with overwhelming probability. This property establishes special soundness of the protocol.

Notably, the system in eq. (13) induces the bijection:

$$(a, r) \in \mathbb{Z}_p^{*2} \longleftrightarrow (\eta_0, \eta_1) \in \mathbb{Z}_p^2 \setminus (\text{Line}(z_0, z_1) \cup \text{Line}(1, c))$$

ensuring that, similar to the Okamoto protocol, Lin2 satisfies sHVZK.

We formalize the above observations below. Before proceeding, two additional remarks. First, $\mathcal{V}$ rejects if $a = 0$, which originates from the rejection and acceptance conditions specified in Figure 2. Second, regarding the trivial statement $Z = 0$, we define the protocol over the punctured domain $Z \in \mathbb{G}^*$, $(z_0, z_1) \in \mathbb{Z}_p^{2*}$; our analysis is henceforth restricted to this domain. For each protocol hereafter, we provide a dedicated lemma stating that $Z = 0$ is rejected by the verifier with overwhelming probability.

Lemma 4 (Lin2 completeness). *Let $(G_0, G_1, Z; z_0, z_1) \in \mathcal{R}_{\text{Lin2}}$. A single execution of the Lin2 protocol (Figure 2) with an honest prover is accepted by the verifier with overwhelming probability, at least $1 - 3/p$.*

Proof. Deferred to Appendix A-4.

Lemma 5 (Lin2 2-special soundness). *Assume $\text{LI}(G_0, G_1)$. Given any two accepting Lin2 transcripts sharing the same statement Z and first message H , and having distinct challenges $c \neq c'$, an extractor recovers a witness (z_0, z_1) satisfying $Z = z_0 G_0 + z_1 G_1$, unless the DLR assumption is broken for G_0, G_1 .*

Proof. Deferred to Appendix A-5.

Lemma 6 (Lin2 sHVZK). *The Lin2 protocol satisfies special honest-verifier zero-knowledge for $\mathcal{R}_{\text{Lin2}}$: given a statement (G_0, G_1, Z) and a challenge $c \in \mathbb{Z}_p$, there exists a PPT simulator that outputs an accepting transcript distributed identically to an honest transcript conditioned on that challenge.*

Proof. Deferred to Appendix A-6.

Theorem 2 (Lin2 is a zero-knowledge argument of knowledge). *Assume $\text{LI}(G_0, G_1)$. The Lin2 protocol in Figure 2 is a zero-knowledge argument of knowledge for the relation $\mathcal{R}_{\text{Lin2}}$.*

Proof. Completeness is Lemma 4. Computational witness-extended emulation follows immediately from Lemma 5. Special honest-verifier zero-knowledge is Lemma 6. Combining these yields the theorem. $\square$

Since the domain of $\mathcal{R}_{\text{Lin2}}$ is punctured at zero, the following lemma is useful, as it establishes that the protocol is well-defined away from that point. Namely, the lemma asserts that if an honest Lin2 verifier accepts, then it is convinced that the excluded statement $Z = 0$ or witness $(z_0, z_1) = (0, 0)$ was not used by the prover.

Lemma 7 (Lin2: the trivial witness and statement are rejected). *Assume $\text{LI}(G_0, G_1)$. Unless an adversary breaks the DLR for G_0, G_1 , the following holds with overwhelming probability: the Lin2 verifier rejects the trivial statement $Z = 0$ and, furthermore, rejects any statement produced using the trivial witness $(z_0, z_1) = (0, 0)$.*

Proof. Deferred to Appendix A-7.

IX. LIN2-XOR: ONE OUT OF TWO LIN2'S

Given four generators $(G_0, G_1, G_2, G_3) \in \mathbb{G}^{*4}$ satisfying $\text{LI}(G_0, G_1, G_2, G_3)$, the relation for the Lin2Xor protocol is

$$\mathcal{R}_{\text{Lin2Xor}} = \{ (G_0, G_1, G_2, G_3, Z) \in \mathbb{G}^{*5} ; (z_0, z_1, s) \in \mathbb{Z}_p^{2*} \times \{0, 1\} \mid Z = z_0 G_{2s} + z_1 G_{2s+1} \}. \quad (15)$$

The protocol is shown in Figure 3. Its common input is (G_0, G_1, G_2, G_3, Z) and the witness is (z_0, z_1, s) . It convinces the verifier that the prover knows a representation of Z with respect to either the pair of generators (G_0, G_1) or the pair (G_2, G_3) . The domain of Lin2Xor is punctured at $Z = 0$.

Observe that Lin2Xor (Figure 3) parallels the structure of Lin2 (Figure 2), albeit with an extended witness domain to accommodate $s \in \{0, 1\}$. For an honest flow of Lin2Xor, the case $s = 0$ implies that $a_1 = 0$, and the protocol reduces identically to Lin2 over the generators (G_0, G_1) with the challenge c . The remaining case $s = 1$ implies that $a_0 = 0$, and the protocol reduces to Lin2 over (G_2, G_3) with the challenge c^2 .

In addition, we note that similarly to Lin2, the verifier rejects if $(a_0, a_1) = (0, 0)$. Thus, any statement involving a witness outside the $\mathcal{R}_{\text{Lin2Xor}}$ domain is accepted with only negligible probability.

Prover $\mathcal{P}$	Verifier $\mathcal{V}$
Common input: statement $(G_0, G_1, G_2, G_3, Z) \in \mathbb{G}^{*5}$ with $Z = z_0 G_{2s} + z_1 G_{2s+1}$. $\mathcal{P}$'s private input: witness $(z_0, z_1, s) \in \mathbb{Z}_p^{2*} \times \{0, 1\}$.	
1: $(\eta_0, \eta_1) \xleftarrow{\$} \mathbb{Z}_p^2 \setminus \text{Line}(z_0, z_1)$	
2: $H \leftarrow \eta_0 G_{2s} + \eta_1 G_{2s+1}$	$\xrightarrow{H}$
3:	$\xleftarrow{c}$ $c \xleftarrow{\\$} \mathbb{Z}_p$
4: $e \leftarrow c^{1+s}$	
5: Halt if $z_1 = e z_0 \vee \eta_1 = e \eta_0$	
6: $\begin{bmatrix} a \\ r \end{bmatrix} \leftarrow \begin{bmatrix} 1 & -\eta_0 \\ e & -\eta_1 \end{bmatrix}^{-1} \cdot \begin{bmatrix} z_0 \\ z_1 \end{bmatrix}$	
7: $(a_s, a_{1-s}) \leftarrow (a, 0)$	$\xrightarrow{(a_0, a_1, r)}$
8:	Reject if $Z + rH = 0$
9:	Accept iff $a_0(G_0 + cG_1) + a_1(G_2 + c^2G_3) = Z + rH$.

Fig. 3: Lin2Xor protocol for $\mathcal{R}_{\text{Lin2Xor}}$. Note, $\mathcal{V}$ rejects if $(a_0, a_1) = (0, 0)$.

Lemma 8 (Lin2Xor completeness). *Let $(G_0, G_1, G_2, G_3, Z; z_0, z_1, s) \in \mathcal{R}_{\text{Lin2Xor}}$. In a single execution of the Lin2Xor protocol (Figure 3) with an honest prover, the verifier accepts with overwhelming probability, at least $1 - 6/p$.*

Proof. Deferred to Appendix A-8.

The response values r in accepting transcripts are distinct, a property we formalize in the following lemma.

Lemma 9 (Lin2Xor distinct r 's). *Assume $\text{LI}(G_0, G_1, G_2, G_3)$. For any two accepting Lin2Xor transcripts sharing the same prefix $(G_0, G_1, G_2, G_3, Z, H)$ and having challenges with distinct squares $c^2 \neq (c')^2$, it must hold that $r \neq r'$ unless the DLR is broken for G_0, G_1, G_2, G_3 .*

Proof. Deferred to Appendix A-9.

Two accepting Lin2Xor transcripts with a common prefix and sufficiently distinct challenges permit an efficient recovery of the scalar coefficients in the representation of $Z \in \text{Span}(G_0, G_1, G_2, G_3)$. While not explicitly required for our main reduction, we formalize this property in the next lemma.

Lemma 10 (Lin2Xor: determining representation). *Assume $\text{LI}(G_0, G_1, G_2, G_3)$. Given any two accepting Lin2Xor transcripts sharing the same prefix $(G_0, G_1, G_2, G_3, Z, H)$ while having challenges with distinct squares $c^2 \neq (c')^2$, an*

extractor uniquely determines four scalars $(\beta_0, \beta_1, \beta_2, \beta_3)$ such that $Z = \beta_0 G_0 + \beta_1 G_1 + \beta_2 G_2 + \beta_3 G_3$, unless the DLR is broken for G_0, G_1, G_2, G_3 .

Proof. Deferred to Appendix A-10.

In the following two lemmas, we establish that Lin2Xor satisfies k -special soundness for $\mathcal{R}_{\text{Lin2Xor}}$. The first lemma, a helper, asserts that the event $(a_0 \neq 0 \wedge a_1 \neq 0)$ in an accepting transcript occurs only with negligible probability. The second is our main lemma; using the helper to reduce Lin2Xor to Lin2 , it establishes that an extractor can efficiently recover $s \in \{0, 1\}$ alongside a representation of $Z \in \text{Span}(G_{2s}, G_{2s+1})$. Thus, Lin2Xor achieves the required k -special soundness (computational witness-extended emulation), with $k = 7$.

Lemma 11 (Lin2Xor : two nonzero a 's break DLR). *Assume $\text{LI}(G_0, G_1, G_2, G_3)$. There exists an efficient algorithm $\mathcal{B}$ that, given two accepting Lin2Xor transcripts τ', τ'' sharing the same prefix $(G_0, G_1, G_2, G_3, Z, H)$ and having challenges with distinct squares $(c')^2 \neq (c'')^2$, breaks the DLR for G_0, G_1, G_2, G_3 upon receiving a third accepting transcript τ with the same prefix $(G_0, G_1, G_2, G_3, Z, H)$ and freshly sampled challenge $c \xleftarrow{\$} \mathbb{Z}_p$, except with negligible probability, provided that all values a_0, a'_1, a''_1 in τ', τ'' are nonzero.*

Proof. Deferred to Appendix A-11.

Lemma 12 (Lin2-Xor lemma, Lin2Xor 7-special soundness). *Assume $\text{LI}(G_0, G_1, G_2, G_3)$. Given any seven accepting Lin2Xor transcripts sharing the same prefix $(G_0, G_1, G_2, G_3, Z, H)$ and having independent uniformly sampled challenges, an extractor recovers a witness (z_0, z_1, s) satisfying $Z = z_0 G_{2s} + z_1 G_{2s+1}$ except with negligible probability, unless the DLR is broken for G_0, G_1, G_2, G_3 .*

Proof. Deferred to Appendix A-12.

We do not claim that the special soundness of Lin2Xor necessitates seven accepting transcripts. Rather, we utilize this number for simplicity in the proof. Crucially, the number of required transcripts is bounded by a constant, which suffices to ensure the efficiency of the extraction procedure.

Theorem 3 (Lin2Xor is an argument of knowledge). *Assume $\text{LI}(G_0, G_1, G_2, G_3)$. The Lin2Xor protocol in Figure 3 is an argument of knowledge for the relation $\mathcal{R}_{\text{Lin2Xor}}$.*

Proof. Completeness is Lemma 8. Computational witness-extended emulation is a direct consequence of Lemma 12. $\square$

Although Lin2Xor is not sHVZK in its basic form, in the next section we show that the sHVZK property is achieved when the prover keeps the scalars a_0, a_1 private, and instead outputs a zero-knowledge proof of their knowledge.

For the excluded case of $Z = 0$, the following lemma implies, similarly to Lemma 7 for Lin2 , that if an honest Lin2Xor verifier accepts, then the excluded witness $z_0 = z_1 = 0$ was not used.

Lemma 13 (Lin2Xor : the trivial statement and witness are rejected). *Assume $\text{LI}(G_0, G_1, G_2, G_3)$. Unless an adversary breaks the DLR for G_0, G_1, G_2, G_3 , the following holds with overwhelming probability: the Lin2Xor verifier rejects the trivial statement $Z = 0$ and, furthermore, rejects any statement produced using the trivial witness $(z_0, z_1, s) \in \mathbb{Z}_p^2 \times \{0, 1\}$.*

Proof. Deferred to Appendix A-13.

In this section, we have introduced the Lin2Xor protocol and established its security through the Lin2-Xor lemma (Lemma 12), which is the primary and most technically demanding result of this paper. Consequently, Lin2Xor constitutes a formal argument of knowledge for the relation $\mathcal{R}_{\text{Lin2Xor}}$. In the following section, we use the core mechanics of Lin2Xor as a base to derive our multi-round membership proof.

X. LIN2SELECTOR: ONE OUT OF n GENERATOR PAIRS IN LOGARITHMIC SIZE

Let $n \in \mathbb{N}$ be a power of 2, i.e., $n := 2^L$, $L \in \mathbb{N}$, and let $n \ll p$ where p is the group order. Let $\mathbf{G} := (G_0, \dots, G_{2n-1}) \in \mathbb{G}^{*2n}$ be a list of generators grouped into n pairs (G_{2j}, G_{2j+1}) for $j \in [0, n-1]$. Assume $\text{LI}(\mathbf{G})$. The relation is that Z admits a decomposition by *one* of these pairs.

$$\mathcal{R}_{\text{Lin2sel}} = \left\{ ((G_0, \dots, G_{2n-1}), Z) \in \mathbb{G}^{*2n} \times \mathbb{G}^* ; (z_0, z_1, s) \in \mathbb{Z}_p^{2*} \times [0, n-1] \mid Z = z_0 G_{2s} + z_1 G_{2s+1} \right\}. \quad (16)$$

In Figure 4, we present an $(L+1)$ -round protocol Lin2Selector for the relation $\mathcal{R}_{\text{Lin2sel}}$, which operates as follows. Define the level- i generator list

$$\mathbf{G}^{(i)} := (G_0^{(i)}, \dots, G_{2n/2^i-1}^{(i)}), \quad \text{with } \mathbf{G}^{(0)} := \mathbf{G}.$$

For each $i \in [0, L-1]$ and each $j \in [0, n/2^{i+1} - 1]$, given a challenge $c_i \in \mathbb{Z}_p$, define the folding rule:

$$\begin{aligned} G_{2j}^{(i+1)} &:= G_{4j}^{(i)} + c_i G_{4j+1}^{(i)}, \\ G_{2j+1}^{(i+1)} &:= G_{4j+2}^{(i)} + c_i^2 G_{4j+3}^{(i)}. \end{aligned} \quad (17)$$

According to this rule, at each i -th level, the generator list $\mathbf{G}^{(i)}$ is split into a sequence of four-tuples such that each j -th four-tuple folds into a pair of generators at the $(i+1)$ -th level. Thus, the number of generators halves each round; after L folds we obtain $\mathbf{G}^{(L)} = (G_0^{(L)}, G_1^{(L)}) \in \mathbb{G}^{*2}$.

The running statement is updated recursively. Starting with $Z^{(0)} := Z$, each round outputs a group element H_i and a scalar r_i , defining the next statement as

$$Z^{(i+1)} := Z^{(i)} + r_i H_i. \quad (18)$$

By unfolding eq. (18), we obtain the accumulated form

$$Z^{(L)} = Z + \sum_{i=0}^{L-1} r_i H_i. \quad (19)$$

The protocol proceeds in L rounds to fold the generator list down to two generators. A final Lin2 execution is then performed on the remaining pair. The prover's internal witness index s is updated as $s^{(0)} := s$, defining

$$s^{(i+1)} := \lfloor s^{(i)} / 2 \rfloor, \quad \text{and} \quad b_i := s^{(i)} \bmod 2 \in \{0, 1\} \quad (20)$$

for each $i \in [0, L-1]$. Here, b_i represents the i -th choice bit along the folding path.

Prover $\mathcal{P}$	Verifier $\mathcal{V}$
Common input: statement $(n, \mathbf{G}, Z) \in \mathbb{N} \times \mathbb{G}^{*2n} \times \mathbb{G}^*$ with $Z = z_0 G_{2s} + z_1 G_{2s+1}$.	
$\mathcal{P}$'s private input: witness $(z_0, z_1, s) \in \mathbb{Z}_p^{*2} \times [0, n-1]$.	
1: If $n = 1$, Then:	$\mathcal{P}, \mathcal{V}$ play Lin2($G_0, G_1, Z; z_0, z_1$).
2: Else:	
3: $(\eta_0, \eta_1) \xleftarrow{\$} \mathbb{Z}_p^2 \setminus \text{Line}(z_0, z_1)$	
4: $H \leftarrow \eta_0 \mathbf{G}_{[2s]} + \eta_1 \mathbf{G}_{[2s+1]}$	$\xrightarrow{H}$
5:	$\xleftarrow{c} \quad c \xleftarrow{\$} \mathbb{Z}_p$
6: $b \leftarrow s \bmod 2$	
7: $e \leftarrow c^{1+b}$	
8: Halt if $z_1 = ez_0 \vee \eta_1 = e\eta_0$	
9: $\begin{bmatrix} a \\ r \end{bmatrix} \leftarrow \begin{bmatrix} 1 & -\eta_0 \\ e & -\eta_1 \end{bmatrix}^{-1} \cdot \begin{bmatrix} z_0 \\ z_1 \end{bmatrix}$	$\xrightarrow{r}$
10: $(\hat{z}_b, \hat{z}_{1-b}) \leftarrow (a, 0)$	
11: $\hat{s} \leftarrow \lfloor s/2 \rfloor$	
12: $\mathcal{P}, \mathcal{V}$ do:	
13: $\hat{Z} \leftarrow Z + rH$	
14: $\hat{n} \leftarrow n/2$	
15: $\hat{\mathbf{G}}_{[0:\hat{n}:2]} \leftarrow \mathbf{G}_{[0:n:4]} + c\mathbf{G}_{[1:n:4]}$	
16: $\hat{\mathbf{G}}_{[1:\hat{n}:2]} \leftarrow \mathbf{G}_{[2:n:4]} + c^2\mathbf{G}_{[3:n:4]}$	
17: play Lin2Selector($\hat{n}, \hat{\mathbf{G}}, \hat{Z}; \hat{z}_0, \hat{z}_1, \hat{s}$).	

Fig. 4: Lin2Selector protocol for $\mathcal{R}_{\text{Lin2sel}}$.

In each non-final round of the Lin2Selector protocol, the interaction adheres to the three-move execution of the Lin2Xor protocol with the following modification: in the third move, the prover replies only with r and withholds the scalars (a_0, a_1) . These values are treated as private witnesses for the subsequent recursive step, in which the prover demonstrates knowledge of them. The recursion terminates in the final round with a Lin2 execution, where the prover explicitly reveals the remaining response scalar a .

We omit the explicit Lin2Xor verification checks in each intermediate round of Lin2Selector, because these checks hold by construction with overwhelming probability. This approach is sufficient to ensure the integrity and soundness of the overall Lin2Selector protocol, as we show in the following lemmas.

Lemma 14 (Lin2Selector completeness). *Let $(n, \mathbf{G}, Z; z_0, z_1, s) \in \mathcal{R}_{\text{Lin2Sel}}$, where $(\mathbf{G}, Z) \in \mathbb{G}^{*2n} \times \mathbb{G}^*$, n is a power of two such that $n \ll p$, and $(z_0, z_1, s) \in \mathbb{Z}_p^{2*} \times [0, n-1]$. An honest execution of the Lin2Selector protocol (Figure 4) is accepted with overwhelming probability, at least $1 - \frac{6\log_2(n)+3}{p}$.*

Proof. Deferred to Appendix A-14.

Lemma 15 (Lin2Selector cWEE). *For any $n \in \mathbb{N}$ such that n is a power of two and $n \ll p$, for any $\mathbf{G} \in \mathbb{G}^{*2n}$ such that $\text{LI}(\mathbf{G})$ holds, the Lin2Selector protocol for the relation $\mathcal{R}_{\text{Lin2Sel}}$ has computational witness-extended emulation, unless the DLR is broken for $\mathbf{G}$.*

Proof. Deferred to Appendix A-15.

Lemma 16 (Lin2Selector sHVZK). *For any $n \in \mathbb{N}$ such that n is a power of two and $n \ll p$, for any $\mathbf{G} \in \mathbb{G}^{*2n}$ such that $\text{LI}(\mathbf{G})$ holds, the Lin2Selector protocol is special honest-verifier zero-knowledge for $\mathcal{R}_{\text{Lin2Sel}}$. That is, given a statement $Z \in \mathbb{G}^*$ and a set of challenges $\{c_0, c_1, \dots, c_{\log_2(n)}\} \subset \mathbb{Z}_p$, there exists a PPT simulator that outputs an accepting transcript distributed identically to an honest transcript conditioned on these challenges.*

Proof. Deferred to Appendix A-16.

Theorem 4 (Lin2Selector is a zero-knowledge argument of knowledge). *For any $n \in \mathbb{N}$ such that n is a power of two and $n \ll p$, for $\mathbf{G} \in \mathbb{G}^{*2n}$ such that $\text{LI}(\mathbf{G})$ holds, the Lin2Selector protocol in Figure 4 is a zero-knowledge argument of knowledge for the relation $\mathcal{R}_{\text{Lin2Sel}}$.*

Proof. Completeness is Lemma 14. Computational witness-extended emulation is Lemma 15. Special honest-verifier zero-knowledge is Lemma 16. $\square$

Mirroring the result of Lemma 13 for Lin2Xor, an honest Lin2Selector verifier accepts any statement corresponding to an excluded trivial witness, such as the statement $Z = 0$, with negligible probability.

Lemma 17 (Lin2Selector: the trivial statement and witness are rejected). *For any $n \in \mathbb{N}$ such that n is a power of two and $n \ll p$, for $\mathbf{G} \in \mathbb{G}^{*2n}$ such that $\text{LI}(\mathbf{G})$ holds, the following holds with overwhelming probability unless an adversary breaks the DLR for $\mathbf{G}$: the Lin2Selector verifier rejects the trivial statement $Z = 0$ and, furthermore, rejects any statement produced using the trivial witness $(z_0, z_1, s) = (0, 0, s)$ for any $s \in [0, n-1]$.*

Proof. Deferred to Appendix A-17.

Having defined the Lin2Selector membership protocol, we have shown it to be a zero-knowledge argument of knowledge for the relation $\mathcal{R}_{\text{Lin2Sel}}$. The Lin2Selector transcript (excluding public inputs and challenges) is highly compact; for a set of n pairs, the proof consists of only $(\log_2(n) + 1)$ elements in $\mathbb{G}$ and $(\log_2(n) + 2)$ scalars in $\mathbb{Z}_p$.

Verification requires evaluating the multi-exponentiation $-aW + Z + \sum_{i=0}^L r_i H_i$, where W is the final folded generator obtained from the $2n$ initial generators $\mathbf{G}$ via (17) and the final folding step. The resulting verification equations are:

$$\begin{cases} -aW + Z + \sum_{i=0}^L r_i H_i = 0, \\ a \neq 0. \end{cases} \quad (21)$$

Since $W \neq 0$ with overwhelming probability, the check $a \neq 0$ serves as a proxy for the condition $Z + \sum_{i=0}^L r_i H_i \neq 0$. It is not necessary to explicitly verify that the statement Z belongs to the domain of $\mathcal{R}_{\text{Lin2Sel}}$ as, by Lemma 17, the trivial statement is rejected with overwhelming probability.

Thus, the verification process is reduced to only the aforementioned multi-exponentiation and two logical conditions, bypassing the need for computationally expensive per-round validations. Consequently, standard batch verification and multi-exponentiation optimization techniques, such as those in [8]–[10], [12], are directly applicable.

XI. ORACLES AND LRS DEFINITIONS

We adopt the standard linkable ring signature (LRS) definition and its corresponding models, including the random-oracle model (ROM) [26] and the Fiat–Shamir heuristic (FS) [22], [25]. The core security properties of an LRS follow the formal frameworks of Pointcheval and Stern [22], Liu et al. [7], and Bender et al. [36].

While we recognize the existence of stronger security models for LRS, such as those proposed in [6], [37], we primarily adopt the baseline model of Liu et al. (LSAG) [7]. This approach demonstrates that our underlying Lin2Selector

membership protocol is fully compatible with standard LRS security reductions. The question of satisfying stronger security models remains open, although we anticipate no fundamental technical barriers to such a proof beyond the considerable verbosity required.

Definition 9 (Random Oracle Model). Let $\mathcal{H} : \{0, 1\}^* \rightarrow \mathcal{Y}$ be a hash function with range $\mathcal{Y}$. In the *random-oracle model*, $\mathcal{H}$ is idealized as a uniformly random function $\mathcal{RO} : \{0, 1\}^* \rightarrow \mathcal{Y}$ that is *lazy-sampled* on demand: on each query $x \in \{0, 1\}^*$, the oracle returns the previously sampled value if x was queried before, and otherwise samples $y \xleftarrow{\$} \mathcal{Y}$ uniformly, stores (x, y) , and returns y . All algorithms (adversaries, provers, verifiers, and reductions) are given oracle access to $\mathcal{RO}$. In security proofs, a reduction is allowed to *program* $\mathcal{RO}$ on selected inputs by fixing the oracle's answer consistently with the lazy-sampling process.

Definition 10 (Hashes as Random Oracles). Hereafter, $H_{sc} : \{0, 1\}^* \rightarrow \mathbb{Z}_p^*$ is modeled as an ideal hash-to-scalar function and $H_{gr} : \{0, 1\}^* \rightarrow \mathbb{G}^*$ as an ideal hash-to-group (a.k.a. hash-to-curve) function; both are treated as *random oracles* (ROs). The group description GGD and predefined generator $G \in \mathbb{G}^*$ are implicitly provided as inputs to H_{gr} .

To avoid trivial edge cases and simplify verification, we define the codomains of H_{sc} and H_{gr} as the punctured sets $\mathbb{Z}_p^*$ and $\mathbb{G}^*$, respectively. Since the statistical distance between the uniform distribution over these punctured sets and that over the full sets $\mathbb{Z}_p$ and $\mathbb{G}$ is exactly $1/p = \text{negl}(\lambda)$, the distributions are *statistically indistinguishable*.

The following lemma formalizes a standard property of H_{gr} : a fresh evaluation of the oracle on a new query is, except with negligible probability, linearly independent of any pre-specified group elements and evaluations on distinct queries.

Lemma 18 (Fresh H_{gr} outputs resist linear representations). *Let $H_{gr} : \{0, 1\}^* \rightarrow \mathbb{G}^*$ be a random oracle. For $0 < k \leq \text{poly}(\lambda)$, consider the following game between the challenger $\mathcal{C}$ and a PPT adversary $\mathcal{A}$.*

$\mathcal{A}$ outputs $(U_1, \dots, U_k) \in \mathbb{G}^k$ and a target query x after making arbitrary RO queries, excluding x . Then, $\mathcal{A}$ receives $Y = H_{gr}(x)$ from $\mathcal{C}$ and outputs $(a_1, \dots, a_k) \in \mathbb{Z}_p^k$. $\mathcal{A}$ succeeds if $Y = \sum_{i=1}^k a_i U_i$.

Let ϵ be $\mathcal{A}$'s success probability and q_{RO} its RO query bound. There exists a PPT algorithm $\mathcal{B}$ that produces a non-trivial solution to the DLR in $\mathbb{G}$ with probability

$$\Pr[\mathcal{B}(\mathcal{A}) \text{ breaks the DLR}] \geq \epsilon^{k+1} - \frac{k+1}{p - q_{RO} - k - 1}.$$

In particular, non-negligible ϵ implies a violation of the DLR assumption.

Furthermore, if a PPT algorithm selects the generators $\{U_i\} \subset \mathbb{G}$ independently of the H_{gr} evaluation Y in its execution, then finding a linear representation of Y with respect to $\{U_i\}$ is possible only with negligible probability, unless the DLR is broken.

Proof. Deferred to Appendix A-18.

For an $(L+1)$ -round, $(2L+1)$ -move, public-coin interactive protocol $(\text{Setup}, \mathcal{P}, \mathcal{V})$ for an NP relation $\mathcal{R}$, we use the FS transform [22], [25] to produce a non-interactive proof system $(\text{Setup}, \mathcal{P}^{\text{RO}}, \mathcal{V}^{\text{RO}})$, e.g., a signature, in the ROM.

We define a message space $\mathcal{MS}$ and a ring space $\mathcal{RS}$, which are hereafter assumed to be a set of polynomially bounded bitstrings and a set of polynomially bounded subsets of $\mathbb{G}^*$, respectively. The linkable ring signature scheme is then defined as follows.

Definition 11 (Linkable Ring Signature (LRS)). An LRS scheme is a quintuple of polynomial-time algorithms $(\text{Setup}, \text{KeyGen}, \text{Sign}, \text{Verify}, \text{Link})$ defined over a message space $\mathcal{MS}$ and a ring space $\mathcal{RS}$ as follows:

- $\text{Setup}(1^\lambda) \rightarrow \text{pp}$: On input a security parameter λ , this algorithm outputs a global *public parameters* structure pp , which is henceforth available to all other algorithms.
- $\text{KeyGen}(\text{pp}) \rightarrow (pk, sk)$: This algorithm generates a secret key $sk \in \mathbb{Z}_p$ and a corresponding public key $pk \in \mathbb{G}$.
- $\text{Sign}(sk_s, m, \mathbf{R}) \rightarrow \sigma$: Given a secret key sk_s , a message $m \in \mathcal{MS}$, and a ring $\mathbf{R} = \{pk_0, \dots, pk_{n-1}\} \subset \mathbb{G}^*$ of distinct public keys such that $pk_s \in \mathbf{R}$, this algorithm outputs a signature σ containing a deterministic key image $\text{KI} \in \mathbb{G}$.
- $\text{Verify}(m, \mathbf{R}, \sigma) \rightarrow \{0, 1\}$: This deterministic algorithm validates a signature σ against a message m and a ring $\mathbf{R}$, outputting 1 if valid and 0 otherwise.
- $\text{Link}(\sigma_1, \sigma_2) \rightarrow \{0, 1\}$: This deterministic algorithm determines whether two valid signatures σ_1, σ_2 were generated by the same signer. It extracts KI_1 and KI_2 from the signatures and outputs 1 if $\text{KI}_1 = \text{KI}_2$, and 0 otherwise.

For the LRS schemes presented in this paper, Setup and KeyGen are instantiated as:

- ◊ $\text{Setup}(1^\lambda)$ returns the fixed public parameters $\text{pp} := (\text{GGD}, H_{sc}, H_{gr}, G)$, where $G \in \mathbb{G}^*$ is a predefined generator.
- ◊ $\text{KeyGen}(\text{pp})$ samples a secret key $x \xleftarrow{\$} \mathbb{Z}_p$ and outputs the key pair $(pk, sk) = (xG, x)$.

The security of an LRS is defined by the following four canonical properties.

Definition 12 (Correctness). An LRS scheme is *correct*, if any honestly generated signature always verifies. That is, for any σ such that $\sigma \leftarrow \text{Sign}(sk_s, m, \mathbf{R})$, it holds that $(m, \mathbf{R}, \sigma) = 1$, provided that sk_s is an output of KeyGen.

Definition 13 (Existential Unforgeability (EUF-CMA)). Let $\Pi = (\text{Setup}, \text{KeyGen}, \text{Sign}, \text{Verify}, \text{Link})$ be an LRS scheme. For $N \leq \text{poly}(\lambda)$, consider the following game:

- 1) The challenger $\mathcal{C}$ runs $\text{pp} \leftarrow \text{Setup}(1^\lambda)$ and generates a *key universe* $\mathcal{L} = \{pk_0, \dots, pk_{N-1}\}$ by running $(pk_i, sk_i) \leftarrow \text{KeyGen}(\text{pp})$ for each $i \in [0, N-1]$.
- 2) The adversary $\mathcal{A}$ receives $(\text{pp}, \mathcal{L})$ and is given oracle access to a signing oracle **SignOracle**. On a query $(m, \mathbf{R})$ such that $\mathbf{R} \subseteq \mathcal{L}$, the oracle chooses an index j where $pk_j \in \mathbf{R}$ and returns $\sigma \leftarrow \text{Sign}(sk_j, m, \mathbf{R})$.
- 3) Eventually, $\mathcal{A}$ outputs a forgery $(m^*, \mathbf{R}^*, \sigma^*)$.

$\mathcal{A}$ wins if: (i) $\text{Verify}(m^*, \mathbf{R}^*, \sigma^*) = 1$; (ii) $\mathbf{R}^* \subseteq \mathcal{L}$; and (iii) $(m^*, \mathbf{R}^*)$ was never queried to **SignOracle**.

Π is *EUF-CMA secure* if for all PPT $\mathcal{A}$, the probability that $\mathcal{A}$ wins is negligible in λ .

Definition 14 (Linkability). An LRS scheme is *linkable* if it satisfies the following two conditions:

- *No false negatives*: For any valid signatures σ, σ' generated using the same secret key sk , $\text{Link}(\sigma, \sigma') = 1$ with overwhelming probability.
- *No false positives*: For any valid signatures σ, σ' generated using distinct secret keys $sk \neq sk'$, $\text{Link}(\sigma, \sigma') = 0$ with overwhelming probability.

Definition 15 (Signer Ambiguity). Let $\Pi = (\text{Setup}, \text{KeyGen}, \text{Sign}, \text{Verify}, \text{Link})$ be an LRS scheme. For any $n \leq \text{poly}(\lambda)$ and $t < n - 1$, consider the following game: The challenger $\mathcal{C}$ generates a ring $\mathbf{R}$ of n public keys via **KeyGen** and selects a set of t corrupted secret keys D_t . For any message $m \in \mathcal{MS}$, $\mathcal{C}$ samples a signing index s uniformly from the set of honest indices $\mathcal{I}_h = \{i : sk_i \notin D_t\}$ and generates a signature $\sigma = \text{Sign}(sk_s, m, \mathbf{R})$. The adversary $\mathcal{A}$ receives $(m, \mathbf{R}, D_t, \sigma)$ and returns an index $s' \in \mathcal{I}_h$. $\mathcal{A}$ wins if $s' = s$.

Π is *signer-ambiguous* if for all PPT $\mathcal{A}$, the probability that $\mathcal{A}$ wins is negligibly close to random guessing:

$$\Pr[\mathcal{A}(m, \mathbf{R}, D_t, \sigma) = s] \approx \frac{1}{n-t}.$$

To establish signer ambiguity, we rely on the Decisional Diffie–Hellman (DDH) assumption and its equivalent formulation, the Divisible Decisional Diffie–Hellman (DDDH) assumption [38]. Since the DDH assumption is stronger than the DL assumption, it implies the hardness of the DLR.

Definition 16 (Decisional Diffie–Hellman (DDH)). Let $\mathbb{G}$ be a cyclic group of prime order p with generator G . For any PPT adversary $\mathcal{A}$, there is only negligible probability to distinguish between the tuples (G, aG, bG, abG) and (G, aG, bG, cG) , where a, b, c are sampled independently and uniformly. That is,

$$\left| \Pr[a, b \xleftarrow{\$} \mathbb{Z}_p : \mathcal{A}(G, aG, bG, abG) = 1] - \Pr[a, b, c \xleftarrow{\$} \mathbb{Z}_p : \mathcal{A}(G, aG, bG, cG) = 1] \right| \approx 0.$$

XII. A LOGARITHMIC-SIZE LRS FROM Lin2Selector

We construct the LRS scheme L2S-LRS by applying the Fiat–Shamir transform to the Lin2Selector membership proof in the ROM. Intuitively, the signature proves that the signer knows a witness (z_0, z_1, s) such that

$$Z = z_0 G_{2s} + z_1 G_{2s+1}$$

for some hidden index $s \in [0, n-1]$, where $\mathbf{G} \in \mathbb{G}^{*2n}$ is a public array of generators derived from a ring of public keys. In addition, the signature provides a proof that this knowledge of the witness is bound to the signer’s secret key and the correctness of the key image, all within a single non-interactive transcript.

L2S-LRS is shown in Figure 5. Given a ring of distinct public keys $\mathbf{K} = (K_0, \dots, K_{n-1}) \in \mathbb{G}^{*n}$ such that $n = 2^L$, and a message m , we derive the generator array

$$\mathbf{G} := (P_0, Q_0, P_1, Q_1, \dots, P_{n-1}, Q_{n-1}) \in \mathbb{G}^{*2n}, \quad (22)$$

where

$$P_i := K_i + \xi \cdot \text{H}_{\text{gr}}(K_i), \quad Q_i := \text{H}_{\text{gr}}(\text{Kl}, K_i), \quad \xi := \text{H}_{\text{sc}}(m, G, \mathbf{K}, \text{Kl}). \quad (23)$$

Each element in $\mathbf{G}$ is derived from a fresh evaluation of the random oracle H_{gr} . Consequently, these elements are distributed independently and uniformly over $\mathbb{G}$ (cf. [39], [40]). Thus, the elements in $\mathbf{G}$ are subject to the DLR assumption, and $\text{LI}(\mathbf{G})$ holds.

Let the signer, designated as prover $\mathcal{P}$, hold a secret key $x \in \mathbb{Z}_p^*$ such that $K_s = xG$ for a specific index s . The signer publishes $\text{KI} := x^{-1}\text{H}_{\text{gr}}(K_s)$ and, via $\text{Lin2Selector}(n, \mathbf{G}, Z; z_0 = x^{-1}, z_1 = 0, s)$, proves membership for the statement

$$Z := G + \xi \text{KI}. \quad (24)$$

Here, in $\text{Lin2Selector}(\dots)$, the Fiat–Shamir challenges are generated using H_{sc} . Note that the signer thereby proves that $Z := G + \xi \text{KI} = x^{-1}P_s + 0 \cdot Q_s$, which binds the key image to the signer’s secret key.

By the sHVZK property of Lin2Selector , the resulting signature reveals nothing beyond the validity of the statement and the key image KI . Linkability follows because KI is deterministic for a fixed secret key, whereas unforgeability follows from the cWEE property of Lin2Selector together with the hardness of the DL assumption for ring public keys.

Prover $\mathcal{P}$	Verifier $\mathcal{V}$
Public: message m , GGD, $G \in \mathbb{G}^*$, $\text{H}_{\text{sc}}, \text{H}_{\text{gr}}$, $n = 2^L$, ring $\mathbf{K} = (K_0, \dots, K_{n-1}) \in \mathbb{G}^{*n}$ of distinct keys. Secret: $(x, s) \in \mathbb{Z}_p^* \times [0, n-1]$ with $xG = K_s \in \mathbf{K}$.	
1: $\text{KI} \leftarrow x^{-1}\text{H}_{\text{gr}}(K_s)$	$\xrightarrow{\text{KI}}$
2: $\mathcal{P}, \mathcal{V}$ compute $\xi \leftarrow \text{H}_{\text{sc}}(m, G, \mathbf{K}, \text{KI})$	
3: $\forall i \in [0, n-1]: P_i \leftarrow K_i + \xi \text{H}_{\text{gr}}(K_i)$	
4: $Q_i \leftarrow \text{H}_{\text{gr}}(\text{KI}, K_i)$	
5: $\mathbf{G} \leftarrow (P_0, Q_0, \dots, P_{n-1}, Q_{n-1})$, $Z \leftarrow G + \xi \text{KI}$	
6: $\mathcal{P}$ produces a Fiat–Shamir NIZK transcript π for $\text{Lin2Selector}(n, \mathbf{G}, Z; z_0 = x^{-1}, z_1 = 0, s)$	
7: $\sigma \leftarrow (\text{KI}, \pi)$	$\xrightarrow{\sigma}$
8:	$\text{Verify}(m, \mathbf{K}, \sigma)$: check the public input, parse $\sigma = (\text{KI}, \pi)$, recompute $\xi, \mathbf{G}, Z$, and verify π .
9:	$\text{Link}(\sigma, \sigma')$: output 1 iff both verify and the key images coincide.

Fig. 5: L2S-LRS linkable ring signature from Lin2Selector (Fiat–Shamir in ROM).

As shown in Figure 5, the **Sign** procedure is partitioned into public and private computation components. While the signature is generated entirely by the signer $\mathcal{P}$, the public component is also executed by the verifier $\mathcal{V}$ during **Verify**. Complete specifications for **Verify** and **Link** are detailed in Figure 5.

Correctness and linkability of L2S-LRS are immediate from Definitions 12 and 14, respectively.

Lemma 19 (L2S-LRS correctness). *The L2S-LRS scheme in Figure 5 is a correct LRS.*

Proof. Deferred to Appendix A-19.

Lemma 20 (L2S-LRS linkability). *The L2S-LRS scheme is a linkable LRS.*

Proof. Deferred to Appendix A-20.

The following helper lemma binds the signer’s secret input to the witness of the internal membership proof. Informally, Z is bound to P_s through the secret x in zero-knowledge. Since P_s is defined by $(K_s, \text{H}_{\text{gr}}(K_s))$ and Z is symmetrically derived from (G, KI) , this construction effectively binds K_s to G and $\text{H}_{\text{gr}}(K_s)$ to KI via the shared secret x . Meanwhile, Q_s remains algebraically detached and does not constrain the proof.

Lemma 21 (Witness binding for L2S-LRS). *Let $\sigma = (\text{KI}, \pi)$ be an L2S-LRS signature such that $\text{Verify}(m, \mathbf{K}, \sigma) = 1$. Let $\xi, \{P_i\}_{i=0}^{n-1}, \{Q_i\}_{i=0}^{n-1}, \mathbf{G}$, and Z be the values computed during **Verify** according to eqs. (22) to (24) and Figure 5.*

Then, the FS transcript π is an argument of knowledge for $\mathcal{R}_{\text{Lin2Sel}}$ in the ROM, and the corresponding extractor outputs a witness (z_0, z_1, s) such that

$$Z = z_0 P_s + z_1 Q_s \in \mathbb{G}^*. \quad (25)$$

Furthermore, under the DLR in the ROM, the extracted witness satisfies

$$z_1 = 0 \wedge z_0 \neq 0 \wedge G = z_0 K_s \wedge \text{KI} = z_0 \text{H}_{\text{gr}}(K_s),$$

except with negligible probability. Thus, the L2S-LRS scheme satisfies cWEE in the ROM for the relation

$$\mathcal{R}_{\text{L2S-LRS}} = \left\{ ((K_0, \dots, K_{n-1}), G, \text{KI}) \in \mathbb{G}^{*n} \times \mathbb{G}^* \times \mathbb{G} ; (z_0, s) \in \mathbb{Z}_p^* \times [0, n-1] \mid G = z_0 K_s \wedge \text{KI} = z_0 \text{H}_{\text{gr}}(K_s) \right\}. \quad (26)$$

Proof. Deferred to Appendix A-21.

We prove unforgeability and signer ambiguity of L2S-LRS via standard reductionist methods, largely repeating the approach of Liu et al. [7] and using the technique of Shoup [41].

Lemma 22 (Existential unforgeability of L2S-LRS). *Under the DLR assumption in the ROM, the L2S-LRS signature scheme is EUF-CMA secure against adversarially-chosen-key attacks.*

Proof. Deferred to Appendix A-22.

Lemma 23 (L2S-LRS signer ambiguity). *Under the DDH assumption in the ROM, the L2S-LRS signature scheme is signer-ambiguous.*

Proof. Deferred to Appendix A-23.

Finally, the following theorem consolidates the preceding results for the L2S-LRS scheme.

Theorem 5 (L2S-LRS security). *Under the DDH assumption in the ROM, the L2S-LRS scheme (Figure 5) is a correct, linkable, and signer-ambiguous LRS that satisfies cWEE for the relation $\mathcal{R}_{\text{L2S-LRS}}$ (26) and is EUF-CMA secure against adversarially-chosen-key attacks.*

Proof. Correctness is Lemma 19, linkability is Lemma 20, signer-ambiguity is Lemma 23, cWEE for the relation $\mathcal{R}_{\text{L2S-LRS}}$ is Lemma 21, and EUF-CMA is Lemma 22. $\square$

The size of an L2S-LRS signature exceeds that of a Lin2Selector transcript solely by the presence of the key image KI. Assuming that group elements in $\mathbb{G}$ and scalars in $\mathbb{Z}_p$ require equivalent storage, the total L2S-LRS signature length is $2\lceil\log_2(n)\rceil + 4$ elements. Furthermore, the capacity for batch verification and multi-exponentiation is retained.

XIII. A MULTISIG-COMPATIBLE LRS FROM Lin2Selector

This section presents L2S-LRS-MS, an enhanced LRS scheme that utilizes the complete Lin2Selector witness space (z_0, z_1) to enable the standard LSAG-style key image $\text{KI} = xH_{\text{gr}}(K_s)$ [6], [7], [42]. Unlike L2S-LRS, where the target group element Z is derived from the key image, L2S-LRS-MS includes Z as an independent, uniformly distributed randomizer.

L2S-LRS-MS is shown in Figure 6. For the ring $\mathbf{K} = (K_0, \dots, K_{n-1}) \in \mathbb{G}^{*n}$, $n = 2^L$, and message m , the signature includes the key image $\text{KI} \in \mathbb{G}$ and randomizer $Z \in \mathbb{G}$. From $(m, G, \mathbf{K}, \text{KI}, Z)$ we derive an FS challenge

$$\xi := H_{\text{sc}}(m, G, \mathbf{K}, \text{KI}, Z) \in \mathbb{Z}_p^*, \quad (27)$$

and construct the generator array

$$\mathbf{G} := (P_0, Q_0, P_1, Q_1, \dots, P_{n-1}, Q_{n-1}) \in \mathbb{G}^{*2n}, \quad (28)$$

where for every $i \in [0, n-1]$:

$$P_i := H_{\text{gr}}(\text{KI}, K_i) + \xi K_i + \xi^2 \text{KI}, \quad Q_i := \xi G + \xi^2 H_{\text{gr}}(K_i). \quad (29)$$

In the ROM, the H_{gr} evaluations in (29) behave as fresh random group elements for distinct queries; thus, LI($\mathbf{G}$) holds under the DLR assumption.

Let the signer hold a secret key $x \in \mathbb{Z}_p^*$ for some index $s \in [0, n-1]$. The signer computes the key image

$$\text{KI} := x H_{\text{gr}}(K_s),$$

samples a fresh scalar $y \xleftarrow{\$} \mathbb{Z}_p^*$, and sets

$$Z := y H_{\text{gr}}(\text{KI}, K_s).$$

The signer then proves membership for the relation $\mathcal{R}_{\text{Lin2Sel}}$ with witness $(z_0, z_1, s) := (y, -xy, s)$. Namely, it produces a Fiat–Shamir Lin2Selector transcript π attesting that

$$Z = z_0 P_s + z_1 Q_s \quad \text{for } (P_s, Q_s) \text{ defined by (29)}. \quad (30)$$

Correctness follows from a direct expansion using $K_s = xG$ and $\text{KI} = xV_s$, where $V_s = H_{\text{gr}}(K_s)$, $U_s = H_{\text{gr}}(\text{KI}, K_s)$:

$$z_0 P_s + z_1 Q_s = y(U_s + \xi K_s + \xi^2 \text{KI}) - xy(\xi G + \xi^2 V_s) = yU_s = Z.$$

To verify $\sigma = (\text{KI}, Z, \pi)$, the verifier recomputes ξ and $(P_i, Q_i)_i$ from $(m, G, \mathbf{K}, \text{KI}, Z)$, forms $\mathbf{G}$ as above, and checks that π verifies as an FS transcript for Lin2Selector($n, \mathbf{G}, Z$). Since KI is deterministically derived from the signing secret key (as in LSAG), this yields standard linkability.

The statement Z provides fresh per-signature randomness without affecting the linking tag. There is no need to explicitly verify that $Z \in \mathbb{G}^*$ as this property is inherent to the validation of `Lin2Selector`. By Lemma 17, once the transcript π is verified, $Z \in \mathbb{G}^*$ holds with overwhelming probability.

Prover $\mathcal{P}$	Verifier $\mathcal{V}$
Public: message m , GGD, $G \in \mathbb{G}^*$, H_{sc}, H_{gr} , $n = 2^L$, ring $\mathbf{K} = (K_0, \dots, K_{n-1}) \in \mathbb{G}^{*n}$ of distinct keys. Secret: $(x, s) \in \mathbb{Z}_p^* \times [0, n-1]$ with $xG = K_s \in \mathbf{K}$.	
1: $KI \leftarrow xH_{gr}(K_s)$	$\xrightarrow{KI}$
2: $y \xleftarrow{\$} \mathbb{Z}_p^*, Z \leftarrow yH_{gr}(KI, K_s)$	$\xrightarrow{Z}$
3: $\mathcal{P}, \mathcal{V}$ compute $\xi \leftarrow H_{sc}(m, G, \mathbf{K}, KI, Z)$	
4: $\forall i \in [0, n-1]: P_i \leftarrow H_{gr}(KI, K_i) + \xi K_i + \xi^2 KI,$	
5: $Q_i \leftarrow \xi G + \xi^2 H_{gr}(K_i)$	
6: $\mathbf{G} \leftarrow (P_0, Q_0, \dots, P_{n-1}, Q_{n-1})$	
7: $\mathcal{P}$ produces a Fiat-Shamir NIZK transcript π for <code>Lin2Selector</code> ($n, \mathbf{G}, Z; z_0 = y, z_1 = -xy, s$)	
8: $\sigma \leftarrow (KI, Z, \pi)$	$\xrightarrow{\sigma}$
9:	Verify($m, \mathbf{K}, \sigma$): check the public input, parse $\sigma = (KI, Z, \pi)$, recompute $\xi, \mathbf{G}$, and verify π .
10:	Link(σ, σ'): output 1 iff both verify and the key images coincide.

Fig. 6: L2S-LRS-MS: a multisig-compatible LRS.

Lemma 24 (L2S-LRS-MS correctness). *The L2S-LRS-MS scheme in Figure 6 is a correct LRS.*

Proof. Deferred to Appendix A-24.

Lemma 25 (L2S-LRS-MS linkability). *The L2S-LRS-MS scheme is a linkable LRS.*

Proof. Deferred to Appendix A-25.

Lemma 26 (Witness binding for L2S-LRS-MS). *Let $\sigma = (KI, Z, \pi)$ be an L2S-LRS-MS signature such that $\text{Verify}(m, \mathbf{K}, \sigma) = 1$. Let $\xi, \{P_i\}_{i=0}^{n-1}, \{Q_i\}_{i=0}^{n-1}$, and $\mathbf{G}$ be computed during Verify according to eqs. (27) to (29) and Figure 6.*

Then, the FS transcript π is an argument of knowledge for $\mathcal{R}_{\text{Lin2Sel}}$ in the ROM, and the corresponding extractor outputs a witness (z_0, z_1, s) such that

$$Z = z_0 P_s + z_1 Q_s \in \mathbb{G}^*. \quad (31)$$

Furthermore, under the DLR in the ROM, the extracted witness satisfies, except with negligible probability,

$$K_s = xG \wedge KI = xH_{gr}(K_s) \wedge Z = yH_{gr}(KI, K_s),$$

where $x = -(z_1/z_0)$ and $y = z_0$. Thus, the L2S-LRS-MS scheme satisfies cWEE in the ROM for the relation

$$\mathcal{R}_{\text{L2S-LRS-MS}} = \left\{ ((K_0, \dots, K_{n-1}), G, KI) \in \mathbb{G}^{*n} \times \mathbb{G}^* \times \mathbb{G} ; (x, s) \in \mathbb{Z}_p^* \times [0, n-1] \mid K_s = xG \wedge KI = xH_{gr}(K_s) \right\}. \quad (32)$$

Proof. Deferred to Appendix A-26.

The unforgeability and signer ambiguity of L2S-LRS-MS follow the standard approach [7] even more directly than L2S-LRS, as the key image now conforms to the standard form.

Lemma 27 (Existential unforgeability of L2S-LRS-MS). *Under the DLR assumption in the ROM, the L2S-LRS-MS signature scheme is EUF-CMA secure against adversarially-chosen-key attacks.*

Proof. Deferred to Appendix A-27.

Lemma 28 (L2S-LRS-MS signer ambiguity). *Under the DDH assumption in the ROM, the L2S-LRS-MS signature scheme is signer-ambiguous.*

Proof. Deferred to Appendix A-28.

The following consolidating theorem summarizes the security properties of L2S-LRS-MS:

Theorem 6 (L2S-LRS-MS security). *Under the DDH assumption in the ROM, the L2S-LRS-MS scheme (Figure 6) is a correct, linkable, and signer-ambiguous LRS that satisfies cWEE for the relation $\mathcal{R}_{\text{L2SIRSMS}}$ (32) and is EUF-CMA secure against adversarially-chosen-key attacks.*

Proof. Correctness is Lemma 24, linkability is Lemma 25, signer-ambiguity is Lemma 28, cWEE for the relation $\mathcal{R}_{\text{L2SIRSMS}}$ is Lemma 26, and EUF-CMA is Lemma 27. $\square$

The size of an L2S-LRS-MS signature exceeds that of L2S-LRS solely by the presence of the randomizer Z . Assuming that group elements in $\mathbb{G}$ and scalars in $\mathbb{Z}_p$ occupy equivalent storage, the total signature length is $2\lceil\log_2 n\rceil + 5$ elements. For a 256-bit group order (where each element is 32 bytes), this formula yields the signature lengths for various ring sizes presented in Table II.

Ring size n	8	16	64	256	1024	8192	65536
L2S-LRS length (bytes)	320	384	512	640	768	960	1152
L2S-LRS-MS length (bytes)	352	416	544	672	800	992	1184

TABLE II: Signature sizes for L2S-LRS and L2S-LRS-MS across various ring sizes (256-bit group order).

Full support for batch verification and multi-exponentiation is retained, requiring only two final logical checks according to (21). The standard key image format enables the distributed generation of multisignatures.

XIV. OPEN QUESTIONS

This work offers some open research directions, ranging from strengthening the instantiated signatures to generalizing the underlying reduction step.

Confidential transactions and stronger adversarial models. The multisig-friendly scheme L2S-LRS-MS appears amenable to the standard confidential-transaction extensions used in anonymous ledgers (e.g., augmenting the ring proof with balance commitments), following established approaches [6], [10]–[12]. In addition, it is conjectured that L2S-LRS-MS can be analyzed in security models stronger than the baseline LSAG model of Liu et al. [7], such as those proposed in [6], [37].

For instance, a reduction to the CLSAG [6] security model should parallel known CLSAG arguments: aside from the linking tag, which is identical for both, L2S-LRS-MS is bound to the signer’s private key via the relation (32), much like in CLSAG. A complete formalization of this claim is left open.

Generalizing the Lin2-Xor lemma beyond two pairs. The core special-soundness statement for Lin2Xor (the Lin2-Xor lemma) is proven for the acceptance equation eq. (3), restated here:

$$a_0(G_0 + cG_1) + a_1(G_2 + c^2G_3) = Z + rH,$$

and relies on the fact that, except with negligible probability, at most one of a_0, a_1 can be nonzero in any accepting transcript. This argument follows from the fact that the two challenge monomials (c and c^2) are linearly independent and their precise degrees are inessential.

A natural conjecture is that an n -way generalization can be obtained by extending (3) to

$$\sum_{i=0}^{n-1} a_i(G_{2i} + c^{i+1}G_{2i+1}) = Z + rH, \quad (33)$$

where $(G_0, \dots, G_{2n-1})$ are linearly independent and $c \xleftarrow{\$} \mathbb{Z}_p$ is a single public-coin challenge. One would like to show that any accepting transcript for (33) implies that $Z \in \text{Span}(G_{2s}, G_{2s+1})$ for some $s \in [0, n-1]$ (equivalently, that all but one coefficient a_i must be zero, except with negligible probability). Establishing this statement would yield a direct (non-recursive) membership proof for $\mathcal{R}_{\text{Lin2sel}}$ whose communication consists of one group element and $n+1$ scalars. It would also provide a convenient bridge to standard compression techniques (e.g., proving knowledge of the sparse $\{a_i\}$ vector via succinct inner-product arguments) [9], [27], [28], and enable a compact m -out-of- n proof.

A fully rigorous treatment of this generalization remains an open question. However, its validity is evident when evaluating eq. (33) componentwise across the generators.

Toward an even more compact Lin2Selector. Assuming an n -way Lin2-Xor statement as described, it is tempting to compress the $\log_2(n)$ per-round reply scalars $\{r_i\}$ in Lin2Selector (Figure 4) and communicate only the $\log_2(n)$ group

elements $\{H_i\}$, deferring scalar consistency to a single final proof, as in, e.g., [28]. Thereby, the total membership proof size would be reduced to $\log_2(n) + O(\log_2(\log_2(n)))$.

One possible route is to modify the folding rule so that different pairs are weighted by distinct powers of the per-round challenge:

$$\begin{aligned} G_{2j}^{(i+1)} &:= G_{4j}^{(i)} + c_i^{2j+1} G_{4j+1}^{(i)}, \\ G_{2j+1}^{(i+1)} &:= G_{4j+2}^{(i)} + c_i^{2j+2} G_{4j+3}^{(i)}. \end{aligned} \quad (34)$$

The prover would proceed as in **Lin2Selector**, withholding the r_i values and finally proving knowledge of $(a^{(L)}, r_0, \dots, r_L)$ satisfying the equation

$$a^{(L)}W = Z + \sum_{i=0}^L r_i H_i, \quad (35)$$

where W is the final folded generator obtained after $L + 1$ rounds (L foldings via (34) and a final step as in eq. (12)). As a result, the total membership proof size would be reduced to $\log_2(n) + O(\log_2(\log_2(n)))$.

A soundness argument would likely follow a standard branching/extraction strategy: by rewinding the final check to obtain multiple accepting instances and eliminating the H_i terms iteratively, one aims to recover an equation of the form (33) for the initial generators, to which an n -way Lin2-Xor lemma could be applied. Making this outline precise raises several technical questions, including how to ensure the linear independence conditions for the communicated H_i values and how to minimize additional proof overhead. We leave a complete construction as an open question.

Post-quantum variants. Both **Lin2Xor** and **Lin2Selector** are Schnorr/Bulletproofs-style arguments of knowledge built from the standard black-box group interface and linear-independence assumption DLR. It is therefore natural to ask whether a similar reductions can be obtained in post-quantum settings (e.g., lattices) while retaining the same high-level reduction structure. Existing approaches to analyzing and designing post-quantum counterparts of Σ -protocols, recursive protocols, and LRS may provide useful starting points [20], [43]–[45]. A concrete post-quantum instantiation of the Lin2-Xor reduction step remains open.

XV. CONCLUSION

In this paper, we addressed the question of which linkable ring signature is the most efficient in a black-box prime-order group under the simplest standard assumptions. We presented **Lin2Selector**, a novel $(2\lceil \log_2 n \rceil + 1)$ -move logarithmic-size membership proof protocol under *only* the DLR assumption. The protocol is obtained by iterating a two-pair OR-reduction step, formalized as the standalone protocol **Lin2Xor**. Our primary contribution is methodological: the Lin2-Xor lemma provides a clean special-soundness handle for the **Lin2Xor** OR-reduction.

As an application, we constructed two logarithmic-size linkable ring signatures, **L2S-LRS** and **L2S-LRS-MS**, built from **Lin2Selector** via the Fiat–Shamir transform in the ROM. Both are proven EUF-CMA secure and signer-ambiguous under *only* the DDH assumption. Quantitatively, the resulting signatures are the most compact in their class, achieving minimal values for the logarithmic coefficient and constant term. They retain efficient verification via a single aggregated equation, while **L2S-LRS-MS** additionally supports the standard multisig-friendly LSAG-style linking tag.

Several natural research directions have also been identified, including generalizing the Lin2-Xor/Lin2Selector framework, exploring the feasibility of post-quantum counterparts, and further optimizing the resulting schemes.

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AI language models were utilized for the stylistic and grammatical polishing of this paper, including this very sentence.

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APPENDIX A

DEFERRED PROOFS

A-1. Proof of Lemma 1

Proof. An honest prover sends $A = \alpha_0 G_0 + \alpha_1 G_1$ and then responds with $u = \alpha_0 + cz_0$ and $v = \alpha_1 + cz_1$. The verifier checks

$$uG_0 + vG_1 = (\alpha_0 + cz_0)G_0 + (\alpha_1 + cz_1)G_1 = \alpha_0 G_0 + \alpha_1 G_1 + c(z_0 G_0 + z_1 G_1) = A + cZ,$$

so it always accepts. □

A-2. Proof of Lemma 2

Proof. From acceptance we have $uG_0 + vG_1 = A + cZ$ and $u'G_0 + v'G_1 = A + c'Z$. Subtracting yields

$$(u - u')G_0 + (v - v')G_1 = (c - c')Z. \quad (36)$$

Since $c \neq c'$, we can divide by $(c - c')$. Define

$$z_0 := (u - u')/(c - c'), \quad z_1 := (v - v')/(c - c').$$

Then eq. (36) becomes $(c - c')Z = (c - c')(z_0G_0 + z_1G_1)$, and hence $Z = z_0G_0 + z_1G_1$. $\square$

A-3. Proof of Lemma 3

Proof. The simulator samples $u \xleftarrow{\$} \mathbb{Z}_p$ and $v \xleftarrow{\$} \mathbb{Z}_p$ uniformly and sets

$$A := uG_0 + vG_1 - cZ.$$

Then the verifier's check holds by construction: $uG_0 + vG_1 = A + cZ$. For an honest execution, (u, v) are uniform in $\mathbb{Z}_p^2$ because (α_0, α_1) are uniform and the map $(\alpha_0, \alpha_1) \mapsto (u, v) = (\alpha_0 + cz_0, \alpha_1 + cz_1)$ is a bijection for fixed (c, z_0, z_1) . Conditioned on (u, v) , the value $A = uG_0 + vG_1 - cZ$ is uniquely determined, so the simulated and real transcript distributions coincide. $\square$

A-4. Proof of Lemma 4

Proof. Fix $(G_0, G_1, Z) \in \mathbb{G}^{*3}$, $(z_0, z_1) \in \mathbb{Z}_p^{*2}$. The prover samples $(\eta_0, \eta_1) \xleftarrow{\$} \mathbb{Z}_p^2 \setminus \text{Line}(z_0, z_1)$ once and sends

$$H := \eta_0G_0 + \eta_1G_1$$

to the verifier. The verifier samples $c \xleftarrow{\$} \mathbb{Z}_p$ uniformly at random.

For the honest prover and verifier, the protocol in Figure 2 fails in exactly four cases: two cases when the prover halts and two cases when the verifier rejects.

- 1) The prover halts if the challenge c makes the equality $z_1 = cz_0$ hold. Since z_0, z_1 are fixed before c is chosen, this equality holds for at most one value of $c \in \mathbb{Z}_p$. Hence,

$$\Pr[z_1 = cz_0] \leq 1/p.$$

- 2) The prover halts if the challenge c makes $\eta_1 = c\eta_0$ hold. Same as the above, since η_0, η_1 are fixed before c is chosen, this equality holds for at most one value of $c \in \mathbb{Z}_p$. Hence,

$$\Pr[\eta_1 = c\eta_0] \leq 1/p.$$

This case precludes zero determinant in eq. (14) and, hence, the prover always sends the (a, r) reply, unless it halts.

- 3) We observe that, provided the prover's messages are honest, the verifier's acceptance condition (12) together with the rejection rule $Z + rH = 0$ are equivalent to the following: reject if $a = 0 \vee (G_0 + cG_1) = 0$, otherwise accept. Consider the case of $a = 0$: the prover computes (a, r) as

$$\begin{bmatrix} a \\ r \end{bmatrix} \leftarrow \begin{bmatrix} 1 & -\eta_0 \\ c & -\eta_1 \end{bmatrix}^{-1} \begin{bmatrix} z_0 \\ z_1 \end{bmatrix},$$

where the determinant of the matrix equals $\eta_1 - c\eta_0$, which is nonzero in this branch.

Recalling (13), the event $a = 0$ would imply that (z_0, z_1) lies on the line $\text{Line}(\eta_0, \eta_1)$, contradicting the prover's sampling condition. Therefore, this event occurs with probability 0.

- 4) The remaining case $(G_0 + cG_1) = 0$ is that the verifier rejects if the challenge c , by chance, matches the discrete logarithm of G_0 with respect to $-G_1$. This event, which occurs with probability $1/p$, leads to a rejection due to the condition $Z + rH = 0$, regardless of whether $a \neq 0$.

Whenever neither above fail occurs, we have eq. (13) with an invertible matrix. Multiplying by (G_0, G_1) yields meeting the condition (12), so the verifier accepts. The failing events with nonzero probabilities are $z_1 = cz_0$, $\eta_1 = c\eta_0$, and $G_0 = -cG_1$, each occurring with probability at most $1/p$. Thus, the verifier accepts with probability at least $1 - 3/p$. $\square$

A-5. Proof of Lemma 5

Proof. Let two accepting transcripts with the same H be

$$(H, c, a, r) \quad \text{and} \quad (H, c', a', r'),$$

where both have nonzero a 's (otherwise triggering rejection) and satisfy the verifier's acceptance equation:

$$a(G_0 + cG_1) = Z + rH, \quad (37)$$

$$a'(G_0 + c'G_1) = Z + r'H. \quad (38)$$

Subtract (38) from (37):

$$(a - a')G_0 + (ac - a'c')G_1 = (r - r')H. \quad (39)$$

We first argue that $r \neq r'$, unless the DLR is broken. Indeed, if $r = r'$, then (39) becomes

$$(a - a')G_0 + (ac - a'c')G_1 = 0. \quad (40)$$

Since $c \neq c'$ and recalling that $a \neq 0, a' \neq 0$, it holds that $(a - a', ac - a'c') \neq (0, 0)$. Otherwise we would have both $a = a'$ and $ac = a'c'$, forcing $c = c'$. Thus, eq. (40) is a non-trivial solution of the DLR for G_0, G_1 , i.e., the DLR is broken. Hence, unless the DLR is broken, $(r - r') \in \mathbb{Z}_p^*$ and (39) yields an explicit decomposition of H :

$$H = \eta_0 G_0 + \eta_1 G_1 \quad \text{with} \quad \eta_0 := \frac{a - a'}{r - r'}, \quad \eta_1 := \frac{ac - a'c'}{r - r'}. \quad (41)$$

Now plug (41) into (37) and rewrite:

$$Z = a(G_0 + cG_1) - rH = aG_0 + acG_1 - r(\eta_0 G_0 + \eta_1 G_1) = (a - r\eta_0)G_0 + (ac - r\eta_1)G_1.$$

Therefore, the extractor outputs

$$z_0 := a - r\eta_0, \quad z_1 := ac - r\eta_1,$$

which satisfies $Z = z_0 G_0 + z_1 G_1$. □

A-6. Proof of Lemma 6

Proof. Fix (G_0, G_1, Z) and a challenge $c \in \mathbb{Z}_p$. If c satisfies $G_0 + cG_1 = 0$, then the transcript is necessarily rejected by the combined application of the rejection and acceptance rules. Hence,

$$G_0 + cG_1 \neq 0. \quad (42)$$

Let $Z = z_0 G_0 + z_1 G_1$ be a representation of Z using the (unknown to the simulator) witness z_0, z_1 .

The sought PPT simulator samples $a \xleftarrow{\$} \mathbb{Z}_p^*$ and $r \xleftarrow{\$} \mathbb{Z}_p^*$ independently and uniformly, and sets

$$H := (a(G_0 + cG_1) - Z)/r \in \mathbb{G}. \quad (43)$$

It outputs the transcript (H, c, a, r) . By construction, (43) implies

$$Z + rH = Z + (a(G_0 + cG_1) - Z) = a(G_0 + cG_1),$$

so the verifier acceptance check holds. The “Reject if $Z + rH = 0$ ” rule never triggers because of eq. (42) and $a \in \mathbb{Z}_p^*$.

The simulated transcript is distributed identically to an honest one. To prove this, consider an honest execution conditioned on the same challenge c . In an accepting honest transcript, the prover samples (η_0, η_1) from $\mathbb{Z}_p^2 \setminus \text{Line}(z_0, z_1)$, sets $H = \eta_0 G_0 + \eta_1 G_1$, and then computes (a, r) as the unique solution to eq. (13), provided $\eta_1 \neq c\eta_0$.

For the computed (a, r) , the prover's halting condition $z_1 \neq cz_0$ results in $r \neq 0$, and the verifier's reject rule ensures that $a \neq 0$. Thus, $(a, r) \in \mathbb{Z}_p^{*2}$ in an honest transcript. At the same time, due to the halting condition $\eta_1 \neq c\eta_0$, it holds that $(\eta_0, \eta_1) \in \mathbb{Z}_p^2 \setminus (\text{Line}(z_0, z_1) \cup \text{Line}(1, c))$ in it.

Now observe that for any $(a, r) \in \mathbb{Z}_p^{*2}$, the system (13) rewrites as

$$\begin{bmatrix} 1 & -z_0 \\ c & -z_1 \end{bmatrix} \cdot \begin{bmatrix} a/r \\ 1/r \end{bmatrix} = \begin{bmatrix} \eta_0 \\ \eta_1 \end{bmatrix}. \quad (44)$$

Taking into account the halting condition $z_1 \neq cz_0$ (which renders the matrix in (44) non-singular), eq. (44) defines a bijective map, for fixed (z_0, z_1, c) ,

$$(a, r) \in \mathbb{Z}_p^{*2} \longleftrightarrow (\eta_0, \eta_1) \in \mathbb{Z}_p^2 \setminus (\text{Line}(z_0, z_1) \cup \text{Line}(1, c)).$$

Moreover, according to the map (??), it holds that

$$H = \eta_0 G_0 + \eta_1 G_1 = (a(G_0 + cG_1) - Z)/r,$$

which matches the simulator's definition (43) exactly.

Therefore, conditioned on the honest protocol accepting and not halting, the distribution of (H, a, r) induced by the uniform admissible (η_0, η_1) is identical to the distribution obtained by sampling (a, r) uniformly in $\mathbb{Z}_p^{*2}$ and defining H via eq. (43). This is exactly what the sought PPT simulator does. $\square$

A-7. Proof of Lemma 7

Proof. For the first, we construct an efficient algorithm $\mathcal{Z}_{\text{Lin2}}$ that, given two accepting Lin2 transcripts τ', τ'' sharing the same (G_0, G_1, Z, H) such that $Z = 0$, unconditionally breaks the DLR for G_0, G_1 . We assume that τ', τ'' have distinct challenges $c' \neq c''$.

Construction of $\mathcal{Z}_{\text{Lin2}}$. For any Lin2 transcript, if the prover's reply a is zero, the transcript is necessarily rejected by the combined application of the rejection and acceptance rules. Therefore, $a \neq 0$ in an accepting transcript.

When $Z = 0 \wedge r = 0$, transcript is necessarily rejected by the rejection rule. Thus, $r \neq 0$ in an accepting transcript having $Z = 0$. Thus, for the given accepting transcripts τ', τ'' , the algorithm $\mathcal{Z}_{\text{Lin2}}$ has

$$a(G_0 + cG_1) = rH \quad \text{with } a \neq 0, r \neq 0, \quad (45)$$

$$a'(G_0 + c'G_1) = r'H \quad \text{with } a' \neq 0, r' \neq 0. \quad (46)$$

Eliminating H from eqs. (45) and (46), it obtains

$$(ar' - a'r)G_0 + (acr' - a'c'r)G_1 = 0. \quad (47)$$

If $(ar' - a'r) = 0$ in eq. (47), then $(acr' - a'c'r) = ar'(c - c') \neq 0$ in it. Hence, $\mathcal{Z}_{\text{Lin2}}$ returns eq. (47) as a non-trivial solution to the DLR for G_0, G_1 .

Verifier rejects trivial statement. Now, suppose there exists a malicious prover $\mathcal{P}^*$ that convinces the verifier $\mathcal{V}$ to accept the trivial statement $Z = 0$ with non-negligible probability.

By a standard rewinding argument underlying the Forking Lemma, as, e.g., in [22], [23], an adversary $\mathcal{A}$ can obtain, with non-negligible probability, two accepting transcripts τ', τ'' that share the same prefix (Z, H) but feature distinct challenges. By invoking $\mathcal{Z}_{\text{Lin2}}$ on these two transcripts, $\mathcal{A}$ directly obtains a non-trivial solution to the DLR for G_0, G_1 .

Hence, unless the DLR is broken, there is only negligible probability for $\mathcal{V}$ to accept the statement $Z = 0$, which concludes the proof for this case.

Verifier rejects trivial witness. Finally, we consider a malicious prover $\mathcal{P}^*$ using the trivial witness $(z_0, z_1) = (0, 0)$. Because this witness corresponds solely to the statement $Z = 0$ (i.e., $Z = 0 \cdot G_0 + 0 \cdot G_1$), this scenario reduces to the case $Z = 0$ resolved above. Thus, unless the DLR is broken, $\mathcal{V}$ accepts such a transcript with only negligible probability. $\square$

A-8. Proof of Lemma 8

Proof. It should be observed that for an honest prover in Figure 3, when $s = 0$, the protocol Lin2Xor reduces to the protocol Lin2 in Figure 2. Both protocols share the same execution flow and checks. The only difference is the additional transmission of the value $a_1 = 0$, which has no further impact on the verification. Hence, for $s = 0$, by Lemma 4, the verifier accepts with probability at least $1 - 3/p$.

In the remaining case, when $s = 1$, the protocol Lin2Xor has $a_0 = 0$ and collapses to Lin2, where (G_2, G_3) is substituted for (G_0, G_1) and $e \leftarrow c^2$ serves as the challenge. To evaluate the acceptance probability, we follow the same steps as those in the proof of Lemma 4 in A-4. As before, the only failing events we have are $z_1 = ez_0$, $\eta_1 = e\eta_0$, and $G_2 = -eG_3$, with the distinction that each now occurs with probability at most $2/p$, as $e = c^2$ for $c \xleftarrow{\$} \mathbb{Z}_p$. Thus, the verifier accepts with probability at least $1 - 6/p$ in this case.

As the honest prover operates in exactly one of the two cases $s \in \{0, 1\}$, the final lower bound on the success probability is the minimum of the two results, which is $1 - 6/p$. $\square$

A-9. Proof of Lemma 9

Proof. By the premise, there are two accepting transcripts with $c \neq \pm c'$

$$\tau = (H, c, a_0, a_1, r) \quad \text{satisfying} \quad a_0(G_0 + cG_1) + a_1(G_2 + c^2G_3) = Z + rH \quad (48)$$

$$\tau' = (H, c', a'_0, a'_1, r') \quad \text{satisfying} \quad a'_0(G_0 + c'G_1) + a'_1(G_2 + c'^2G_3) = Z + r'H. \quad (49)$$

Consider the case $r = r'$. Subtracting the acceptance equalities eq. (48) and eq. (49), we have

$$a_0(G_0 + cG_1) + a_1(G_2 + c^2G_3) - a'_0(G_0 + c'G_1) + a'_1(G_2 + c'^2G_3) = 0, \quad (50)$$

which rewrites as the following linear relation

$$\alpha_0 G_0 + \alpha_1 G_1 + \alpha_2 G_2 + \alpha_3 G_3 = 0 \quad (51)$$

with coefficients

$$\alpha_0 := a_0 - a'_0, \quad \alpha_1 := a_0 c - a'_0 c', \quad \alpha_2 := a_1 - a'_1, \quad \alpha_3 := a_1 c^2 - a'_1 c'^2,$$

all of which must be zero unless the DLR for G_0, G_1, G_2, G_3 is broken.

From $\alpha_0 = \alpha_2 = 0$ we have $a_0 = a'_0$ and $a_1 = a'_1$, and therefore

$$\alpha_1 = a_0(c - c'), \quad \alpha_3 = a_1(c^2 - c'^2). \quad (52)$$

Since $c \neq \pm c'$, it follows that $(c - c') \neq 0$ and $(c^2 - c'^2) \neq 0$. Moreover, at least one of a_0, a_1 must be nonzero; otherwise $\mathcal{V}$ would have rejected τ . Thus, according to (52), at least one of α_1, α_3 is nonzero, which makes the linear combination in (51) a non-trivial solution to the DLR for G_0, G_1, G_2, G_3 . This proves the lemma. $\square$

A-10. Proof of Lemma 10

Proof. By the premise, an extractor has two accepting transcripts where $c \neq \pm c'$

$$\tau = (H, c, a_0, a_1, r) \text{ satisfying } a_0(G_0 + cG_1) + a_1(G_2 + c^2G_3) = Z + rH \quad (53)$$

$$\tau' = (H, c', a'_0, a'_1, r') \text{ satisfying } a'_0(G_0 + c'G_1) + a'_1(G_2 + c'^2G_3) = Z + r'H. \quad (54)$$

Subtracting the acceptance equalities eq. (53) and eq. (54), the extractor has

$$a_0(G_0 + cG_1) + a_1(G_2 + c^2G_3) - a'_0(G_0 + c'G_1) + a'_1(G_2 + c'^2G_3) = (r - r')H, \quad (55)$$

where $(r - r') \neq 0$ by Lemma 9, unless the DLR is efficiently broken. Thus, from eq. (55), the extractor obtains

$$H = \alpha_0 G_0 + \alpha_1 G_1 + \alpha_2 G_2 + \alpha_3 G_3, \text{ with} \quad (56)$$

$$\alpha_0 := \frac{a_0 - a'_0}{r - r'}, \quad \alpha_1 := \frac{a_0 c - a'_0 c'}{r - r'}, \quad \alpha_2 := \frac{a_1 - a'_1}{r - r'}, \quad \alpha_3 := \frac{a_1 c^2 - a'_1 c'^2}{r - r'}.$$

By substituting the representation of H from eq. (56) into eq. (53), the extractor recovers the scalars $\beta_0, \beta_1, \beta_2, \beta_3$ forming a representation of Z :

$$Z = \beta_0 G_0 + \beta_1 G_1 + \beta_2 G_2 + \beta_3 G_3, \text{ with} \quad (57)$$

$$\beta_0 := \frac{a'_0 r - a_0 r'}{r - r'}, \quad \beta_1 := \frac{a'_0 c' r - a_0 c r'}{r - r'}, \quad \beta_2 := \frac{a'_1 r - a_1 r'}{r - r'}, \quad \beta_3 := \frac{a'_1 c'^2 r - a_1 c^2 r'}{r - r'}.$$

Uniqueness of $\beta_0, \beta_1, \beta_2, \beta_3$ holds unless the DLR for G_0, G_1, G_2, G_3 is broken: a non-trivial solution for the DLR follows immediately as soon as another representation of Z is demonstrated. $\square$

A-11. Proof of Lemma 11

Proof. The design of algorithm $\mathcal{B}$ is detailed below. $\mathcal{B}$ builds a linear relation among the generators G_0, G_1, G_2, G_3 , which we will show to be non-trivial except with negligible probability over the choice of the challenge c in the third transcript τ .

For each transcript, acceptance means

$$a_0(G_0 + cG_1) + a_1(G_2 + c^2G_3) = Z + rH, \quad (58)$$

and similarly for (c', a'_0, a'_1, r') and (c'', a''_0, a''_1, r'') .

Step 1: eliminate (Z, H) by a linear combination. Upon receiving τ', τ'', τ as input, $\mathcal{B}$ picks $\mu', \mu'' \in \mathbb{Z}_p$ such that

$$\mu' + \mu'' - 1 = 0, \quad \mu' r' + \mu'' r'' - r = 0, \quad \text{which rewrites in the matrix form as}$$

$$\begin{bmatrix} 1 & 1 \\ r' & r'' \end{bmatrix} \cdot \begin{bmatrix} \mu' \\ \mu'' \end{bmatrix} = \begin{bmatrix} 1 \\ r \end{bmatrix}. \quad (59)$$

By Lemma 9, and since $(c')^2 \neq (c'')^2$ by the premise, it follows that $r' \neq r''$ unless the DLR is efficiently broken. Consequently, the matrix in (59) is non-singular, and hence μ', μ'' are uniquely determined for any r in the input τ .

Now, $\mathcal{B}$ linearly combines the acceptance equations (58) for transcripts τ', τ'', τ using the weights $(\mu', \mu'', -1)$. This combination annihilates the right-hand sides, effectively eliminating Z, H from the expression. Thus, $\mathcal{B}$ arrives at the following equality

$$\begin{aligned} &(\mu' a'_0 + \mu'' a''_0 - a_0)G_0 + (\mu' a'_0 c' + \mu'' a''_0 c'' - a_0 c)G_1 \\ &+ (\mu' a'_1 + \mu'' a''_1 - a_1)G_2 + (\mu' a'_1 c'^2 + \mu'' a''_1 c''^2 - a_1 c^2)G_3 = 0, \end{aligned} \quad (60)$$

which has the form

$$\gamma_0 G_0 + \gamma_1 G_1 + \gamma_2 G_2 + \gamma_3 G_3 = 0 \quad (61)$$

with the coefficients γ_i efficiently computed by $\mathcal{B}$ as

$$\gamma_0 := \mu' a'_0 + \mu'' a''_0 - a_0, \quad (62)$$

$$\gamma_1 := \mu' a'_0 c' + \mu'' a''_0 c'' - a_0 c, \quad (63)$$

$$\gamma_2 := \mu' a'_1 + \mu'' a''_1 - a_1, \quad (64)$$

$$\gamma_3 := \mu' a'_1 c'^2 + \mu'' a''_1 c''^2 - a_1 c^2. \quad (65)$$

Step 2: consider the case where all coefficients of G_0, G_1, G_2, G_3 are zero. Our goal is to show that probability of the event $(\gamma_0, \gamma_1, \gamma_2, \gamma_3) = (0, 0, 0, 0)$ is negligible.

Define the two-dimensional vectors

$$v := \begin{bmatrix} a_0 \\ a_0 c \end{bmatrix}, \quad v' := \begin{bmatrix} a'_0 \\ a'_0 c' \end{bmatrix}, \quad v'' := \begin{bmatrix} a''_0 \\ a''_0 c'' \end{bmatrix}, \quad w := \begin{bmatrix} a_1 \\ a_1 c^2 \end{bmatrix}, \quad w' := \begin{bmatrix} a'_1 \\ a'_1 c'^2 \end{bmatrix}, \quad w'' := \begin{bmatrix} a''_1 \\ a''_1 c''^2 \end{bmatrix}.$$

Suppose the event $(\gamma_0, \gamma_1, \gamma_2, \gamma_3) = (0, 0, 0, 0)$ occurs. Then (62)–(63) and (64)–(65) become

$$\mu' v' + \mu'' v'' - v = 0, \quad \mu' w' + \mu'' w'' - w = 0. \quad (66)$$

Rewrite (66) as two 2×2 systems:

$$\underbrace{\begin{bmatrix} v' & v'' \end{bmatrix}}_{=:V} \begin{bmatrix} \mu' \\ \mu'' \end{bmatrix} = v, \quad \underbrace{\begin{bmatrix} w' & w'' \end{bmatrix}}_{=:W} \begin{bmatrix} \mu' \\ \mu'' \end{bmatrix} = w. \quad (67)$$

As $a'_0 \neq 0, a'_1 \neq 0, a''_0 \neq 0, a''_1 \neq 0$ and $(c')^2 \neq (c'')^2$ by the premise, both matrices V, W in (67) are invertible. Therefore, the first system in (67) uniquely determines $\mu := [\mu', \mu'']^T$ as $\mu = -V^{-1}v$. Substituting this into the second system yields

$$w = WV^{-1}v. \quad (68)$$

Define the matrix

$$T := WV^{-1} \in \mathbb{Z}_p^{2 \times 2}. \quad (69)$$

Crucially, the matrix T depends only on the first two transcripts (τ', τ'') , that is, on $c', c'', a'_0, a''_0, a'_1, a''_1$, and is independent of the *third* transcript τ . Thus, T can be computed only once by $\mathcal{B}$ during a preprocessing phase using (τ', τ'') . Moreover, as both V, W are invertible, T is also invertible.

Step 3: the third transcript must satisfy $w = Tv$. According to eq. (68) and eq. (69), any *third* accepting transcript τ for the same (Z, H) and a fresh challenge c , whenever the event $(\gamma_0, \gamma_1, \gamma_2, \gamma_3) = (0, 0, 0, 0)$ occurs for it, must satisfy

$$a_1 \begin{bmatrix} 1 \\ c^2 \end{bmatrix} = a_0 \cdot T \begin{bmatrix} 1 \\ c \end{bmatrix}, \quad \text{where } T \text{ is an invertible matrix that is invariant across all input } \tau. \quad (70)$$

Step 4: estimate probability that (70) holds for an accepting τ . Due to the invertibility of T in (70), it holds that $a_0 = 0$ iff $a_1 = 0$ for any values of a_0, a_1, c . The case of $a_0 = a_1 = 0$ is precluded, as $\mathcal{V}$ rejects such a transcript τ . Thus, the only possible case is $a_0 \neq 0 \wedge a_1 \neq 0$.

Write $T = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix}$. Since both of a_0, a_1 are nonzero, the two coordinates of (70) imply that:

$$\frac{t_{21} + ct_{22}}{t_{11} + ct_{12}} = c^2, \quad \text{where the denominator is nonzero.} \quad (71)$$

Rearranging gives a polynomial equation in c :

$$t_{21} + t_{22}c - t_{11}c^2 - t_{12}c^3 = 0. \quad (72)$$

For fixed T (i.e., fixed τ', τ''), the left-hand side is a degree- ≤ 3 polynomial in c . Unless it is the *zero polynomial*, it has at most 3 roots in $\mathbb{Z}_p$, hence

$$\Pr_{c \xleftarrow{\$} \mathbb{Z}_p} [(72)] \leq \frac{3}{p}.$$

If (72) were the zero polynomial, we would have $t_{12} = t_{11} = t_{22} = t_{21} = 0$, i.e., $T = 0$. However, T is invertible, hence, this is not the case. Thus, for any fresh uniform challenge c , (72) holds with the probability not exceeding $3/p$. Consequently, for any transcript τ with a fresh uniform c , (70) holds with probability at most $3/p$.

Step 5: $\mathcal{B}$ has a non-trivial solution to (61). From the above steps we conclude that under $\text{LI}(G_0, G_1, G_2, G_3)$, for an accepting *third* transcript τ as per the lemma's premise, the algorithm $\mathcal{B}$ efficiently derives the coefficients $\gamma_0, \gamma_1, \gamma_2, \gamma_3$ in the equality (61), unless the DLR is already broken.

Futhermore, we have established that the event $(\gamma_0, \gamma_1, \gamma_2, \gamma_3) = (0, 0, 0, 0)$ implies that the equality eq. (70) holds for τ . Since we have shown that eq. (70) is satisfied with probability at most $3/p$, it follows that the event $(\gamma_0, \gamma_1, \gamma_2, \gamma_3) = (0, 0, 0, 0)$ likewise occurs with probability at most $3/p$.

Summarizing these results, $\mathcal{B}$ efficiently derives the coefficients $\gamma_0, \gamma_1, \gamma_2, \gamma_3$ for a linear relation among the independent generators G_0, G_1, G_2, G_3 , unless the DLR is already efficiently solved for these generators. (In the latter case, $\mathcal{B}$ merely outputs a known non-trivial solution.) The probability that $\gamma_0, \gamma_1, \gamma_2, \gamma_3$ vanish simultaneously is at most $3/p$; consequently, the linear relation (61) is non-trivial with overwhelming probability $1 - 3/p$. Hence, $\mathcal{B}$ outputs this relation, thereby completing the proof of the lemma. $\square$

A-12. Proof of Lemma 12

Proof. We describe the construction of the extractor $\mathcal{X}$ below.

Given seven accepting transcripts $\tau'_0, \tau''_0, \tau'_1, \tau''_1, \tau'_2, \tau''_2, \tau$ sharing the same $(G_0, G_1, G_2, G_3, Z, H)$ and having independent uniformly sampled challenges $c'_0, c''_0, c'_1, c''_1, c'_2, c''_2, c$, the extractor $\mathcal{X}$ fails upon an event of challenge collision, i.e., when $\exists i \in \{0, 1, 2\} : (c'_i)^2 = (c''_i)^2$. This event occurs only with negligible probability (not exceeding $6/p$). Otherwise, $\mathcal{X}$ proceeds as follows.

For each $i \in \{0, 1, 2\}$, $\mathcal{X}$ checks if all of four a 's, namely, a'_0, a'_1, a''_0, a''_1 , are nonzero for i -th transcript pair (τ'_i, τ''_i) . If this is the case for some i , then $\mathcal{X}$ appends τ to the i -th pair and runs the algorithm $\mathcal{B}$ from Lemma 11 for the triplet $(\tau'_i, \tau''_i, \tau)$. By Lemma 11, $\mathcal{B}$ breaks the DLR with overwhelming probability. In the complementary event of negligible probability, where $\mathcal{B}$ fails to reduce the input to a DLR breach, $\mathcal{X}$ also fails.

At this stage, $\mathcal{X}$ has either successfully broken the DLR or identified at least one zero-valued a in each transcript pair (τ'_i, τ''_i) for $i \in \{0, 1, 2\}$, failing only with negligible probability.

In the overwhelmingly probable event that zero-valued a 's are identified across all three transcript pairs $(\tau'_i, \tau''_i), i \in \{0, 1, 2\}$, $\mathcal{X}$ selects a transcript with zero-valued a from each pair, thereby obtaining three transcripts, each containing a zero-valued a . From these three transcripts, $\mathcal{X}$ picks those two which have zero-valued a 's of the same subscript index $x \in \{0, 1\}$. Thus, $\mathcal{X}$ obtains two distinct transcripts, denote them as θ and θ' , which for some $x \in \{0, 1\}$ carry zero-valued a_x and a'_x , respectively.

Since all challenges of the input transcripts are sampled independently and uniformly, there is only negligible probability ($\leq 2/p$) of the event that the obtained θ, θ' have challenges with equal squares, which let $\mathcal{X}$ fail. Consequently, with overwhelming probability $\mathcal{X}$ proceeds with θ, θ' having challenges with distinct squares.

In the case of $x = 1$, the transcripts θ, θ' carry $a_1 = a'_1 = 0$. Comparing the protocol designs in Figures 2 and 3, we find that Lin2Xor reduces identically to Lin2 where $a \leftarrow a_0$. As θ, θ' contain distinct challenges, $\mathcal{X}$ invokes the witness extractor from Lemma 5 to extract (z_0, z_1) satisfying $Z = z_0 G_0 + z_1 G_1$ (unless the DLR is broken). Given these values, $\mathcal{X}$ outputs the witness $(z_0, z_1, 0)$ for the relation $\mathcal{R}_{\text{Lin2xor}}$.

In the remaining case, when $x = 0$, the transcripts θ, θ' carry $a_0 = a'_0 = 0$. Comparing the designs in Figures 2 and 3, we find that Lin2Xor reduces to Lin2 where $a \leftarrow a_1, (G_0, G_1) \leftarrow (G_2, G_3), c \leftarrow c^2$. As θ, θ' contain challenges with distinct squares, $\mathcal{X}$ invokes the witness extractor from Lemma 5, thus extracting (z_0, z_1) such that $Z = z_0 G_2 + z_1 G_3$ (unless the DLR is broken). Given these values, $\mathcal{X}$ outputs the witness $(z_0, z_1, 1)$ for the relation $\mathcal{R}_{\text{Lin2xor}}$.

Thus, $\mathcal{X}$ successfully outputs the witness (z_0, z_1, s) for the relation $\mathcal{R}_{\text{Lin2xor}}$ except with negligible probability, unless the DLR is broken. This proves the lemma. $\square$

A-13. Proof of Lemma 13

Proof. We start by constructing an efficient algorithm $\mathcal{Z}_{\text{Lin2Xor}}$ that, given two accepting Lin2Xor transcripts τ, τ' sharing the same prefix $(G_0, G_1, G_2, G_3, Z, H)$ where $Z = 0$, unconditionally breaks the DLR for G_0, G_1, G_2, G_3 , provided that τ, τ' have challenges with distinct squares $c^2 \neq c'^2$.

Construction of $\mathcal{Z}_{\text{Lin2Xor}}$. For any Lin2Xor transcript, if it holds that $(a_0, a_1) = (0, 0)$ in the prover's reply, the transcript is necessarily rejected by the combined application of the rejection and acceptance rules (Figure 3). Therefore, $a_0 \neq 0 \vee a_1 \neq 0$ in an accepting transcript.

When $Z = 0 \wedge r = 0$, transcript is necessarily rejected by the rejection rule. Thus, $r \neq 0$ in an accepting transcript with $Z = 0$.

By the above, for the given accepting transcripts τ, τ' , the algorithm $\mathcal{Z}_{\text{Lin2Xor}}$ has

$$a_0(G_0 + cG_1) + a_1(G_2 + c^2G_3) = rH \quad \text{with } (a_0 \neq 0 \vee a_1 \neq 0), r \neq 0, \quad (73)$$

$$a'_0(G_0 + c'G_1) + a'_1(G_2 + c'^2G_3) = r'H \quad \text{with } (a'_0 \neq 0 \vee a'_1 \neq 0), r' \neq 0. \quad (74)$$

Eliminating H from eqs. (73) and (74), the algorithm obtains

$$(a_0r' - a'_0r)G_0 + (a_0cr' - a'_0c'r)G_1 + (a_1r' - a'_1r)G_2 + (a_1c^2r' - a'_1c'^2r)G_3 = 0. \quad (75)$$

For the case $a_0 \neq 0$, suppose the coefficient of G_0 in eq. (75) is zero. It follows that the coefficient of G_1 becomes $(a_0cr' - a'_0c'r) = a_0r'(c - c') \neq 0$, provided the challenges c, c' are distinct which follows from $c^2 \neq c'^2$.

For the remaining case $a_1 \neq 0$, suppose the coefficient of G_2 in eq. (75) is zero. It follows that the coefficient of G_3 becomes $(a_1c^2r' - a'_1c'^2r) = a_1r'(c^2 - c'^2) \neq 0$, provided the challenges have distinct squares $c^2 \neq c'^2$.

Thus, in each case, at least one of the coefficients in eq. (75) is nonzero; hence, $\mathcal{Z}_{\text{Lin2Xor}}$ outputs eq. (75) as a non-trivial solution to the DLR among G_0, G_1, G_2, G_3 .

Verifier rejects trivial statement. Now, suppose there exists a malicious prover $\mathcal{P}^*$ that convinces the verifier $\mathcal{V}$ to accept the trivial statement $Z = 0$ with non-negligible probability.

By a standard rewinding argument underlying the Forking Lemma (e.g., [22], [23]), an adversary $\mathcal{A}$ can obtain, with non-negligible probability, two accepting transcripts τ, τ' that share a common prefix (Z, H) but have distinct challenges such that $c^2 \neq c'^2$. By invoking $\mathcal{Z}_{\text{Lin2Xor}}$ on these two transcripts, $\mathcal{A}$ directly obtains a non-trivial solution to the DLR for G_0, G_1, G_2, G_3 .

Hence, unless the DLR is broken, there is only negligible probability for $\mathcal{V}$ to accept the statement $Z = 0$, which concludes the proof for this case.

Verifier rejects trivial witness. Finally, we consider a malicious prover $\mathcal{P}^*$ using the trivial witness scalars $(z_0, z_1) = (0, 0)$ along with any witness index s . Because such a witness corresponds solely to the statement $Z = 0$ (i.e., $Z = 0 \cdot G_{2s} + 0 \cdot G_{2s+1}$), this scenario reduces to the case $Z = 0$ resolved above. Thus, unless the DLR is broken, $\mathcal{V}$ accepts this transcript only with negligible probability. □

A-14. Proof of Lemma 14

Proof. The protocol consists of $L = \log_2(n)$ recursive rounds and one base round (the call to Lin2 when $n = 1$). In accordance with Definition 4, completeness is evaluated assuming all participants are honest.

We observe that augmenting a protocol with additional termination conditions (verification checks) cannot increase its success probability. Thus, to derive a lower bound for acceptance probability of Lin2Selector (Figure 4), we consider an augmented execution where the omitted Lin2Xor (Figure 3) checks and the transmission of a_0, a_1 are restored. Under this transformation, each recursive round of Lin2Selector becomes functionally identical to Lin2Xor. Consequently, if a round satisfies the augmented acceptance criteria, Lin2Selector proceeds to the subsequent iteration, thereby establishing the success probability of the augmented protocol as a lower bound for Lin2Selector.

Note that, in Figure 4, a_0, a_1 are denoted by $\hat{z}_0, \hat{z}_1$, respectively, and the restored checks are the following two:

$$\begin{cases} \text{Reject if} & Z + rH = 0, \\ \text{Accept iff} & \hat{z}_0(G_0 + cG_1) + \hat{z}_1(G_2 + c^2G_3) = Z + rH. \end{cases} \quad (76)$$

Fix any recursive round with the current instance $(n, \mathbf{G}, Z)$ and witness (z_0, z_1, s) satisfying

$$Z = z_0 G_{2s} + z_1 G_{2s+1}, \quad Z \neq 0, \quad (z_0, z_1) \neq (0, 0). \quad (77)$$

By Lemma 8, the augmented counterpart of this round succeeds with probability at least $(1 - 6/p)$. Hence, the actual round proceeds to the subsequent iteration with at least the same probability.

According to Figure 4, the subsequent round takes as input $Z \leftarrow Z + rH$ and $(z_0, z_1) \leftarrow (\hat{z}_0, \hat{z}_1)$. The conditions (77) are satisfied due to the acceptance checks (76) in the previous round. Hence, every intermediate recursive round succeeds with probability $\geq (1 - 6/p)$. The final base round, by Lemma 4, succeeds with probability $\geq (1 - 3/p)$.

In total, the Lin2Xor protocol has L recursive rounds, each succeeding with probability $\geq (1 - 6/p)$, and the final base round with accepting probability $\geq (1 - 3/p)$. Applying the union bound over the failure probabilities of all rounds, the total success probability is conservatively bounded by $(1 - \frac{6 \log_2(n)+3}{p})$, for $n \ll p$. $\square$

A-15. Proof of Lemma 15

Proof. Let $L = \log_2(n)$. We will construct a witness extractor $\mathcal{X}$ for the relation $\mathcal{R}_{\text{Lin2sel}}$ that operates on an accepting transcript tree $\mathcal{T}$ with branching profile $(n_1, \dots, n_L, n_{L+1}) = (\underbrace{7, \dots, 7}_L, 2)$.

Given $\mathcal{T}$, the extractor $\mathcal{X}$ recovers the witness (z_0, z_1, s) with overwhelming probability, unless the DLR is broken. Regarding the branching profile, it satisfies the condition that $\prod_{i=1}^{L+1} n_i \leq \text{poly}(\lambda)$ for $n \ll p$.

In the initial stage, $\mathcal{X}$ extracts witnesses (z_0, z_1) for all 7^L leaf pairs in $\mathcal{T}$. These pairs consist of Lin2 transcripts over the generators $\mathbf{G}^{(L)}$, as defined by the recursive formulas in (17). Because the transcripts within each pair share a common reply H , $\mathcal{X}$ invokes the extraction procedure from Lemma 5. By Lemma 5, $\mathcal{X}$ succeeds with overwhelming probability, failing only in the negligible event of a challenge collision, conditioned on the DLR for $\mathbf{G}^{(L)}$ (and consequently for $\mathbf{G}$). Furthermore, Lemma 7 ensures that all extracted witnesses are non-trivial, i.e., $(z_0, z_1) \neq (0, 0)$, as a trivial witness would yield a solution to the DLR. To conclude this stage, $\mathcal{X}$ prunes the leaf nodes from $\mathcal{T}$, reducing the tree height by one for the subsequent recursive phase.

$\mathcal{X}$ then initializes the state for the recursive process: $(\hat{z}_0, \hat{z}_1) \leftarrow (z_0, z_1)$, the integral witness index $s \leftarrow 0$, and the current level $l \leftarrow L$. The extraction then proceeds recursively for L iterations. At each level l , $\mathcal{X}$ processes 7^l nodes in $\mathcal{T}$, each associated with a non-trivial witness $(\hat{z}_0, \hat{z}_1) \neq (0, 0)$ established in the preceding iteration.

$\mathcal{X}$ treats each of these 7^l nodes as an accepting Lin2Xor transcript over the four generators $\mathbf{G}_{[4s:4s+4]}^{(l-1)}$. Given that $(\hat{z}_0, \hat{z}_1) \neq (0, 0)$, the acceptance conditions in (76) are satisfied with overwhelming probability. Using the extraction procedure from Lemma 12, $\mathcal{X}$ extracts witnesses (z_0, z_1, b) from each of the 7^{l-1} seven-tuples. By Lemma 12 and Lemma 13, this extraction succeeds with overwhelming probability and yields a non-trivial witnesses $(z_0, z_1) \neq (0, 0)$, provided the DLR for $\mathbf{G}$ remains hard. Each iteration concludes by updating the state: $(\hat{z}_0, \hat{z}_1) \leftarrow (z_0, z_1)$, $s \leftarrow 2s + b$, and $l \leftarrow l - 1$.

Overall, $\mathcal{X}$ traverses the steps of the Lin2Selector protocol (Figure 4) in reverse order. In each iteration, it recovers an intermediate witness via the corresponding extractor and accumulates the witness index in s . Upon reaching $l = 0$, $\mathcal{X}$ outputs the tuple (z_0, z_1, s) as the final extracted witness.

Having established the existence of $\mathcal{X}$, we invoke the Forking Lemma [9] to conclude that Lin2Selector satisfies computational witness-extended emulation. Specifically, this ensures that any prover capable of convincing the verifier with non-negligible probability must possess the corresponding witness in $\mathcal{R}_{\text{Lin2sel}}$. $\square$

A-16. Proof of Lemma 16

Proof. We describe a simulator $\mathcal{S}$ that, given the common input $(n, \mathbf{G}, Z)$ where $n = 2^L$, $\mathbf{G} \in (\mathbb{G}^*)^{2n}$, and challenges $c_0, c_1, \dots, c_L$, outputs a full transcript distributed identically to an honest accepting transcript, under $\text{LI}(\mathbf{G})$.

Base case ($n = 1$). If $n = 1$, $\mathcal{S}$ invokes the sHVZK simulator of Lin2 on the common input (G_0, G_1, Z) and challenge c_0 . This yields an accepting transcript distributed identically to an honest one by the sHVZK property of Lin2 (Lemma 6).

Recursive case ($n > 1$). When $n > 1$, $\mathcal{S}$ samples $r \xleftarrow{\$} \mathbb{Z}_p^*$ and $H \xleftarrow{\$} \mathbb{G}^*$. It then outputs the simulation (H, c_0, r) followed by a recursive call to $\mathcal{S}(n/2, \hat{\mathbf{G}}, \hat{Z} = Z + rH, c_1, \dots, c_L)$, where $\hat{\mathbf{G}}$ is the halved generator set computed via (17) with the challenge c_0 .

Distributions of the simulated $r \xleftarrow{\$} \mathbb{Z}_p^*$, $H \xleftarrow{\$} \mathbb{G}^*$ are identical to those in a real conversation.

Step 1: witness transformation. Let us consider the transformation of the witness scalars (z_0, z_1) in each recursive round of an honest Lin2Selector interaction. Following the protocol in Figure 4, the prover computes the pair (a, r) and updated witness scalars $(\hat{z}_b, \hat{z}_{1-b})$ as follows:

$$\begin{bmatrix} a \\ r \end{bmatrix} \leftarrow \begin{bmatrix} 1 & -\eta_0 \\ e & -\eta_1 \end{bmatrix}^{-1} \cdot \begin{bmatrix} z_0 \\ z_1 \end{bmatrix}, \quad (78)$$

$$(\hat{z}_b, \hat{z}_{1-b}) \leftarrow (a, 0), \quad (79)$$

where b is the least significant bit of the index s in the current round, and the effective challenge $e \in \{c, c^2\}$ is determined by b . The prover's internal randomness (η_0, η_1) is effectively (due to the halting criterion $\eta_1 = e\eta_0$) sampled from $\mathbb{Z}_p^2 \setminus (\text{Line}(z_0, z_1) \cup \text{Line}(1, e))$, and therefore the matrix in (78) is non-singular.

Note that, for any input witness scalars $(z_0, z_1) \in \mathcal{R}_{\text{Lin2Sel}}$, i.e., for $(z_0, z_1) \neq (0, 0)$, the updated witness scalars $(\hat{z}_0, \hat{z}_1)$ remains nonzero across each round of an honest accepting **Lin2Selector** interaction. This invariant follows from the prover's halting criterion $z_1 = ez_0$.

Step 2: bijection $(a, r) \leftrightarrow (\eta_0, \eta_1)$. As in the proof of Lemma 6, we establish that, for any recursive round of the **Lin2Selector** protocol, for any fixed witness $(z_0, z_1) \neq 0$ and challenge c , there exists a bijective mapping:

$$(a, r) \in \mathbb{Z}_p^{*2} \longleftrightarrow (\eta_0, \eta_1) \in \mathbb{Z}_p^2 \setminus (\text{Line}(z_0, z_1) \cup \text{Line}(1, e)). \quad (80)$$

In fact, for any $(a, r) \in \mathbb{Z}_p^{*2}$, (78) where the matrix is non-singular, rewrites as

$$\begin{bmatrix} 1 & -z_0 \\ e & -z_1 \end{bmatrix} \cdot \begin{bmatrix} a/r \\ 1/r \end{bmatrix} = \begin{bmatrix} \eta_0 \\ \eta_1 \end{bmatrix}. \quad (81)$$

Taking into account the prover's halting criterion $z_1 \neq ez_0$, which makes the matrix in (81) non-singular, eq. (81) defines the bijective map eq. (80), for fixed (z_0, z_1, c) . Thus, conditioned on the honest protocol accepting and not halting, the distribution of (a, r) induced by uniform admissible (η_0, η_1) is identical to the distribution obtained by sampling (a, r) uniformly in $\mathbb{Z}_p^{*2}$.

Step 3: distribution of H . Fix any secret index s and consider the honest distribution of H in a single execution level. Write the (unknown) discrete-log relation between the two bases as $G_{2s+1} = \kappa G_{2s}$ for some $\kappa \in \mathbb{Z}_p^*$ (existence holds since the group is cyclic). Then

$$H = \eta_0 G_{2s} + \eta_1 G_{2s+1} = (\eta_0 + \kappa \eta_1) G_{2s}.$$

Assuming that $(a, r) \xleftarrow{\$} \mathbb{Z}_p^{*2}$ and finding (η_0, η_1) from eq. (81), we have

$$H = ((1 + \kappa e)a - (z_0 + \kappa z_1))r^{-1} G_{2s},$$

where (a, r) is independent randomness, and κ, e, z_0, z_1 are fixed. The value $H = 0$ is excluded, as $(\eta_0, \eta_1) \neq (0, 0)$ for $(a, r) \in \mathbb{Z}_p^{*2}$, which follows from eq. (81).

Thus, H has uniform distribution in $\mathbb{G}^*$. Moreover, the distribution of H is independent of r , unless $(1 + \kappa e) = 0$. The latter never holds for any valid transcript under **LI(G)**, as the discrete logarithm κ is assumed to remain unknown.

We conclude that the distributions of r and H in each recursive round of an accepting **Lin2Selector** transcript are independent and uniform in $\mathbb{Z}_p^*$ and $\mathbb{G}^*$, respectively. Consequently, the simulated values $r \xleftarrow{\$} \mathbb{Z}_p^*$ and $H \xleftarrow{\$} \mathbb{G}^*$ are identically distributed to those in a real execution. Since the base case also yields identical distributions by Lemma 6, the proof follows by induction, and **Lin2Selector** is sHVZK. $\square$

A-17. Proof of Lemma 17

Proof. We first prove that the verifier rejects the trivial statement $Z = 0$ with overwhelming probability. The proof is by induction on n , which is a power of 2.

Base case ($n = 1$). If $n = 1$, the result follows directly from Lemma 7.

Recursive case ($n > 1$). The induction hypothesis is that the verifier rejects $Z = 0$ w.o.p. for size $n/2$. By the construction of **Lin2Selector** in Figure 4, it follows that, for size n , the verifier rejects w.o.p. if the folded statement $\hat{Z}$ at the next level is zero.

For $n > 1$, a non-negligible acceptance probability for **Lin2Selector** implies (via a standard rewinding argument underlying the Forking Lemma, e.g., [22], [23]) that an adversary $\mathcal{A}$ can obtain, with non-negligible probability, two accepting transcripts τ, τ' sharing the prefix $(Z = 0, H)$, but featuring distinct challenges c, c' such that $c^2 \neq c'^2$. Thus, $\mathcal{A}$ has

$$\hat{z}_0(G_0 + cG_1) + \hat{z}_1(G_2 + c^2G_3) = rH, \quad (82)$$

$$\hat{z}'_0(G_0 + c'G_1) + \hat{z}'_1(G_2 + c'^2G_3) = r'H, \quad (83)$$

where

$$G_0 = \mathbf{G}_{[4\lfloor s/2 \rfloor]}, G_1 = \mathbf{G}_{[4\lfloor s/2 \rfloor + 1]}, G_2 = \mathbf{G}_{[4\lfloor s/2 \rfloor + 2]}, G_3 = \mathbf{G}_{[4\lfloor s/2 \rfloor + 3]}.$$

Both eqs. (82) and (83) must have nonzero right-hand sides; otherwise $\mathcal{A}$ has $\hat{Z} = Z + rH = 0$ or $\hat{Z}' = Z + r'H = 0$, and the proof reduces immediately to the case for $n/2$. Consequently, it must hold that $(\hat{z}_0, \hat{z}_1) \neq 0$ and $(\hat{z}'_0, \hat{z}'_1) \neq 0$, and also $r \neq 0, r' \neq 0$.

By eliminating H from eqs. (82) and (83), $\mathcal{A}$ obtains

$$(\hat{z}_0 r' - \hat{z}'_0 r)G_0 + (\hat{z}_0 c r' - \hat{z}'_0 c' r)G_1 + (\hat{z}_1 r' - \hat{z}'_1 r)G_2 + (\hat{z}_1 c^2 r' - \hat{z}'_1 c'^2 r)G_3 = 0. \quad (84)$$

If $\hat{z}_0 \neq 0$, then at least one coefficient of G_0 or G_1 in eq. (84) is nonzero, which renders eq. (84) a non-trivial solution to the DLR for $\mathbf{G}$. Otherwise, $\hat{z}_1 \neq 0$ must hold, and at least one coefficient of G_2 or G_3 in eq. (84) is nonzero, which renders eq. (84) a non-trivial solution to the DLR for $\mathbf{G}$, again.

This completes the proof that the trivial statement is rejected: either the adversary obtains a non-trivial solution to the DLR problem, or the claim follows directly from the inductive hypothesis for $n/2$.

Verifier rejects trivial witness. Finally, we consider a malicious prover $\mathcal{P}^*$ that uses the trivial witness, where both scalars z_0, z_1 are zero and the index s is arbitrary. Because such a witness corresponds solely to the statement $Z = 0$ (i.e., $Z = 0 \cdot G_{2s} + 0 \cdot G_{2s+1}$), this scenario reduces to the case $Z = 0$ resolved above. Thus, unless the DLR is broken, the verifier accepts this transcript with only negligible probability. $\square$

A-18. Proof of Lemma 18

Proof. We construct a PPT reduction $\mathcal{B}$ that outputs a non-trivial DLR instance by invoking the Forking Lemma in the ROM k times, following the methodology of [22], [26], [32]. Note that $q_{RO} \leq \text{poly}(\lambda)$ since $\mathcal{A}$ is a PPT algorithm.

$\mathcal{B}$ simulates the lemma's game between $\mathcal{C}$ and $\mathcal{A}$. With probability ϵ , $\mathcal{B}$ obtains the tuple $(x, U_1, \dots, U_k)$ and coefficients $(a_{0,1}, \dots, a_{0,k})$ such that, for $Y_0 = \mathbf{H}_{\text{gr}}(x)$ where $Y_0 \xleftarrow{\$} \mathbb{G}^*$, the following holds:

$$Y_0 = \sum_{i=1}^k a_{0,i} U_i.$$

Then, $\mathcal{B}$ rewinds the game to the point of the oracle query for x and forks. It reprograms $\mathbf{H}_{\text{gr}}$ to return a fresh sample $Y_1 = \mathbf{H}_{\text{gr}}(x)$ where $Y_1 \xleftarrow{\$} \mathbb{G}^*$, and obtains a second vector $(a_{1,1}, \dots, a_{1,k})$ from $\mathcal{A}$ such that:

$$Y_1 = \sum_{i=1}^k a_{1,i} U_i.$$

Crucially, since $\{U_i\}_{i=1}^k$ are generated by $\mathcal{A}$ independently of Y_0 , they remain invariant across rewinds. However, the probability of $\mathcal{A}$ successfully outputting subsequent coefficient vectors is reduced by the probability that a fresh sample Y_j collides with a previously queried oracle output or a prior Y value. $\mathcal{B}$ halts in the event of such a collision, hence the success probability for the first fork is bounded from below by $\epsilon - (p - q_{RO} - 2)^{-1}$.

By iteratively rewinding and sampling fresh outputs $Y_j = \mathbf{H}_{\text{gr}}(x)$ for $j \in [0, k]$, $\mathcal{B}$ obtains $k+1$ tuples $(Y_j, \mathbf{a}_j)$, where $\mathbf{a}_j = (a_{j,1}, \dots, a_{j,k}) \in \mathbb{Z}_p^k$, such that:

$$Y_j = \sum_{i=1}^k a_{j,i} U_i \in \mathbb{G}^*.$$

To ensure the independence of each linear representation, each subsequent Y_j must not collide with any previously sampled values or the q_{RO} pre-existing oracle evaluations. Consequently, the success probability of the j -th fork is bounded from below by $\epsilon - (p - q_{RO} - j - 1)^{-1}$, which is conservatively bounded as $\epsilon - (p - q_{RO} - k - 1)^{-1}$.

Since $\{\mathbf{a}_j\}_{j=0}^k$ is a set of $k+1$ vectors in an k -dimensional space $\mathbb{Z}_p^k$, they are necessarily *linearly dependent*. Furthermore, these vectors are non-trivial; otherwise, some Y_j would equal zero, which is excluded.

Finding the coefficients $\{\alpha_j\}_{j=0}^k \in \mathbb{Z}_p^{k+1}$ (not all zero) of the linear dependence between $\{\mathbf{a}_j\}_{j=0}^k$, which takes $\leq \mathcal{O}(k^3)$ time, e.g., using Gaussian elimination, it follows that $\sum_{j=0}^k \alpha_j \mathbf{a}_j = 0$. Consequently, $\mathcal{B}$ obtains a non-trivial linear relation:

$$\sum_{j=0}^k \alpha_j Y_j = \sum_{j=0}^k \alpha_j \left(\sum_{i=1}^k a_{j,i} U_i \right) = \sum_{i=1}^k \left(\sum_{j=0}^k \alpha_j a_{j,i} \right) U_i = 0.$$

As $\{Y_j\}_{j=0}^k$ are sampled independently and uniformly from $\mathbb{G}^*$, this non-trivial relation is a solution to the DLR for $\{Y_j\}_{j=0}^k$. The joint probability that $\mathcal{B}$ successfully extracts this solution is conservatively bounded from below by:

$$\Pr[\mathcal{B}(\mathcal{A}) \text{ breaks the DLR}] \geq \left(\epsilon - \frac{1}{p - q_{RO} - k - 1} \right)^{k+1} \geq \epsilon^{k+1} - \frac{k+1}{p - q_{RO} - k - 1}. \quad (85)$$

Since the fractional term in eq. (85) is $\text{negl}(\lambda)$, $\mathcal{B}$ violates the DLR assumption whenever ϵ is non-negligible.

For instance, if a PPT algorithm finds, with probability bounded by ϵ , a linear representation of an H_{gr} evaluation by a set of independent of this evaluation group elements, such as pre-specified group elements and evaluations of H_{gr} on pre-specified queries, then $\epsilon \leq \text{negl}(\lambda)$ unless the DLR is broken. $\square$

A-19. Proof of Lemma 19

Proof. Correctness is immediate from the construction in Figure 5 and completeness of `Lin2Selector` (Lemma 14). With $KI := x^{-1}H_{gr}(K_s)$ and $\xi := H_{sc}(m, G, \mathbf{K}, KI)$ we have

$$Z = G + \xi KI = x^{-1}(K_s + \xi H_{gr}(K_s)) = z_0 P_s + 0 \cdot Q_s \quad \text{for } z_0 = x^{-1},$$

so the FS proof π verifies and, by Definition 12, `Lin2Selector` is correct. $\square$

A-20. Proof of Lemma 20

Proof. The proof is by Definition 14. If the same secret key signs twice, the key image is deterministically $KI = x^{-1}H_{gr}(xG)$ and repeats, so $\text{Link} = 1$ always.

$$\Pr[\text{Link}(\sigma_1, \sigma_2) = 0 : x_1 = x_2] = 0.$$

If two different signers $x_1 \neq x_2$ sign, the key images are independent uniform in $\mathbb{G}^*$. In fact, multiplication by a fixed nonzero scalar is a bijection on $\mathbb{G}^*$, so KI_1, KI_2 are independent uniform in $\mathbb{G}^*$. Thus $\Pr[KI_1 = KI_2] = 1/(p-1)$, and hence

$$\Pr[\text{Link}(\sigma_1, \sigma_2) = 1 : x_1 \neq x_2] \leq \frac{1}{p-1} = \text{negl}(\lambda).$$

$\square$

A-21. Proof of Lemma 21

Proof. We stress that this proof does *not* assume that (z_0, z_1, s) are fixed before ξ : the scalars and index are allowed to be chosen *after* seeing ξ and KI .

The FS transcript π is a *non-interactive argument of knowledge* for $\mathcal{R}_{\text{Lin2sel}}$ in the ROM. This assertion follows from the completeness and cWEE properties of `Lin2Selector` (Lemmas 14 and 15) via the *General Forking Lemma*, as is standard for the FS transformation of public-coin protocols in the ROM [22], [32]. Valid π enables the extraction of the witness $(z_0, z_1, s) \in \mathbb{Z}_p^{2*} \times [0, n-1]$ via Lemmas 15 and 17. Note that, for any $G \in \mathbb{G}^*$, $KI \in \mathbb{G}$, and RO evaluation ξ , with overwhelming probability it holds that $Z \neq 0$.

Step 1: show $z_1 = 0$. Suppose for contradiction that $z_1 \neq 0$. Rearranging (25) gives an explicit representation of the hash-to-group output $Q_s = H_{gr}(KI, K_s)$:

$$Q_s = z_1^{-1}(Z - z_0 P_s). \quad (86)$$

Crucially, in `Verify`, it holds that

$$Z = G + \xi KI, \quad P_s = K_s + \xi H_{gr}(K_s), \quad (87)$$

and, hence, the right-hand side of eq. (86) is a *known-coefficient linear combination* of the elements $G, KI, K_s, H_{gr}(K_s)$.

Neither of these four elements can depend on Q_s . By Definition 10, the generator G is an implicit input to H_{gr} , while KI and K_s are the explicit query for Q_s . Consequently, these three elements are independent of Q_s . Furthermore, the evaluation $H_{gr}(K_s)$ is likewise independent of Q_s as an RO's output on a distinct query.

Therefore, by Lemma 18, the event $z_1 \neq 0$ occurs only with negligible probability unless the DLR is broken, and with overwhelming probability we have $z_1 = 0$.

Step 2: show $G = z_0 K_s$ and $KI = z_0 H_{gr}(K_s)$. With $z_1 = 0$, equation (25) becomes

$$Z = z_0 P_s. \quad (88)$$

d Expanding with (87) yields

$$G + \xi KI = z_0 K_s + \xi z_0 H_{gr}(K_s). \quad (89)$$

By rewinding to the invocation of the H_{sc} oracle for the query $(m, G, \mathbf{K}, KI)$, forking with a fresh challenge $\xi' \neq \xi$, extracting a second valid witness (z'_0, z'_1, s') , and taking into account that $z'_1 = 0$ must hold with overwhelming probability, we obtain:

$$Z = z'_0 P_{s'}. \quad (90)$$

As $Z \neq 0$ with overwhelming probability, $z_0 \neq 0$ in eq. (90). Eliminating Z from eqs. (88) and (90), we have a non-trivial solution to the DLR for $P_{s'}, P_s$, for $s' \neq s$:

$$z'_0 P_{s'} - z_0 P_s = 0.$$

Therefore, $s' = s$ must hold with overwhelming probability, and eq. (90) unfolds as

$$G + \xi' \text{KI} = z'_0 K_s + \xi' z'_0 \text{H}_{\text{gr}}(K_s). \quad (91)$$

Recalling that $\xi, \xi' \in \mathbb{G}^*$ by Definition 10, eliminating KI from eqs. (89) and (91) yields

$$(\xi - \xi')G = (\xi z'_0 - \xi' z_0)K_s + \xi \xi' (z'_0 - z_0) \text{H}_{\text{gr}}(K_s). \quad (92)$$

If $z'_0 \neq z_0$, then eq. (92) becomes a *known-coefficients linear representation* of the H_{gr} (Definition 10) evaluation at (K_s, G) with respect to the basis $\{K_s, G\}$. Such an event is ruled out by the DLR assumption, as it occurs with only negligible probability by Lemma 18. Hence, $z'_0 = z_0$ holds with overwhelming probability, and eq. (92) rewrites as

$$G = z_0 K_s. \quad (93)$$

Inserting eq. (93) into (87), we obtain

$$\text{KI} = z_0 \text{H}_{\text{gr}}(K_s). \quad (94)$$

Step 3: witness-extended emulation in the ROM. In total, a valid L2S-LRS signature enables the extraction of the witness (z_0, z_1, s) satisfying eq. (25). Additionally, we have shown that a further fork yields w.o.p. the constraints eqs. (93) and (94) within the ROM. Therefore, the L2S-LRS scheme satisfies cWEE in the ROM for the relation $\mathcal{R}_{\text{L2S/LRS}}$ (26). This completes the proof of the lemma. $\square$

A-22. Proof of Lemma 22

Proof. Let $\mathcal{A}$ be a PPT adversary that wins the EUF-CMA game (Definition 13) with non-negligible probability ϵ . We construct a simulator $\mathcal{B}$ that invokes $\mathcal{A}$ to solve the DLR problem for a challenge $\hat{K} = \hat{x}G \in \mathbb{G}^*$, where the discrete logarithm $\hat{x} \in \mathbb{Z}_p^*$ is unknown to $\mathcal{B}$. The simulation proceeds as follows.

Challenge embedding. Upon initialization, $\mathcal{B}$ receives the challenge public key $\hat{K}$, which is sampled independently from the public key distribution of KeyGen . $\mathcal{B}$ acts as the challenger $\mathcal{C}$ in the EUF-CMA game against $\mathcal{A}$. It samples a target index $\hat{j} \xleftarrow{\$} [0, N-1]$ and generates the key universe $\mathcal{L}$ by invoking KeyGen , with the exception of the $\hat{j}$ -th entry which it sets to $\hat{K}$. Thus, for $\mathcal{L} = \{K_i\}_{i=0}^{N-1}$, $\hat{K} = K_{\hat{j}} \in \mathcal{L}$.

RO simulation (lazy sampling and programming). $\mathcal{B}$ maintains tables for H_{sc} and H_{gr} and answers consistently:

- $\text{H}_{\text{sc}}(\dots) \in \mathbb{Z}_p^*$: on a new query, sample uniformly from $\mathbb{Z}_p^*$ and store.
- $\text{H}_{\text{gr}}(\dots) \in \mathbb{G}^*$: on the first query $\hat{K}$ (normally invoked when P_i 's or KI 's are constructed), sample $\hat{\alpha} \xleftarrow{\$} \mathbb{Z}_p^*$, store $\hat{\alpha}$, and return $\text{H}_{\text{gr}}(\hat{K}) := \hat{\alpha}\hat{K}$. This is uniform in $\mathbb{G}^*$. For all other queries, forward the request to the standard oracle implementation and return the resulting uniformly random value from $\mathbb{G}^*$.

Programming is used only for FS challenge queries inside the simulated proofs, exactly as allowed by Definition 9.

Signing oracle simulation. $\mathcal{B}$ maintains a simulated signing oracle, which lets the underlying original SignOracle answer queries $(m, \mathbf{R})$, with the sole exception regarding $K_{\hat{j}}$.

Specifically, for a query where $\mathbf{R} \subseteq \mathcal{L}$, SignOracle deterministically chooses j such that $K_j \in \mathbf{R}$ and returns $\sigma \leftarrow \text{Sign}(x_j, m, \mathbf{R})$. However, if the chosen index j coincides with the target index $\hat{j}$, the simulator instead outputs a simulated signature $\hat{\sigma}$ produced without knowledge of the secret key, as follows.

- 1) Identify $s \in [0, |\mathbf{R}| - 1]$ such that $\hat{K} = K_{\hat{j}} = \mathbf{R}_{[s]}$ and treat s as the signing index.
- 2) Ensure the oracle value $\text{H}_{\text{gr}}(\hat{K})$ is defined. By the simulation above, it holds that $\text{H}_{\text{gr}}(\hat{K}) = \hat{\alpha}\hat{K}$ for known $\hat{\alpha} \in \mathbb{Z}_p^*$. Set the simulated key image

$$\text{KI} := \hat{\alpha}G.$$

The simulated KI has the *same distribution* as a real key image $\hat{x}^{-1}\text{H}_{\text{gr}}(\hat{K})$, because $\text{H}_{\text{gr}}(\hat{K})$ is uniform in $\mathbb{G}^*$ and scaling by $x^{-1} \in \mathbb{Z}_p^*$ is a permutation.

- 3) Let $n = |\mathbf{R}|$. Query $\xi := \text{H}_{\text{sc}}(m, G, \mathbf{R}, \text{KI})$ and compute all P_i 's, Q_i 's, and $Z := G + \xi \text{KI}$ exactly as in Figure 5.

- 4) Produce an FS transcript π for $\text{Lin2Selector}(n, \mathbf{G}, Z)$ using the sHVZK simulator of Lin2Selector round-by-round (Lemma 16), while *programming* H_{sc} on the corresponding FS challenge inputs so that verification recomputes the simulator-chosen challenges via the standard technique [22], [25].
- 5) Output $\hat{\sigma} := (\text{KI}, \pi)$.

The simulated signature $\hat{\sigma}$ is statistically indistinguishable from a real signature. This follows from the fact that the simulated KI is identically distributed to a genuine key image, while the remaining component π is statistically identical to that of a valid signature conditioned on this KI.

Discovering unknown discrete logarithm. $\mathcal{B}$ lets $\mathcal{A}$ run within the simulated environment, i.e., using the simulated H_{sc}, H_{gr} , and signing oracle. Since these simulations are indistinguishable from the authentic oracles and produce identical distributions, $\mathcal{A}$ outputs a forgery with its original probability ϵ .

Once $\mathcal{A}$ outputs a successful forgery $\sigma^* = (m^*, \mathbf{K}^*, \text{KI}^*, \pi^*)$ such that $\mathbf{K}^* = \{K_i\}_{i=0}^{|\mathbf{K}^*|-1} \subseteq \mathcal{L}$ and $\text{Verify}(\sigma^*) = 1$, $\mathcal{B}$ invokes the cWEE extractor for the accepting FS transcript π^* of the Lin2Selector protocol in the ROM. This extractor, which exists by Lemma 15 following the methodology in [22], [32], is guaranteed to succeed except with negligible probability. Given that Lin2Selector is complete and rejects inputs outside the $\mathcal{R}_{\text{Lin2Sel}}$ relation (Lemmas 14 and 17), the extractor outputs the valid witness (z_0, z_1, s) .

Thus, $\mathcal{B}$ has the valid witness (z_0, z_1, s) obtained from π via the corresponding extractor with the probability, in sum, $(\epsilon - \text{negl}(\lambda))$. By Lemma 21, this witness satisfies

$$G = z_0 K_s, \quad (95)$$

except with negligible probability.

As $\mathbf{K}^* \subseteq \mathcal{L}$, the public key K_s is an element of the key universe $\mathcal{L}$. Given that the challenge key $\hat{K}$ is uniformly distributed across $\mathcal{L}$, the probability of the event $K_s = \hat{K}$ is at least $1/N$. Consequently, the overall success probability for $\mathcal{B}$ to extract z_0 satisfying eq. (117) for $K_s = \hat{K}$ is $(\epsilon/N - \text{negl}(\lambda))$. This probability is *non-negligible* for any non-negligible ϵ ; thus, the existence of the adversary $\mathcal{A}$ contradicts the hardness of the DLR, and we conclude that by Definition 13, under the DLR assumption, L2S-LRS is EUF-CMA secure against adversarially-chosen-key attacks. $\square$

A-23. Proof of Lemma 23

Proof. Fix any message m , ring size n , and corrupted key set size $t < n - 1$. Using the *sequence-of-games technique* [41] we show that any PPT adversary $\mathcal{A}$ winning the anonymity game (Definition 15) with a non-negligible advantage over random guessing (among uncorrupted indices) can be reduced to a DDDH distinguisher [38], thereby violating the DDH assumption (Definition 16).

First game. Let us denote by $\mathcal{G}_1$ the anonymity game in Definition 15, and suppose that a PPT $\mathcal{A}$ wins it with a non-negligible advantage ϵ over random guessing among $(n - t)$ uncorrupted indices.

Second game. Denoted by $\mathcal{G}_2$, it is a slightly modified $\mathcal{G}_1$: prior to sending $(\mathbf{R}, D_t, \sigma = (\pi, \text{KI}))$ to $\mathcal{A}$, the challenger $\mathcal{C}$ replaces the Lin2Selector FS transcript π within σ with a simulated sHVZK transcript π^* , which exists by Lemma 16. For the simulation π^* , $\mathcal{C}$ uses the same public inputs and RO challenges (programming H_{sc} as needed) as for π . Then, completing the modification, $\mathcal{C}$ hands over the triplet $(\mathbf{R}, D_t, \sigma^* = (\pi^*, \text{KI}))$ to $\mathcal{A}$.

In $\mathcal{G}_2$, $\mathcal{A}$ maintains its success probability with a non-negligible advantage $\epsilon - \text{negl}(\lambda)$. Any non-negligible shift in the advantage would enable $\mathcal{A}$ to distinguish between real and simulated transcripts in the sHVZK game (Definition 7), contradicting the sHVZK property of Lin2Selector established in Lemma 16.

Third game. Now, consider game $\mathcal{G}_3$, which is a slight modification of $\mathcal{G}_2$: instead of generating the honest key image $\text{KI} = x_s^{-1} H_{gr}(x_s G)$, the challenger $\mathcal{C}$ fabricates it as $\phi \xleftarrow{\$} \mathbb{Z}_p^*$, $\text{KI}^* = \phi G$. After that, $\mathcal{C}$ proceeds as in $\mathcal{G}_2$; namely, it generates a simulated sHVZK transcript π^* and hands over $(\mathbf{R}, D_t, \sigma^* = (\pi^*, \text{KI}^*))$ to $\mathcal{A}$.

In $\mathcal{G}_3$, the signature σ^* is statistically independent of the signing index s . In other words, σ^* is void of any values related to s , x_s , or $H_{gr}(\dots, x_s G)$, because it is simulated for the fabricated key image KI^* . Consequently, in $\mathcal{G}_3$, $\mathcal{A}$ attains zero advantage in identifying s , which is chosen uniformly at random from the set of $n - t$ uncorrupted indices.

Distinguisher. The DDDH distinguisher algorithm $\mathcal{D}$ 'interpolates' between games $\mathcal{G}_2$ and $\mathcal{G}_3$ in the sense of [41]. Upon receiving a DDDH tuple (G, A, B, C) instantiated outside in one of two forms: $(G, \alpha G, \beta G, \beta \alpha^{-1} G)$ or $(G, \alpha G, \beta G, \phi G)$ for $\alpha, \beta, \phi \xleftarrow{\$} \mathbb{Z}_p^*$, the distinguisher proceeds as follows.

$\mathcal{D}$ generates $\mathbf{R}$ and samples an uncorrupted index s , in line with $\mathcal{C}$ in $\mathcal{G}_2$ and $\mathcal{G}_3$. It then substitutes A for the s -th entry in $\mathbf{R}$, programs the random oracle H_{gr} to return $H_{gr}(A) = B$, and sets $\text{KI} = C$. Note that, by this, KI actually gets

either $\beta\alpha^{-1}G$, which makes it equal to KI in $\mathcal{G}_2$, or ϕG , which makes it equal to KI^* in $\mathcal{G}_3$. Subsequently, in line with $\mathcal{C}$ in $\mathcal{G}_2$ and $\mathcal{G}_3$, $\mathcal{D}$ produces the simulated π^* based on KI , and hands over $(\mathbf{R}, D_t, \sigma^* = (\pi^*, \text{KI}))$ to $\mathcal{A}$, which outputs s' .

Up to this point, $\mathcal{D}$ faithfully simulates one of the games $\mathcal{G}_2$ or $\mathcal{G}_3$; which one is determined solely by the input tuple form. Consequently, $\mathcal{A}$ retains the advantage of $\epsilon - \text{negl}(\lambda)$ in identifying the correct index s provided that the input is $(G, \alpha G, \beta G, \beta\alpha^{-1}G)$, whereas it attains zero advantage when the input is $(G, \alpha G, \beta G, \phi G)$.

$\mathcal{D}$ outputs 1 if $s' = s$ and 0 otherwise. Accordingly, the probability of $\mathcal{D}$ distinguishing the input tuples is

$$\left| \Pr[\alpha, \beta \xleftarrow{\$} \mathbb{Z}_p^* : \mathcal{A}(G, \alpha G, \beta G, \beta\alpha^{-1}G) = 1] - \Pr[\alpha, \beta, \phi \xleftarrow{\$} \mathbb{Z}_p^* : \mathcal{A}(G, \alpha G, \beta G, \phi G) = 1] \right| \approx \epsilon.$$

Thus, under the assumption of non-negligible ϵ , $\mathcal{D}$ distinguishes DDDH tuples with a non-negligible advantage, which violates the DDH assumption. It follows that, under the hardness of DDH, no PPT adversary $\mathcal{A}$ can achieve a non-negligible advantage in the signer anonymity game (Definition 15); therefore, L2S-LRS satisfies signer anonymity. $\square$

A-24. Proof of Lemma 24

Proof. Fix $(m, \mathbf{K})$ and an honest secret (x, s) with $K_s = xG$. In **Sign**, the signer computes the key image

$$\text{KI} := x \text{H}_{\text{gr}}(K_s) \in \mathbb{G}^*,$$

samples a fresh scalar $y \xleftarrow{\$} \mathbb{Z}_p^*$, and sets

$$Z := y \text{H}_{\text{gr}}(\text{KI}, K_s).$$

Both parties recompute the Fiat–Shamir scalar

$$\xi := \text{H}_{\text{sc}}(m, G, \mathbf{K}, \text{KI}, Z),$$

and then define, for every $i \in [0, n-1]$,

$$P_i := \text{H}_{\text{gr}}(\text{KI}, K_i) + \xi K_i + \xi^2 \text{KI}, \quad Q_i := \xi G + \xi^2 \text{H}_{\text{gr}}(K_i), \quad \mathbf{G} := (P_0, Q_0, \dots, P_{n-1}, Q_{n-1}).$$

The signer outputs a Fiat–Shamir transcript π for

$$\text{Lin2Selector}(n, \mathbf{G}, Z; z_0 = y, z_1 = -xy, s).$$

We show that the witness $(z_0, z_1, s) = (y, -xy, s)$ is valid for the derived statement $(n, \mathbf{G}, Z)$. Write the shorthands

$$U_i := \text{H}_{\text{gr}}(\text{KI}, K_i), \quad V_i := \text{H}_{\text{gr}}(K_i),$$

so that $U_s = \text{H}_{\text{gr}}(\text{KI}, K_s)$ and $V_s = \text{H}_{\text{gr}}(K_s)$. Using $K_s = xG$ and $\text{KI} = xV_s$, we expand at index s :

$$\begin{aligned} z_0 P_s + z_1 Q_s &= y(U_s + \xi K_s + \xi^2 \text{KI}) + (-xy)(\xi G + \xi^2 V_s) \\ &= yU_s + y\xi(xG) + y\xi^2(xV_s) - xy\xi G - xy\xi^2 V_s \\ &= yU_s \\ &= Z. \end{aligned}$$

Thus $(n, \mathbf{G}, Z; y, -xy, s) \in \mathcal{R}_{\text{Lin2Sel}}$ and an honest execution of **Lin2Selector** accepts w.o.p. by completeness (Lemma 14).

The verifier's recomputation of $\xi, \mathbf{G}$ matches the signer's. Since π is produced by applying FS to a public-coin protocol with completeness (namely, **Lin2Selector**), the NIZK verification of π succeeds except with negligible probability arising from the standard ROM edge cases (e.g., excluded RO outputs if H_{sc} is modeled into $\mathbb{Z}_p^*$). Hence $\text{Verify}(m, \mathbf{K}, \sigma) = 1$ except with negligible probability. This proves correctness of L2S-LRS-MS by Definition 12. $\square$

A-25. Proof of Lemma 25

Proof. This proof mirrors the proof of Lemma 20, for x moved into the nominator.

The proof is by Definition 14. If the same secret key signs twice, the key image is deterministically $\text{KI} = x \text{H}_{\text{gr}}(xG)$ and repeats, so **Link** = 1 always.

$$\Pr[\text{Link}(\sigma_1, \sigma_2) = 0 : x_1 = x_2] = 0.$$

If two different signers $x_1 \neq x_2$ sign, the key images are independent uniform in $\mathbb{G}^*$. In fact, multiplication by a fixed nonzero scalar is a bijection on $\mathbb{G}^*$, so KI_1, KI_2 are independent uniform in $\mathbb{G}^*$. Thus $\Pr[\text{KI}_1 = \text{KI}_2] = 1/(p-1)$, and hence

$$\Pr[\text{Link}(\sigma_1, \sigma_2) = 1 : x_1 \neq x_2] \leq \frac{1}{p-1} = \text{negl}(\lambda).$$

□

A-26. Proof of Lemma 26

Proof. Similar to the proof of Lemma 21, the FS transcript π is a non-interactive argument of knowledge for $\mathcal{R}_{\text{Lin2sel}}$ in the ROM (by Lemmas 14 and 15, via the *General Forking Lemma*, as is standard for the FS transformations [22]). Valid π enables the extraction of the witness $(z_0, z_1, s) \in \mathbb{Z}_p^{2*} \times [0, n-1]$ via Lemma 15, and also the validity of π implies $Z \neq 0$ by Lemma 17.

Throughout, for every $i \in [0, n-1]$, we denote

$$U_i := H_{\text{gr}}(\text{Kl}, K_i) \text{ and } V_i := H_{\text{gr}}(K_i).$$

The extractor is allowed to rewind *only* the unique random-oracle query to H_{sc} on input $(m, G, \mathbf{K}, \text{Kl}, Z)$, thereby producing forks with fresh independent ξ -values while keeping *all* H_{gr} -outputs (hence all U_i, V_i) fixed.

Step 0 (expand one extracted equation). Let (z_0, z_1, s) be extracted from π so that eq. (31) holds. Writing $U_s := H_{\text{gr}}(\text{Kl}, K_s)$ and $V_s := H_{\text{gr}}(K_s)$, and expanding P_s, Q_s , we obtain

$$Z = z_0 U_s + \xi \cdot (z_0 K_s + z_1 G) + \xi^2 \cdot (z_0 \text{Kl} + z_1 V_s). \quad (96)$$

Step 1 (prove $s = s'$ and also $z_0 \neq 0 \vee z'_0 \neq 0$ using two transcripts). Fork the RO answer for ξ to obtain a second accepting transcript π' on the same public input $(m, G, \mathbf{K}, \text{Kl}, Z)$, hence the same $\{U_i, V_i\}$, but with a fresh $\xi' \neq \xi$. Extract (z'_0, z'_1, s') from π' so that

$$Z = z'_0 U_{s'} + \xi' \cdot (z'_0 K_{s'} + z'_1 G) + \xi'^2 \cdot (z'_0 \text{Kl} + z'_1 V_{s'}). \quad (97)$$

Suppose toward a contradiction that $s \neq s'$. Subtract (97) from (96) and isolate the term $z_0 U_s - z'_0 U_{s'}$:

$$z_0 U_s - z'_0 U_{s'} = \xi' \cdot (z'_0 K_{s'} + z'_1 G) - \xi \cdot (z_0 K_s + z_1 G) + \xi'^2 \cdot (z'_0 \text{Kl} + z'_1 V_{s'}) - \xi^2 \cdot (z_0 \text{Kl} + z_1 V_s). \quad (98)$$

The event, when both of z_0, z'_0 are zero, has only negligible probability. To see why, insert $z_0 = z'_0 = 0$ into eq. (98):

$$(\xi' z'_1 - \xi z_1)G + \xi'^2 z'_1 V_{s'} - \xi^2 z_1 V_s = 0. \quad (99)$$

When $z_0 = z'_0 = 0$, the case of $z_1 = 0 \vee z'_1 = 0$ is excluded by the verifier's check $Z \neq 0$. Thus, eq. (99) yields a non-trivial known-coefficient linear relation between the fresh RO output V_s and $G, V_{s'}$, which occurs with only negligible probability by Lemma 18, for $s \neq s'$. Therefore, $z_0 \neq 0 \vee z'_0 \neq 0$ holds w.o.p.

Without loss of generality let $z_0 \neq 0$. Move $z'_0 U_{s'}$ in eq. (98) to the right-hand side and divide by z_0

$$U_s = z_0^{-1} \left(\xi' \cdot (z'_0 K_{s'} + z'_1 G) - \xi \cdot (z_0 K_s + z_1 G) + \xi'^2 \cdot (z'_0 \text{Kl} + z'_1 V_{s'}) - \xi^2 \cdot (z_0 \text{Kl} + z_1 V_s) + z'_0 U_{s'} \right). \quad (100)$$

The left-hand side of (100) is the fresh RO output U_s conditioned on Kl, K_s, G , whereas the right-hand side of (100) is a known-coefficients linear combination of group elements that are independent of U_s . Thus, (100) yields a known-coefficients linear representation of U_s with the elements independent of U_s , which by Lemma 18 occurs only with negligible probability unless DLR is broken. Therefore, it holds that $s = s'$ w.o.p.

Step 2 (prove $z_0 = z'_0 \neq 0$ using the same two transcripts). Having established $s = s'$, rewrite (98) as

$$(z_0 - z'_0)U_s = \xi' \cdot (z'_0 K_s + z'_1 G) - \xi \cdot (z_0 K_s + z_1 G) + \xi'^2 \cdot (z'_0 \text{Kl} + z'_1 V_s) - \xi^2 \cdot (z_0 \text{Kl} + z_1 V_s). \quad (101)$$

If $z_0 \neq z'_0$, divide by $(z_0 - z'_0) \in \mathbb{Z}_p^*$ and obtain an explicit known-coefficients representation of U_s in the span of $\{K_s, G, \text{Kl}, V_s\}$ with coefficients computable from $(\xi, \xi', z_0, z'_0, z_1, z'_1)$. Since G, Kl , and K_s are fixed *before* the RO samples U_s , this again contradicts Lemma 18 except with negligible probability. Hence, $z_0 = z'_0$ w.o.p.

Since $z_0 \neq 0 \vee z'_0 \neq 0$ w.o.p. by Step 1, this yields $z_0 = z'_0 \neq 0$ w.o.p.

Step 3 (prove $z_1 = z'_1$ using four transcripts). Now produce *four* accepting forks at once, with pairwise distinct challenges ξ, ξ', ξ'', ξ''' for the same fixed $(m, G, \mathbf{K}, \text{Kl}, Z)$, and extract witnesses

$$(z_0, z_1, s), \quad (z_0, z'_1, s), \quad (z_0, z''_1, s), \quad (z_0, z'''_1, s),$$

where s and z_0 are shared by Steps 1–2 (except with negligible probability). For each fork, (96) holds with the corresponding ξ and z_1 .

Introduce the vectors (in $\mathbb{Z}_p^4$)

$$\xi := (\xi, \xi', \xi'', \xi''')^T, \quad z := (z_1, z'_1, z''_1, z'''_1)^T, \quad \mathbf{1} := (1, 1, 1, 1)^T,$$

and elementwise powers/products $\xi^{\circ 2} := (\xi^2, \xi'^2, \xi''^2, \xi'''^2)^\top$, $\xi \odot z := (\xi z_1, \xi' z'_1, \xi'' z''_1, \xi''' z'''_1)^\top$, etc.

Let $\mu \in \mathbb{Z}_p^{4*}$ be a nonzero vector satisfying the Vandermonde-orthogonality conditions

$$\langle \mu, \mathbf{1} \rangle = 0, \quad \langle \mu, \xi \rangle = 0, \quad \langle \mu, \xi^{\circ 2} \rangle = 0. \quad (102)$$

The vector μ exists uniquely up to scaling, as ξ, ξ', ξ'', ξ''' are pairwise distinct, and can be found in a polynomial time.

Multiply each of the four expanded equations (96) (with their own ξ and z_1) by the corresponding μ -weight and sum them. By (102), the terms involving Z , U_s , K_s , and Kl vanish, leaving

$$\langle \mu, \xi \odot z \rangle G + \langle \mu, \xi^{\circ 2} \odot z \rangle V_s = 0. \quad (103)$$

Since V_s is sampled as an RO output on input G, K_s (implicit and explicit, fixed before sampling), the existence of a non-trivial known-coefficients relation between G and V_s as in (103) would violate the DLR by Lemma 18, except with negligible probability. Therefore, w.o.p. both coefficients must vanish:

$$\langle \mu, \xi \odot z \rangle = 0, \quad \langle \mu, \xi^{\circ 2} \odot z \rangle = 0. \quad (104)$$

Now use the linear-algebra characterization of μ : because (102) states that μ spans the orthogonal complement of $\text{span}\{\mathbf{1}, \xi, \xi^{\circ 2}\}$, the equalities (104) are equivalent to the existence of scalars $\alpha_0, \alpha_1, \alpha_2, \beta_0, \beta_1, \beta_2 \in \mathbb{Z}_p$, which can be found in a polynomial time, that satisfy

$$\xi \odot z = \alpha_0 \mathbf{1} + \alpha_1 \xi + \alpha_2 \xi^{\circ 2}, \quad \xi^{\circ 2} \odot z = \beta_0 \mathbf{1} + \beta_1 \xi + \beta_2 \xi^{\circ 2}. \quad (105)$$

Multiplying the first equality in (105) by ξ and equating both yields:

$$\xi^{\circ 2} \odot z = \alpha_0 \xi + \alpha_1 \xi^{\circ 2} + \alpha_2 \xi^{\circ 3} = \beta_0 \mathbf{1} + \beta_1 \xi + \beta_2 \xi^{\circ 2}, \quad (106)$$

$$\beta_0 \mathbf{1} + (\beta_1 - \alpha_0) \xi + (\beta_2 - \alpha_1) \xi^{\circ 2} + \alpha_2 \xi^{\circ 3} = 0. \quad (107)$$

The equality eq. (107) is a linear relation between four linearly independent vectors $\mathbf{1}, \xi, \xi^{\circ 2}, \xi^{\circ 3} \in \mathbb{Z}_p^{4*}$, which by linear algebra must be only trivial; this results in

$$\beta_0 = 0, \quad \beta_1 = \alpha_0, \quad \beta_2 = \alpha_1, \quad \alpha_2 = 0. \quad (108)$$

By inserting the coefficients from (108) to the first equality in (105), obtain $\mathbf{z}$:

$$\mathbf{z} = \alpha_0 \xi^{\circ -1} + \alpha_1 \mathbf{1}, \quad (109)$$

or, componentwise,

$$z_1^{(j)} = \alpha_0 / \xi_j + \alpha_1.$$

Substitute this form into the *difference* of two expanded equations (e.g., (96) minus (97), where s and z_0 are shared):

$$0 = z_0(\xi - \xi')K_s + z_0(\xi^2 - \xi'^2)\text{Kl} + (\xi z_1 - \xi' z'_1)G + (\xi^2 z_1 - \xi'^2 z'_1)V_s.$$

Using $\xi z_1 = \alpha_0 + \alpha_1 \xi$ and $\xi^2 z_1 = \alpha_0 \xi + \alpha_1 \xi^2$ (and similarly for ξ', z'_1), divide by $(\xi - \xi') \neq 0$ and obtain

$$z_0 K_s + z_0(\xi + \xi')\text{Kl} + \alpha_1 G + (\alpha_0 + \alpha_1(\xi + \xi'))V_s = 0. \quad (110)$$

Repeating the same derivation with (ξ, ξ'') in place of (ξ, ξ')

$$z_0 K_s + z_0(\xi + \xi'')\text{Kl} + \alpha_1 G + (\alpha_0 + \alpha_1(\xi + \xi''))V_s = 0. \quad (111)$$

Subtracting the two resulting eqs. (110) and (111) eliminates K_s and G and yields

$$z_0 \text{Kl} + \alpha_1 V_s = 0, \quad \text{which rewrites, recalling that } z_0 \neq 0, \text{ as } \text{Kl} = -(\alpha_1 / z_0) V_s. \quad (112)$$

Plugging eq. (112) back into eq. (110) collapses it to

$$z_0 K_s + \alpha_1 G + \alpha_0 V_s = 0. \quad (113)$$

If $\alpha_0 \neq 0$, then this is a known-coefficients linear representation of the random-oracle output V_s in the span of $\{K_s, G\}$, contradicting Lemma 18 except with negligible probability. Hence

$$\alpha_0 = 0 \quad \text{w.o.p.}, \quad (114)$$

and therefore eq. (109) becomes $\mathbf{z} = \alpha_1 \mathbf{1}$, that is $z_1^{(j)} = \alpha_1$ for all four forks, w.o.p. In particular,

$$\alpha_1 = z_1 = z'_1 \quad \text{w.o.p.} \quad (115)$$

Step 4 (recover $Z = yU_s$). By defining $x := -\alpha_1/z_0 \in \mathbb{Z}_p^*$, we have simultaneously from eqs. (112) to (115)

$$K_s = xG, \quad \text{KI} = xV_s = xH_{\text{gr}}(K_s), \quad z_1 = -xz_0. \quad (116)$$

Substitute (116) into the expanded equation (96):

$$Z = z_0U_s + \xi(z_0xG + z_1G) + \xi^2(z_0xV_s + z_1V_s) = z_0U_s.$$

By letting $y := z_0 \in \mathbb{Z}_p^*$, we obtain

$$Z = yU_s = yH_{\text{gr}}(\text{KI}, K_s),$$

and together with (116) this yields the claimed witness-binding statement.

Step 5: witness-extended emulation in the ROM. In total, a valid L2S-LRS-MS signature enables the extraction of the witness (z_0, z_1, s) satisfying eq. (31). Additionally, we have shown that five further forks yield w.o.p. the constraints (116) within the ROM. Therefore, the L2S-LRS-MS scheme satisfies cWEE in the ROM for the relation $\mathcal{R}_{\text{L2SIRS-MS}}$ (32). This completes the proof of the lemma. $\square$

A-27. Proof of Lemma 27

Proof. This proof mirrors the proof of Lemma 22, with the sole exception that the H_{gr} simulation returns $\hat{\alpha}G$ for the query $\hat{K}$, whereas the signing oracle simulation sets $\text{KI} = \hat{\alpha}\hat{K}$ accordingly. The new element Z is computed identically to its definition in the real signing procedure.

Let $\mathcal{A}$ be a PPT adversary that wins the EUF-CMA game (Definition 13) with non-negligible probability ϵ . We construct a simulator $\mathcal{B}$ that invokes $\mathcal{A}$ to solve the DLR problem for a challenge $\hat{K} = \hat{x}G$, where the discrete logarithm $\hat{x}$ is unknown to $\mathcal{B}$. The simulation is performed as follows.

Challenge embedding. Upon initialization, $\mathcal{B}$ receives the challenge public key $\hat{K}$, which is sampled independently from the public key distribution of KeyGen . $\mathcal{B}$ acts as the challenger $\mathcal{C}$ in the EUF-CMA game against $\mathcal{A}$. It samples a target index $\hat{j} \xleftarrow{\$} [0, N-1]$ and generates the key universe $\mathcal{L}$ by invoking KeyGen , with the exception of the $\hat{j}$ -th entry that it sets to $\hat{K}$. Thus, for $\mathcal{L} = \{K_i\}_{i=0}^{N-1}$, $\hat{K} = K_{\hat{j}} \in \mathcal{L}$.

RO simulation (lazy sampling and programming). $\mathcal{B}$ maintains tables for H_{sc} and H_{gr} and answers consistently:

- $H_{\text{sc}}(\dots) \in \mathbb{Z}_p^*$: on a new query, sample uniformly from $\mathbb{Z}_p^*$ and store.
- $H_{\text{gr}}(\dots) \in \mathbb{G}^*$: on the first query $\hat{K}$ (normally invoked when P_i 's or KI 's are constructed), sample $\hat{\alpha} \xleftarrow{\$} \mathbb{Z}_p^*$, store $\hat{\alpha}$, and return $H_{\text{gr}}(\hat{K}) := \hat{\alpha}G$, which is uniform in $\mathbb{G}^*$. For all other queries, forward the request to the standard oracle implementation and return the resulting uniformly random value from $\mathbb{G}^*$.

Programming is used only for FS challenge queries inside the simulated proofs, exactly as allowed by Definition 9.

Signing oracle simulation. $\mathcal{B}$ maintains a simulated signing oracle, which lets the underlying original SignOracle answer queries $(m, \mathbf{R})$, with the sole exception regarding $K_{\hat{j}}$.

Specifically, for a query where $\mathbf{R} \subseteq \mathcal{L}$, SignOracle deterministically chooses j such that $K_j \in \mathbf{R}$ and returns $\sigma \leftarrow \text{Sign}(x_j, m, \mathbf{R})$. However, if the chosen index j coincides with the target index $\hat{j}$, the simulator instead outputs a simulated signature $\hat{\sigma}$ produced without knowledge of the secret key, as follows.

- 1) Identify $s \in [0, |\mathbf{R}| - 1]$ such that $\hat{K} = K_{\hat{j}} = \mathbf{R}_{[s]}$ and treat s as the signing index.
- 2) Ensure the oracle value $H_{\text{gr}}(\hat{K})$ is defined. By the simulation above, it holds that $H_{\text{gr}}(\hat{K}) = \hat{\alpha}G$ for known $\hat{\alpha} \in \mathbb{Z}_p^*$. Set the simulated key image

$$\text{KI} := \hat{\alpha}\hat{K}.$$

It has identical distribution to a real key image $\hat{x}H_{\text{gr}}(\hat{K})$, because both of them are equal to $\hat{x}\hat{\alpha}G$.

- 3) Let $n = |\mathbf{R}|$. Query $\xi := H_{\text{sc}}(m, G, \mathbf{R}, \text{KI})$ and compute all P_i 's, Q_i 's exactly as in Figure 6. Furthermore, sample Z via the same formula as in Figure 6: $y \xleftarrow{\$} \mathbb{G}^*, Z \leftarrow yH_{\text{gr}}(\text{KI}, \hat{K})$.
- 4) Produce an FS transcript π for $\text{Lin2Selector}(n, \mathbf{G}, Z)$ using the sHVZK simulator of Lin2Selector round-by-round (Lemma 16), while *programming* H_{sc} on the corresponding FS challenge inputs so that verification recomputes the simulator-chosen challenges via the standard technique [22], [25].
- 5) Output $\hat{\sigma} := (\text{KI}, Z, \pi)$.

The simulated signature $\hat{\sigma}$ is statistically indistinguishable from a real signature. This follows from the fact that the simulated KI, Z are identically distributed to the genuine values, while the remaining component π is statistically identical to that of a valid signature conditioned on these KI, Z .

Discovering unknown discrete logarithm. $\mathcal{B}$ lets $\mathcal{A}$ run within the simulated environment, i.e., using the simulated $\text{H}_{\text{sc}}, \text{H}_{\text{gr}}$, and signing oracle. Since these simulations are indistinguishable from the authentic oracles and produce identical distributions, $\mathcal{A}$ outputs a forgery with its original probability ϵ .

Once $\mathcal{A}$ outputs a successful forgery $\sigma^* = (m^*, \mathbf{K}^*, \text{KI}^*, Z^*, \pi^*)$ such that $\mathbf{K}^* = \{K_i\}_{i=0}^{|\mathbf{K}^*|-1} \subseteq \mathcal{L}$ and $\text{Verify}(\sigma^*) = 1$, $\mathcal{B}$ invokes the cWEE extractor for the accepting FS transcript π^* of the Lin2Selector protocol in the ROM. This extractor, which exists by Lemma 15 following the methodology in [22], [32], is guaranteed to succeed except with negligible probability. Given that Lin2Selector is complete and rejects inputs outside the $\mathcal{R}_{\text{Lin2Sel}}$ relation (Lemmas 14 and 17), the extractor outputs the valid witness (z_0, z_1, s) .

Thus, $\mathcal{B}$ has the valid witness (z_0, z_1, s) obtained from π via the corresponding extractor with the probability, in sum, $(\epsilon - \text{negl}(\lambda))$. By Lemma 26, this witness satisfies

$$K_s = xG, \text{ where } x = -z_1/z_0, \quad (117)$$

except with negligible probability.

As $\mathbf{K}^* \subseteq \mathcal{L}$, the public key K_s is an element of the key universe $\mathcal{L}$. Given that the challenge key $\hat{K}$ is uniformly distributed across $\mathcal{L}$, the probability of the event $K_s = \hat{K}$ is at least $1/N$. Consequently, the overall success probability for $\mathcal{B}$ to extract x satisfying eq. (117) for $K_s = \hat{K}$ is $(\epsilon/N - \text{negl}(\lambda))$. This probability is *non-negligible* for any non-negligible ϵ ; thus, the existence of the adversary $\mathcal{A}$ contradicts the hardness of the DLR, and we conclude that by Definition 13, under the DLR assumption, L2S-LRS-MS is EUF-CMA secure against adversarially-chosen-key attacks. $\square$

A-28. Proof of Lemma 28

Proof. This proof is similar to that of Lemma 23; noting that now KI takes the form $x\text{H}_{\text{gr}}(xG)$, the reduction relies on the standard DDH rather than DDDH tuples. Furthermore, the introduction of the element Z into the signature necessitates an additional game and a corresponding DDH distinguisher.

Fix any message m , ring size n , and corrupted key set size $t < n - 1$. Using the sequence-of-games technique [41] we demonstrate that any PPT adversary $\mathcal{A}$ winning the anonymity game (Definition 15) with a non-negligible advantage over random guessing can be reduced to a DDH distinguisher (Definition 16).

First game. Let us denote by $\mathcal{G}_1$ the anonymity game in Definition 15, and suppose that a PPT $\mathcal{A}$ wins it with a non-negligible advantage ϵ over random guessing among $(n - t)$ uncorrupted indices.

Second game. Denoted by $\mathcal{G}_2$, it is a slightly modified $\mathcal{G}_1$: prior to sending $(\mathbf{R}, D_t, \sigma = (\pi, \text{KI}, Z))$ to $\mathcal{A}$, the challenger $\mathcal{C}$ replaces the Lin2Selector FS transcript π within σ with a simulated sHVZK transcript π^* , which exists by Lemma 16. For the simulation π^* , $\mathcal{C}$ uses the same public inputs and RO challenges (programming H_{sc} as needed) as for π . Then, completing the modification, $\mathcal{C}$ hands over the triplet $(\mathbf{R}, D_t, \sigma^* = (\pi^*, \text{KI}, Z))$ to $\mathcal{A}$.

In $\mathcal{G}_2$, $\mathcal{A}$ maintains its success probability with a non-negligible advantage $\epsilon - \text{negl}(\lambda)$. Any non-negligible shift in the advantage would enable $\mathcal{A}$ to distinguish between real and simulated transcripts in the sHVZK game (Definition 7), contradicting the sHVZK property of Lin2Selector established in Lemma 16.

Third game. Now, consider game $\mathcal{G}_3$, which is a slight modification of $\mathcal{G}_2$: instead of generating the honest key image $\text{KI} = x_s\text{H}_{\text{gr}}(x_sG)$, the challenger $\mathcal{C}$ fabricates it as $\phi \xleftarrow{\$} \mathbb{Z}_p^*$, $\text{KI}^* = \phi G$. Then $\mathcal{C}$ honestly computes Z by sampling $y \xleftarrow{\$} \mathbb{Z}_p^*$ and setting $Z \leftarrow y\text{H}_{\text{gr}}(\text{KI}, x_sG)$. After that, $\mathcal{C}$ proceeds as in $\mathcal{G}_2$; namely, it generates a simulated sHVZK transcript π^* and hands over $(\mathbf{R}, D_t, \sigma^* = (\pi^*, \text{KI}^*, Z))$ to $\mathcal{A}$.

Suppose toward a contradiction that, in $\mathcal{G}_3$, $\mathcal{A}$ has an advantage ϵ' over random guessing among the $n - t$ uncorrupted indices, such that $|\epsilon' - \epsilon|$ is non-negligible. In this case, we can construct the following DDH distinguisher.

First distinguisher. The DDH distinguisher algorithm $\mathcal{D}_1$ 'interpolates' between the games $\mathcal{G}_2$ and $\mathcal{G}_3$ in the sense of [41]. Upon receiving a DDH tuple (G, A, B, C) instantiated outside in one of two forms: $(G, \alpha G, \beta G, \beta \alpha G)$ or $(G, \alpha G, \beta G, \phi G)$ for $\alpha, \beta, \phi \xleftarrow{\$} \mathbb{Z}_p^*$, the distinguisher proceeds as follows.

$\mathcal{D}_1$ generates $\mathbf{R}$ and samples an uncorrupted index s , in line with $\mathcal{C}$ in $\mathcal{G}_2$ and $\mathcal{G}_3$. It then substitutes A for the s -th entry in $\mathbf{R}$, programs the random oracle H_{gr} to return $\text{H}_{\text{gr}}(A) = B$, and sets $\text{KI} = C$. Note that, by this, KI actually gets either $\beta \alpha G$, which makes it equal to KI in $\mathcal{G}_2$, or ϕG , which makes it equal to KI^* in $\mathcal{G}_3$. Subsequently, in line with $\mathcal{C}$ in $\mathcal{G}_2$ and $\mathcal{G}_3$, $\mathcal{D}_1$ produces the honest Z along with simulated π^* based on KI, Z , and hands over $(\mathbf{R}, D_t, \sigma^* = (\pi^*, \text{KI}, Z))$ to $\mathcal{A}$, which outputs s' .

Up to this point, $\mathcal{D}_1$ faithfully simulates one of the games $\mathcal{G}_2$ or $\mathcal{G}_3$; which one is determined solely by the input tuple form. Therefore, $\mathcal{A}$ retains the advantage of $\epsilon - \text{negl}(\lambda)$ in identifying the correct index s provided the input is $(G, \alpha G, \beta G, \beta \alpha G)$, whereas it attains the advantage ϵ' when the input is $(G, \alpha G, \beta G, \phi G)$.

$\mathcal{D}_1$ outputs 1 if $s' = s$ and 0 otherwise. Accordingly, the probability of $\mathcal{D}_1$ distinguishing the input tuples is

$$\left| \Pr[\alpha, \beta \xleftarrow{\$} \mathbb{Z}_p^* : \mathcal{A}(G, \alpha G, \beta G, \beta \alpha G) = 1] - \Pr[\alpha, \beta, \phi \xleftarrow{\$} \mathbb{Z}_p^* : \mathcal{A}(G, \alpha G, \beta G, \phi G) = 1] \right| \approx |\epsilon' - \epsilon|.$$

Thus, under the DDH assumption (Definition 16), $|\epsilon' - \epsilon|$ is negligible, which implies that $\mathcal{A}$'s advantages over random guessing in $\mathcal{G}_3$ and $\mathcal{G}_1$ differ only negligible: $\epsilon' \approx \epsilon$.

Fourth game. Game $\mathcal{G}_4$ is a slight modification of $\mathcal{G}_3$: instead of generating Z honestly as $Z \leftarrow yH_{\text{gr}}(\text{KI}^*, x_s G)$, where $y \xleftarrow{\$} \mathbb{Z}_p^*$, the challenger $\mathcal{C}$ fabricates it as $\gamma \xleftarrow{\$} \mathbb{Z}_p^*$, $Z^* = \gamma G$. Then $\mathcal{C}$ proceeds as in $\mathcal{G}_3$: it generates a simulated sHVZK transcript π^* and hands over $(\mathbf{R}, D_t, \sigma^* = (\pi^*, \text{KI}^*, Z^*))$ to $\mathcal{A}$.

In $\mathcal{G}_4$, the signature σ^* is statistically independent of the signing index s , i.e., σ^* is void of any values related to s , x_s , or $H_{\text{gr}}(\dots, x_s G)$, because it is simulated for the fabricated key image KI^* and Z^* . Consequently, in $\mathcal{G}_4$, $\mathcal{A}$ attains zero advantage in identifying s , which is chosen uniformly at random from the set of $n - t$ uncorrupted indices.

Second distinguisher. The DDH distinguisher algorithm $\mathcal{D}_2$ 'interpolates' between the games $\mathcal{G}_3$ and $\mathcal{G}_4$. Upon receiving a DDH tuple (G, A, B, C) instantiated in one of two forms: $(G, \alpha G, \beta G, \beta \alpha G)$ or $(G, \alpha G, \beta G, \phi G)$ for $\alpha, \beta, \phi \xleftarrow{\$} \mathbb{Z}_p^*$, the distinguisher proceeds as follows.

$\mathcal{D}_2$ generates $\mathbf{R}$, samples an uncorrupted index s , and does everything in line with $\mathcal{C}$ in $\mathcal{G}_3$ and $\mathcal{G}_4$, except for that $\mathcal{D}_2$ programs H_{gr} to return $H_{\text{gr}}(\text{KI}^*, K_s) = A$ and sets $Z = C$. Note, $\mathcal{D}_2$ does not use B at all. $\mathcal{D}_2$ hands over $(\mathbf{R}, D_t, \sigma^* = (\pi^*, \text{KI}^*, Z))$ to $\mathcal{A}$, which outputs s' .

Up to this point, $\mathcal{D}_2$ faithfully simulates one of the games $\mathcal{G}_3$ or $\mathcal{G}_4$; which one is determined solely by the input tuple form. Consequently, $\mathcal{A}$ retains the advantage of ϵ' in identifying the correct index s provided the input is $(G, \alpha G, \beta G, \beta \alpha G)$, whereas it attains zero advantage when the input is $(G, \alpha G, \beta G, \phi G)$.

$\mathcal{D}_2$ outputs 1 if $s' = s$ and 0 otherwise. Accordingly, the probability of $\mathcal{D}_2$ distinguishing the input tuples is

$$\left| \Pr[\alpha, \beta \xleftarrow{\$} \mathbb{Z}_p^* : \mathcal{A}(G, \alpha G, \beta G, \beta \alpha G) = 1] - \Pr[\alpha, \beta, \phi \xleftarrow{\$} \mathbb{Z}_p^* : \mathcal{A}(G, \alpha G, \beta G, \phi G) = 1] \right| \approx \epsilon' \approx \epsilon.$$

Thus, under the supposition of non-negligible ϵ , $\mathcal{D}_2$ distinguishes the DDH tuples with a non-negligible advantage, which violates the DDH assumption. It follows that, under the hardness of DDH, no PPT adversary $\mathcal{A}$ can achieve a non-negligible advantage in the signer anonymity game (Definition 15); therefore, L2S-LRS-MS satisfies signer anonymity. $\square$